



Arab Academy for Science, Technology and Maritime Transport

College of Artificial Intelligence

Intelligent Systems Department

B. Sc. Final Year Project

NADEL - FULLY AUTONOMOUS SERVICE ROBOT

An Autonomous Restaurant Service Robot for Food Delivery

Presented by :

Youssef Ibrahim

Abdallah Elsayed

Abdelrahman Mostafa

Abdelrahman Eid

Youssef Elfeshawy

Yusuf Osama

Supervised by :

Dr. Ahmed Abouelfarag

Eng. Mohamed Elsayed

DECLARATION

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ARAB ACADEMY FOR SCIENCE, TECHNOLOGY AND MARITIME TRANSPORT

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Senior Project Summary Report

Project Title	Nadel - Fully Autonomous Service Waiter Robot	
Supervisor(s)	Dr. Ahmed Abouelfarag Eng. Mohamed Elsayed	
Team members:	Youssef Ibrahim Abdallah Elsayed Abdelrahman Mostafa Abdelrahman Eid Youssef Elfeshawy Yusuf Osama	221000718 221003814 221004536 221010111 221010399 20100041
Project Deliverables	<i>Nadel is a waiter robot prototype with autonomous navigation, Arabic/English interaction</i>	
Abstract	<i>Nadel is an autonomous waiter robot designed to support restaurant and café operations by delivering food and drinks to customer tables. It uses Jetson Orin Nano, sensors, motors, ESP32 modules, and an ESP32-based Call/Pay unit that sends the table ID and request type so the robot can navigate to the customer.</i> <i>The robot uses ROS2 Nav2 for navigation and obstacle avoidance, along with an Arabic/English interactive interface supported by Whisper Small, Gemma 31B, and speech output. Nadel aims to</i>	

Project Title	Nadel - Fully Autonomous Service Waiter Robot
	<i>reduce staff workload, improve service speed, minimize delivery errors, and enhance customer experience.</i>
System Standards	<i>The project follows ROS2-based design, structured hardware integration, map-based navigation.</i>
System Constraints	<i>The main constraints include prototype cost, indoor operating environment, sensor reliability, odometry drift, voice interaction reliability in noisy environments, dynamic obstacles in restaurant spaces, and the need for safe human-aware motion.</i>
Project Impact	<i>The project supports staff, improves service efficiency, and demonstrates the integration of AI, robotics, embedded systems, and restaurant automation.</i>
Team Organisation	<i>The team covers mechanical design, hardware integration, ROS2 navigation, perception, UI, voice interaction, testing, documentation, and business analysis.</i>
Ethics /Safety	<i>The system is designed for safe indoor operation through obstacle detection, stable design, sensor placement.</i>

Main Supervisor Signature

.....

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ABSTRACT

The hospitality industry faces significant operational challenges, including high labor costs, staff shortages during peak hours, slow service, human errors, and inconsistent service quality caused by fatigue or lack of experience. This project presents Nadel, a fully autonomous service waiter robot designed to assist restaurant and café service by delivering food and drinks from the kitchen to customer tables safely and efficiently. The robot aims to reduce repetitive staff workload, improve service speed, minimize wrong deliveries, and enhance the overall customer experience.

Nadel combines autonomous navigation, obstacle avoidance, mapping, localization, human-robot interaction, and smart restaurant workflow integration. The hardware architecture is based on the NVIDIA Jetson Orin Nano for high-performance processing, the RPLIDAR A1M8 for 360-degree environment scanning, ultrasonic sensors for short-range obstacle detection, motor encoders for odometry, an ESP32 microcontroller for low-level communication, and DC motors with motor drivers for movement. The software architecture uses ROS2 Nav2 for navigation and path planning, table-based goal selection, a web-based facial interface implemented using HTML, CSS, and JavaScript, UI support, and speech-to-text, Gemma 31B request processing, and speech output. The proposed workflow connects order placement, robot delivery assignment, table identification, delivery tracking. Overall, the project demonstrates how an autonomous service robot can support human workers and improve restaurant service efficiency.

In addition, the proposed system is designed as a modular and scalable platform that can be adapted to different restaurant layouts and service requirements. The navigation system allows the robot to move between the kitchen area and predefined customer tables while continuously detecting obstacles and updating its movement path when necessary. The interaction interface enables customers to communicate with the robot through touch, voice, and multilingual visual feedback, making the system suitable for diverse users.

Keywords: Autonomous Service Robot; ROS2 Navigation; Computer Vision; SLAM; Human-Robot Interaction; Mobile Robotics; Smart Restaurant System.

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LIST OF ACRONYMS/ABBREVIATIONS

Acronym	Definition
AASTMT	Arab Academy for Science, Technology and Maritime Transport
AI	Artificial Intelligence
ESP32	Espressif 32-bit Microcontroller
Gemma	Large Language Model used for request interpretation and response
HRI	Human-Robot Interaction
IoT	Internet of Things
KDS	Kitchen Display System
LiDAR	Light Detection and Ranging
LLM	Large Language Model
Nav2	ROS2 Navigation Stack
NUC	Next Unit of Computing
ROS2	Robot Operating System 2
SLAM	Simultaneous Localization and Mapping
SOL	Speech Output Layer
STT	Speech-to-Text
UI	User Interface
Whisper	Speech Recognition Model

Chapter One

1 INTRODUCTION

This section introduces the background, motivation, and main challenges behind the Nadel project. Restaurants and cafés often face service delays, repetitive staff workload, and difficulty maintaining consistent service quality during busy periods. Nadel is proposed as an autonomous waiter robot that supports indoor delivery tasks through navigation, table-side service requests, touch-screen and voice interaction, and sensor-based obstacle awareness. The following subsections explain the project context, the motivation for developing Nadel, and the practical challenges addressed by the proposed system.

1.1 BACKGROUND, MOTIVATION, AND CHALLENGES

The rapid development of artificial intelligence, robotics, embedded systems, autonomous navigation, and smart service technologies has created new opportunities for improving service-based industries. Restaurants, cafés, hotels, and food-service environments are no longer limited to traditional manual workflows. They are increasingly adopting customer self-service interfaces, and intelligent monitoring tools. Within this digital transformation, autonomous service robots represent an important direction because they connect digital restaurant management with physical service execution [1].

In restaurant environments, service quality is strongly linked to speed, accuracy, consistency, and customer satisfaction. A restaurant may have a good menu and a well-designed interior, but slow service, repeated delays, or wrong deliveries can negatively affect the customer experience. During peak hours, staff members may have to receive orders, move between tables, communicate with the kitchen, deliver food, answer customer requests. These tasks are necessary, but many of them are repetitive and physically demanding. In addition to the touch screen interface, and voice-based interaction, Nadel includes an ESP32-based table-side interaction unit installed on customer tables. This unit contains two physical buttons: Call and Pay. These buttons do not directly perform service or payment operations. Instead, they send the table ID and

request intent to the system. When the Call button is pressed, the system understands that the customer requires general service interaction. When the Pay button is pressed, the system understands that the customer requires payment-related assistance. In both cases, Nadel can be assigned to navigate to the requesting table, and the customer interaction continues after the robot arrives [16].

This project introduces Nadel - Fully Autonomous Service Waiter Robot, an autonomous mobile service robot designed to support restaurant and café operations by delivering food and drinks from the kitchen area to customer tables. The robot is designed to operate inside indoor service environments, move safely between tables, avoid obstacles, interact with users through a visual interface and voice output. The aim is not to remove human service from restaurants, but to assist human staff by handling repetitive delivery tasks and reducing operational pressure during busy service periods.

Nadel is developed as a multidisciplinary graduation project that combines artificial intelligence, robotics, mechanical design, embedded control, web technologies, human-robot interaction, and business analysis. From a technical perspective, the robot integrates sensing, processing, planning, control, and user interaction. From an operational perspective, it supports restaurant order flow by connecting table identification, kitchen preparation, and delivery assignment. From a business perspective, it targets restaurants and cafés that need to improve service speed, reduce repeated staff movement, and create a modern customer experience.

The physical prototype of Nadel demonstrates the transformation of the project from an initial design model into a functional robotic platform. The prototype includes a mobile base, service body, tray structure, customer-facing display area, and enclosed body design. The robot appearance is intentionally friendly because it is expected to operate near customers rather than in an isolated industrial environment. A restaurant service robot must be technically reliable, but it must also look acceptable and comfortable for customers and staff.



Figure 1-1: Front view of Nadel robot prototype.

Figure 1-1 shows the front view of the Nadel prototype. This view represents the customer-facing side of the robot and shows the upper display section, the main service body, the lower mobile base, and the project identity on the front panel. The figure is important because the front appearance is the first part that customers see when the robot approaches a table. A friendly visual design helps improve user acceptance and supports the human-robot interaction goals of the project.

From the hardware perspective, Nadel is based on a set of integrated components. The main processing unit is the NVIDIA Jetson Orin Nano, which is used to support navigation, perception, and control algorithms. For environmental perception, the robot uses an RPLIDAR A1M8 laser scanner. The robot also uses ultrasonic sensors for short-range obstacle detection, motor encoders for odometry, DC brushed motors for movement, motor drivers for motor control, and an ESP32 microcontroller for wireless communication and low-level motion control [13], [14].

From the software perspective, Nadel is designed around ROS2 Nav2, which provides the navigation framework for autonomous movement. ROS2 Nav2 supports map-based navigation, goal selection, path planning, obstacle avoidance, and motion execution. The robot can use an occupancy grid map of the restaurant environment, where fixed elements such as walls, furniture, and tables are represented. Each table can be linked to a unique table ID and coordinates, allowing the system to translate restaurant orders into robot navigation goals [9], [12].

The robot includes a screen-based interface that can display a robot face and restaurant menu. The facial interface is implemented using web technologies such as HTML, CSS, and JavaScript. It can be deployed within the Ubuntu and ROS environment and controlled through ROS topic messages. This allows the face display to reflect system states and create a more interactive customer experience. The project also includes multilingual interface support in Arabic and English, which improves usability in local restaurant environments and supports wider deployment possibilities.

Speech-to-text and voice-based interaction are important components of the human-robot interaction layer in Nadel. The current implementation uses Whisper Small as the speech-to-text module to convert customer voice input into text. This

allows the robot to receive spoken customer requests and transform them into a format that can be processed by the system [18].

The transcribed text is then processed using Gemma 31B, which helps interpret the meaning of the customer's request and generate an appropriate response. This enables the robot to understand basic service-related interactions, such as customer requests, order-related commands, confirmation messages, and simple conversational inputs within the restaurant workflow [20].

By combining Whisper Small and Gemma 31B, Nadel provides a more natural interaction layer between the customer and the robot. This integration improves the robot's ability to understand spoken input, support hands-free communication, and respond intelligently according to the service context [18], [20].

Nadel is also connected conceptually and functionally to a smart restaurant workflow. A waiter or customer can place an order through a tablet, user interface, or planned voice command. When the order is ready, Nadel receives information such as table number and item details, moves toward the correct table, delivers the food, and supports order tracking. This closes the gap between digital order management and physical service delivery.

High-Level System Concept of Nadel Robot

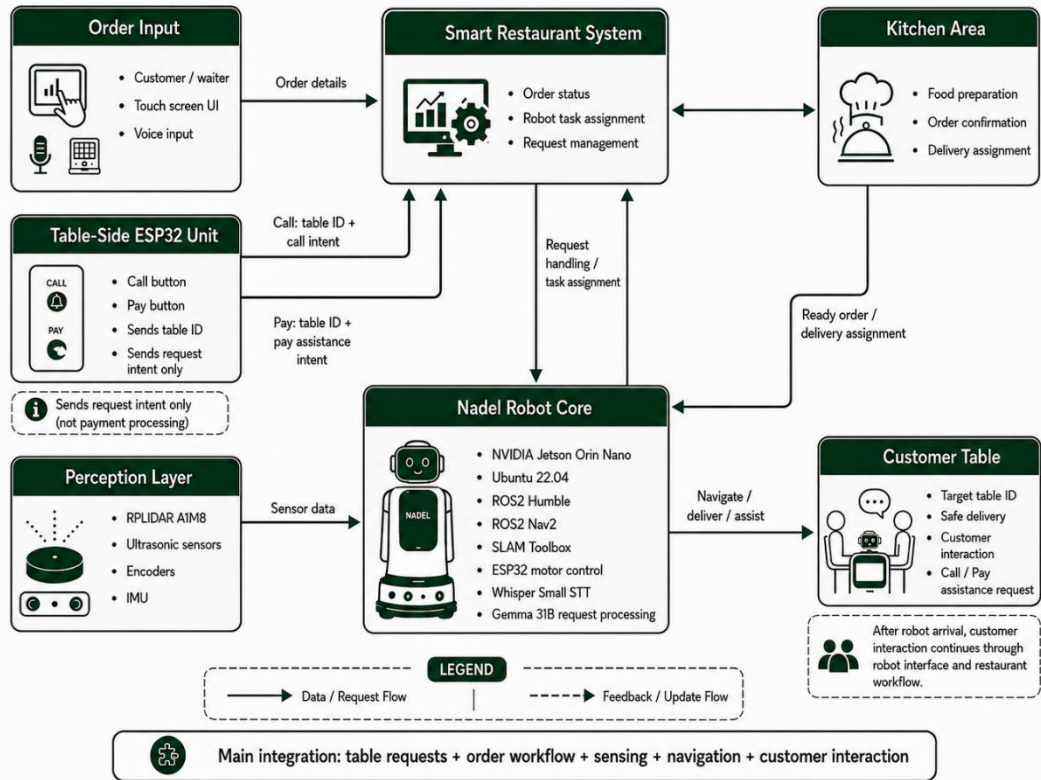


Figure 1-2: High-level system concept of Nadel robot.

Figure 1-2 presents the high-level system concept of Nadel robot and shows that the robot operates as part of an integrated restaurant service ecosystem rather than as a standalone mobile platform. The diagram includes the order input layer, table-side ESP32 unit, smart restaurant system, kitchen area, perception layer, Nadel robot core, and customer table [16].

The order input layer allows a customer or waiter to interact with the system through the touch screen interface, or voice input. In addition, the table-side ESP32 unit provides two physical buttons: Call and Pay. These buttons send the table ID and request intent only. The Call button indicates that the customer needs service interaction, while the Pay button indicates that the customer needs payment-related assistance. The buttons do not directly perform service or payment operations [16].

The smart restaurant system receives order information and table-side requests, manages request status, assigns robot tasks, and connects the robot with the restaurant workflow. The kitchen area prepares and confirms orders before delivery assignment. The perception layer provides sensor data through the RPLIDAR A1M8, Intel RealSense D435i, ultrasonic sensors, encoders, and IMU. The Nadel robot core processes navigation, control, request interpretation, and interaction using the Jetson Orin Nano, ROS2 Humble, ROS2 Nav2, SLAM Toolbox, ESP32 motor control, Whisper Small, Gemma 31B, and the speech output layer [9], [12].

The diagram highlights the main integration of table-side requests, sensing, navigation, and customer interaction. It also clarifies that after Nadel arrives at the customer table, the interaction continues through the robot interface and restaurant/POS workflow.

Table 1-1: General overview of the Nadel project.

Element	Description
Project name	Nadel - Fully Autonomous Service Waiter Robot
Main field	Autonomous service robotics for restaurants and cafés
Main function	Delivering food and drinks from the kitchen area to customer tables, responding to table-side Call/Pay assistance requests, and supporting customer interaction after robot arrival.
Core technologies	ROS2 Humble, ROS2 Nav2, SLAM Toolbox, LiDAR mapping, ultrasonic sensing, encoder odometry, IMU feedback, embedded control, touch screen UI, ESP32-based table-side Call/Pay unit, Whisper Small STT, Gemma 31B, and speech output layer.
Target users	Restaurants, cafés, hospitality businesses, and technology-based service environments
Main goal	Improve service speed, reduce repetitive workload, minimize errors, and enhance customer experience

The motivation behind this project comes from the operational challenges faced by restaurants and cafés and from the growing role of robotics in service environments. Restaurants depend on speed, accuracy, and customer satisfaction. However, many restaurants experience high workload during peak hours, especially when the number of customers increases and staff members must respond to several service requests at the same time. This creates pressure on workers and may lead to slower service and repeated mistakes.

One of the main motivations for developing Nadel is reducing repetitive physical workload. In a typical restaurant, waiters repeatedly move between the kitchen and customer tables. This movement may seem simple, but it consumes time and effort throughout the day. A waiter may walk many times to deliver food, return empty trays, check table status, and respond to customer requests. When these repetitive movements are handled by a robot, human staff can focus more on customer communication, problem solving, hospitality, and quality control [1].

Another motivation is reducing operational cost. Labor cost is one of the important expenses in restaurant operations. Although robots require initial investment, maintenance, and technical support, they can provide long-term value by improving efficiency and supporting staff during busy hours. Nadel can reduce the need for additional temporary workers during peak periods and can make service processes more organized. This does not mean replacing staff completely, but using automation to support staff productivity.

Service consistency is also a major motivation. Human performance can change depending on fatigue, workload, experience, and stress. During crowded periods, staff members may deliver the wrong order, forget items, delay delivery, or serve tables in the wrong sequence. Nadel can help reduce these problems by following a structured digital workflow. When an order is linked to a table ID, the robot receives the destination and follows the planned route to the correct table. This makes the delivery process more repeatable and less dependent on human memory.

The project is also motivated by customer experience. Many customers are interested in modern, technology-enhanced service experiences. A waiter robot can make a restaurant more attractive, especially for customers who enjoy innovative technologies. The screen-based face, multilingual interface, and voice output can create a friendly interaction. This can support restaurant branding and provide an experience that differentiates the restaurant from competitors.

Nadel has educational motivation as well. The project allows students to apply theoretical knowledge in a practical engineering system. It requires understanding of robot kinematics, odometry, mapping, path planning, sensor integration, embedded systems, web development, and business planning. Instead of studying these topics separately, the project combines them into one complete prototype. This makes it suitable for a final year project in the College of Artificial Intelligence [12].

The primary application of Nadel is restaurant food delivery. In this application, the robot receives order details from the restaurant system, identifies the correct table, moves through the environment, avoids obstacles, and delivers the food. This can be useful in large restaurants, busy cafés, hotel restaurants, food courts, and service halls. The robot can be especially useful in places where the distance between kitchen and tables is large or where staff members experience high walking load.

Another application is café service. Cafés often have repeated orders such as coffee, tea, desserts, and light meals. Nadel can transport these items from the preparation area to the customer seating zone. With a suitable tray structure and stable movement, the robot can support beverage and dessert delivery while staff members focus on preparation and customer relations.

Hotels can also benefit from similar robotic service concepts. Robots can be used for lobby delivery, event service, conference support, and room service in controlled indoor areas. Although Nadel is designed mainly for restaurants and cafés, the same core technologies can be extended to hotel service and indoor delivery applications.

Nadel can also be used in exhibitions and technology events. The prototype appearance and autonomous behavior make it suitable for demonstration. A service robot can attract visitors, show practical artificial intelligence applications, and demonstrate how robotics can be integrated into business processes. This is useful for academic exhibitions, graduation project fairs, innovation events, and industrial demonstrations.

Another possible application is controlled delivery in healthcare or institutional settings. In hospitals, clinics, laboratories, or care facilities, delivery robots can transport items while reducing direct contact and repeated staff movement. Although Nadel is not designed as a medical robot, its indoor navigation, obstacle avoidance, and delivery workflow can inspire similar systems for non-clinical item transportation.

The motivation and applications of Nadel can therefore be understood from several perspectives. Operationally, it improves restaurant workflow. Technically, it demonstrates autonomous robot integration. Socially, it supports staff and improves customer interaction. Economically, it can reduce repeated workload and service delays. Educationally, it provides a practical platform for applying intelligent systems knowledge.

Another motivation is improving customer access to service. In traditional restaurants, customers may need to wait until a waiter notices them before requesting assistance or asking for payment support. The ESP32-based table-side Call/Pay unit reduces this waiting time by allowing customers to send direct request intent from their table. The Call button requests general service interaction, while the Pay button requests payment-related assistance. In both cases, the robot can be sent to the correct table, improving responsiveness and making the restaurant workflow more organized [16].

Restaurant Service Workflow Supported by Nadel

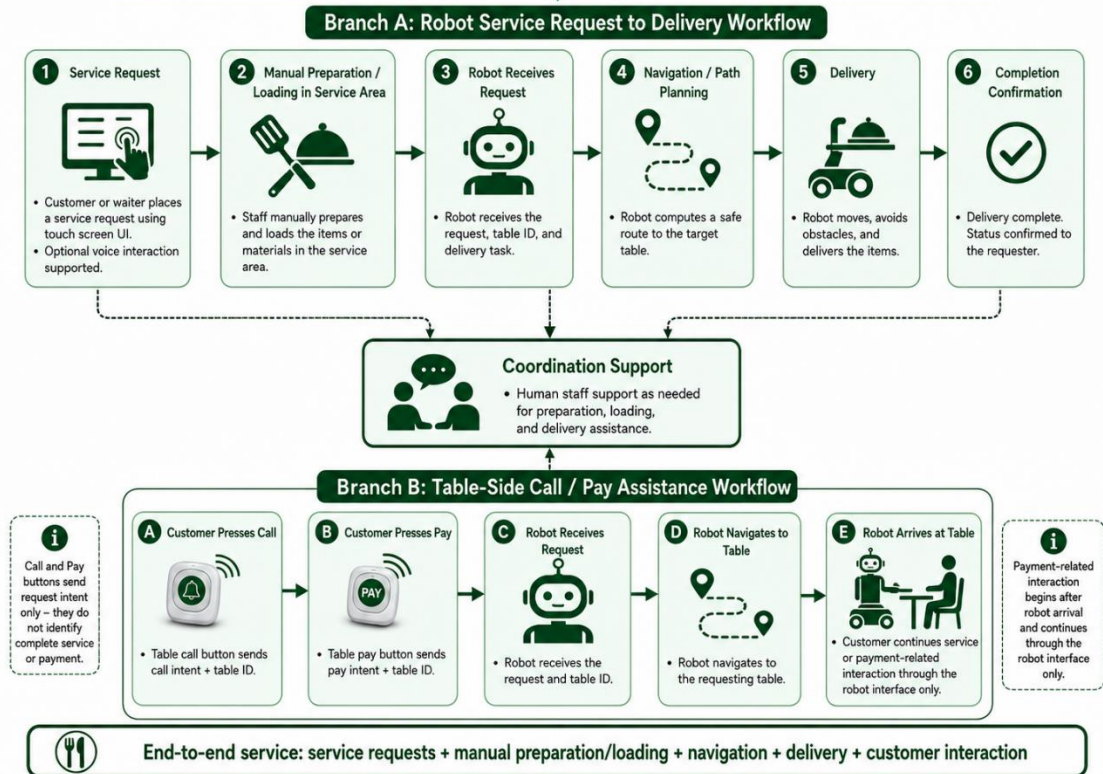


Figure 1-3: Restaurant service workflow supported by Nadel.

Figure 1-3 summarizes the restaurant service workflow supported by Nadel. The diagram shows two main workflow branches. The first branch is the order and delivery workflow, which starts when a customer or waiter places an order using the touch screen interface, POS system, or restaurant interface. The order is then sent to the kitchen display system for preparation. After the order is ready, Nadel receives the table ID, order items, and delivery task. ROS2 Nav2 computes a safe route to the target table, and the robot moves through the restaurant while avoiding obstacles until it delivers the order. The order status is then updated in the restaurant system [9].

The second branch is the table-side Call/Pay assistance workflow. When the customer presses the Call button, the ESP32 table unit sends the table ID and call intent to the smart restaurant system. When the customer presses the Pay button, the table unit sends the table ID and pay assistance intent. In both cases, the system identifies the table and request type, assigns Nadel to navigate to the requesting table, and allows the customer to continue the service or payment-related interaction after the robot arrives [16].

Table 1-2: Motivation and possible applications of Nadel.

Motivation / Application	Explanation
Reducing workload	The robot can handle repeated delivery movement between kitchen and tables.
Improving service speed	Autonomous delivery can support faster table service during peak hours.
Reducing errors	Digital table ID and order assignment can reduce wrong deliveries.
Enhancing customer experience	Friendly robot appearance, screen face, and voice output can create a modern experience.
Restaurant and café use	The main application is food and drink delivery in indoor hospitality environments.
Hotel and event support	The concept can be extended to controlled indoor delivery in hotels and exhibitions.
Table-side service access	The Call button allows customers to request robot attendance for general service interaction.
Payment assistance request	The Pay button allows customers to request Nadel for payment-related interaction without directly processing payment.

Developing an autonomous waiter robot is challenging because restaurant environments are dynamic, crowded, and less predictable than controlled industrial spaces. In a factory, robots may move in restricted areas with fixed paths and limited human interaction. In a restaurant, the robot must move close to customers, staff, tables, chairs, and temporary objects. People may walk suddenly in front of the robot, chairs may be moved from their original positions, and service paths may become partially blocked. This makes safe autonomous navigation one of the central challenges of the project.

The first major challenge is autonomous navigation. Nadel must move from one point to another without direct human control. This requires the robot to know where it is, understand the environment map, choose a destination, plan a path, and follow that path while avoiding obstacles. In restaurant service, the destination is usually a table.

Therefore, table locations must be represented clearly in the map and linked to unique table IDs. If table identification is incorrect, the robot may deliver the order to the wrong table.

Localization is another important challenge. Localization refers to estimating the position and orientation of the robot inside the mapped environment. Nadel uses encoder-based odometry, where wheel encoder ticks are used to estimate movement. This method is useful because it gives continuous information about wheel rotation, linear distance, and rotational displacement. However, odometry can accumulate error over time. Wheel slipping, uneven floors, mechanical errors, or inaccurate wheel diameter can cause drift. This means that the estimated position may slowly become different from the true position.

Mapping is also challenging. The robot requires a representation of the restaurant environment. Using LiDAR data, the system can create an occupancy grid map. In this map, each cell represents free space, occupied space, or unknown space. Walls, fixed furniture, and large obstacles can be represented as occupied cells. However, restaurant environments change frequently. Customers move, chairs are pulled out, tables may be rearranged, and temporary objects may appear. A map that is accurate in the morning may not represent the environment perfectly during peak hours. Therefore, dynamic obstacle handling is necessary [12], [13].

Obstacle avoidance is a critical challenge for safety. Static obstacles such as walls and fixed furniture can be handled by global and local planners. Dynamic obstacles such as customers, waiters, children, and moving chairs require real-time detection and reaction. Nadel uses sensors such as LiDAR, and ultrasonic sensors to detect obstacles. The robot may pause when a person is detected, request path clearance, or replan locally around unexpected objects. However, this behavior must be smooth and safe because sudden movement or abrupt stopping may disturb customers [13], [14].

Human safety is one of the most important considerations. Since the robot operates near people, the system must reduce collision risk and move at a safe speed. The robot body should also be designed to avoid sharp edges and protect internal components. Short-range sensors can help detect objects close to the robot, especially in

blind spots. Future versions should include an emergency stop button, safer bumpers, speed control strategies, and stronger fault detection mechanisms.

Human-robot interaction is another challenge. Customers may not know how to interact with a robot waiter. Some may feel excited, while others may be confused or uncomfortable. The interface must therefore be simple, friendly, and clear. A screen-based face can make the robot appear more approachable, while voice interaction and speech output help the robot communicate with customers.

Speech interaction is difficult in real restaurant environments because background conversations, kitchen sounds, music, and user distance can affect recognition quality. In the current implementation, Nadel uses Whisper Small for speech-to-text, Gemma 31B for request interpretation and response generation, and a speech output layer for robot voice replies. Therefore, voice interaction is treated as an implemented interaction layer, while future work focuses on improving robustness in noisy and crowded restaurant conditions [18], [20].

Hardware integration is another significant challenge. Nadel combines several components, including Jetson Orin Nano, ESP32, RPLIDAR, ultrasonic sensors, encoders, motors, motor drivers, screen, speakers, microphone, and batteries. These components must communicate reliably and fit physically inside the robot body. Each part has specific power requirements and communication protocols. A problem in wiring, power distribution, sensor mounting, or driver configuration can affect the whole system [13], [14].

Mechanical design is also a challenge. A waiter robot must be stable while moving and carrying food or drinks. The body must provide enough space for trays while keeping the center of gravity safe. If the robot is too tall or too narrow, it may become unstable when moving or turning. If the robot is too wide, it may not fit between tables. The prototype must balance tray capacity, stability, size, appearance, and manufacturability.

Cost is a practical challenge. The prototype includes relatively advanced components such as an RPLIDAR A1M8, NVIDIA Jetson Orin Nano, touch display, motors, drivers, wheels, caster wheels, and 3D-printed body. The estimated prototype cost is approximately 1700 to 2500 USD. This cost may be acceptable for a graduation

prototype and demonstration, but a commercial version must be optimized for manufacturing, maintenance, reliability, and affordability [13], [14].

Reliability is also critical. A restaurant robot should not fail frequently during service. It must handle communication errors, navigation failures, low battery, blocked paths, sensor noise, and unexpected behavior. Future improvements may include fault tolerance and recovery mechanisms, automatic return to charging station, system health monitoring, and clear error reporting to staff.

Another challenge is the reliability of the ESP32-based table-side Call/Pay unit. Since the unit sends table ID and request intent wirelessly, the system must ensure that each request is received correctly, linked to the correct table, and interpreted accurately. A delayed, duplicated, or incorrect request could affect robot assignment and restaurant workflow. Therefore, the table-side unit must be integrated carefully with the smart restaurant system and robot navigation logic [16].

Finally, customer and staff acceptance must be considered. Some workers may worry that robots will replace their jobs. However, the project presents Nadel as an assistive service robot that supports staff, reduces repeated movement, and allows humans to focus on tasks that require communication and judgment. Successful adoption depends on showing that the robot is a helpful tool rather than a replacement for human service.



Figure 1-4: Side view of Nadel prototype showing tray structure and mobile base

Figure 1-4 shows the side view of the prototype and highlights the tray structure, body height, lower base, and wheel placement. This figure supports the discussion of mechanical design challenges because the robot must carry items safely while remaining stable and compact enough for indoor movement.

Table 1-3: Main challenges addressed by the project.

Challenge	Project response
Navigation in indoor spaces	Use ROS2 Nav2, map-based planning, local planning, and table coordinates.
Dynamic obstacles	Use LiDAR, ultrasonic sensors, pausing behavior, and local replanning.
Localization drift	Use encoder odometry and future sensor fusion or recalibration.
Customer interaction	Use touch screen UI, table-side Call/Pay buttons, Whisper Small STT, Gemma 31B request processing, and speech output layer.
Hardware integration	Combine Jetson, ESP32, sensors, motors, drivers, and display into one platform.
Cost and feasibility	Estimate prototype cost and identify future optimization paths.
Wireless table-side requests	Use ESP32-based table units to send table ID and request intent for Call or Pay assistance.

1.2 PROBLEM STATEMENT

Restaurants and cafés face repeated operational problems that affect service quality, cost, and customer satisfaction. During busy periods, staff members must handle multiple tasks at the same time. They receive orders, communicate with customers, move between the kitchen and dining area, deliver food, respond to requests, and update order status. When the workload increases, the probability of delays, wrong deliveries, forgotten items, and inconsistent service also increases.

Traditional service depends heavily on human movement. Even when restaurants use the physical delivery of food from the kitchen to the table is still usually performed manually. This creates a gap between digital order management and physical service execution. The restaurant may be digitally organized at the order level, but the actual delivery task still depends on staff availability and physical effort.

The main problem addressed by this project is the absence of an integrated autonomous service support system that can receive delivery tasks, respond to table-side service and payment-assistance requests, identify the correct table, navigate safely inside the restaurant, avoid obstacles, and support customer interaction after arrival while remaining connected to the restaurant workflow.

The problem is not limited to building a moving robot. A successful waiter robot must integrate several system layers. It needs a hardware platform that can move and carry items. It needs sensors to understand the environment. It needs navigation software to plan and follow paths. It needs table representation to know where to go. It needs interaction tools to communicate with customers. It also needs connection to the restaurant order workflow so that robot movement is linked to real service tasks.

Therefore, the problem can be stated as follows: restaurants need an intelligent autonomous service robot that can support staff by delivering food and drinks to customer tables, responding to table-side Call and Pay assistance requests, reducing repetitive workload, minimizing delivery errors, improving service speed, and operating safely in dynamic indoor environments while integrating with a smart restaurant workflow.

Nadel is proposed as a practical solution to this problem. It combines autonomous navigation, mapping, table-based goal selection, obstacle avoidance, speech-to-text, Gemma 31B-based request interpretation, speech output, multilingual interface design. This makes the project a complete service automation concept rather than only a mobile robot prototype [12], [18].

The importance of solving this problem is clear from both operational and customer perspectives. Operationally, the robot can reduce repeated walking, support staff during peak hours, and help organize delivery tasks. From the customer perspective, faster and more accurate delivery can improve satisfaction. From the business perspective, a service robot can create a modern image and potentially reduce long-term operating pressure.



Figure 1-5: Rear view of Nadel prototype showing enclosed body design and mobile structure

Figure 1-5 shows the rear side of the robot. The enclosed structure helps protect internal components and gives the robot a clean external appearance. This is important for a public service environment because exposed wiring or unprotected components may reduce safety and professional appearance.

1.3 OBJECTIVE

The main objective of this project is to design and develop Nadel, a fully autonomous service waiter robot that can assist restaurant staff by delivering food and drinks from the kitchen area to customer tables. The robot aims to improve service speed, reduce repeated human effort, minimize delivery errors, and enhance customer experience through safe autonomous navigation and user-friendly interaction.

This general objective is divided into several specific objectives that describe the technical, operational, and business goals of the project.

The first objective is to design and build a physical robot prototype suitable for indoor restaurant environments. The prototype must include a stable mobile base, a service body, tray space for food and drinks, and a customer-facing display area. The body design should be compact enough for indoor movement while providing enough space for service items and internal components.

The prototype should also have a friendly and professional appearance. Since the robot operates near customers, it should not appear as an industrial machine. The face display, clean body surface, and enclosed design help the robot fit into a restaurant or café environment.

The second objective is to implement autonomous navigation using ROS2 Nav2. The robot should be able to move from its current position to a target table without direct manual control. This requires map-based localization, goal selection, path planning, local planning, and motion execution. Navigation is one of the core functions of the project because the robot must deliver orders to the correct location [9].

The third objective is to represent the restaurant environment using a map. The robot uses LiDAR scanning to support occupancy grid generation. In the map, free spaces, walls, furniture, and obstacles can be represented. Tables are identified as fixed locations, and each table is linked to a unique ID and coordinates. This allows the robot to convert a table number from the order system into a navigation target [12], [13].

The fourth objective is to integrate sensors that support perception, obstacle detection, and safe navigation. The RPLIDAR A1M8 provides 360-degree scanning for mapping and environment awareness. Ultrasonic sensors provide short-range detection to cover blind spots. Motor encoders provide wheel movement feedback for odometry [12], [13].

The fifth objective is to use encoder-based odometry and differential drive motion control. Wheel encoder ticks from the left and right motors allow the system to estimate movement. By calculating the linear and rotational displacement, the robot can estimate its pose in the form of x , y , and θ . This supports localization, control, and path following.

The sixth objective is to develop a clear and friendly human-robot interaction interface. The robot uses a screen-based face to improve social acceptance and make interaction easier. The facial interface is implemented using HTML, CSS, and JavaScript and can be connected to ROS topic messages. This allows facial expressions to change according to robot state or interaction needs.

The seventh objective is to support Arabic and English in the user interface and provide a two-way voice interaction layer. Whisper Small converts customer speech into text, Gemma 31B interprets the request and generates an appropriate response, and the speech output layer allows Nadel to reply by voice. This improves accessibility and gives customers a natural interaction option in addition to the touch screen interface [18], [20].

The eighth objective is to connect Nadel with a smart restaurant workflow. The workflow includes order placement, kitchen display, delivery assignment, order tracking, and billing updates. The robot receives task information such as table ID and order items, then performs the delivery task. This integration is important because the robot must operate as part of the restaurant service process, not as an isolated device.

The final objective is to evaluate the prototype and identify limitations that should be improved in future versions. Evaluation should consider navigation behavior, obstacle avoidance, mapping, table delivery workflow, user interface, voice interaction, speech output, hardware stability, and prototype cost.

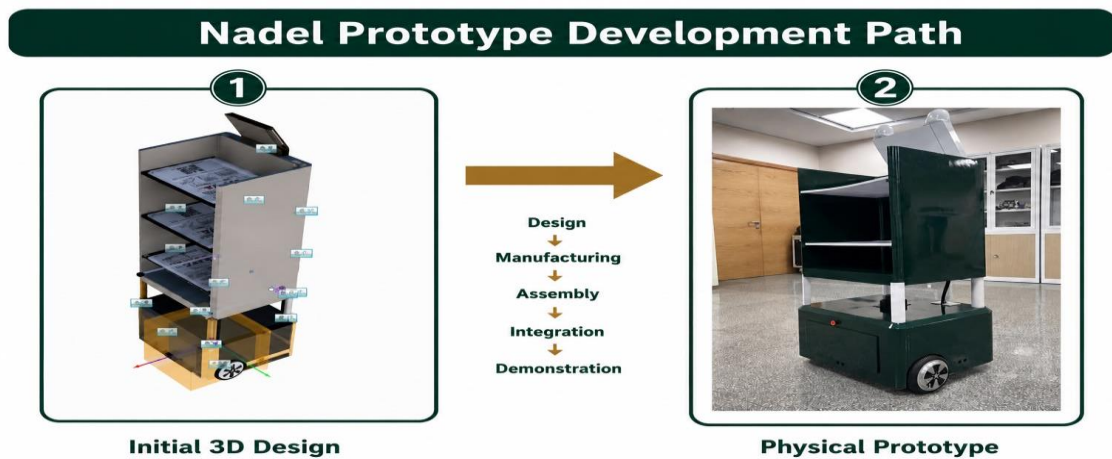


Figure 1-6: Development path from initial 3D model to physical prototype

Table 1-4: Project objectives and expected outcomes.

Objective	Expected outcome
Build physical prototype	A mobile service robot body with tray structure, screen, wheels, and enclosed design.
Implement autonomous navigation	Robot can navigate from start position to selected table using ROS2 Nav2.
Create map and table representation	Restaurant tables are linked to IDs and coordinates in the map.
Integrate perception sensors	Robot can detect environment features, obstacles, and nearby objects.
Develop interaction interface	Robot can display a face, provide touch screen support, process speech using Whisper Small, interpret requests using Gemma 31B, and reply by voice through the speech output layer.
Evaluate and improve	Limitations are identified and future enhancements are proposed.
Integrate table-side Call/Pay unit	Customers can request Nadel from the table using an ESP32-based unit that sends table ID and request intent only.
Develop interaction interface	Robot can display a face, provide touch screen support, process speech using Whisper Small, interpret requests using Gemma 31B, and reply by voice through the speech output layer.

1.4 THESIS OUTLINE

This report is organized into seven chapters. Each chapter presents a specific part of the project, starting from the general background and ending with the conclusion and future work. The structure follows the graduation project report template and is designed to move from general explanation to technical design, evaluation, and business analysis.

Chapter One introduces the project background and explains the need for autonomous service robots in restaurant environments. It presents the overview, motivation, applications, challenges, problem statement, objectives, and thesis outline. This chapter explains why Nadel is needed and how it can support restaurant service [1].

Chapter Two reviews related work in service robots, waiter robots, autonomous mobile robots, indoor navigation, SLAM, LiDAR-based mapping, human-robot interaction, and smart restaurant systems. It compares existing solutions and identifies the gap that Nadel aims to address [12], [13].

Chapter Three explains the main technical terms used in the project. These include ROS2, Nav2, SLAM Toolbox, LiDAR, RealSense depth camera, odometry, IMU, motor encoders, differential drive, occupancy grid, path planning, obstacle avoidance, ESP32, ESP32-based table-side Call/Pay unit, Whisper Small, Gemma 31B, speech output layer, and human-robot interaction [9], [12].

Chapter Four presents the proposed Nadel system in detail. It describes the hardware architecture, software architecture, robot body design, sensor integration, navigation system, mapping process, table representation, control system, touch screen interface, voice interaction using Whisper Small and Gemma 31B, speech output layer, ESP32-based table-side Call/Pay interaction module, and POS workflow [12], [16].

Chapter Five discusses the testing and evaluation of the project. It presents the testing environment, navigation behavior, obstacle avoidance behavior, mapping results, delivery workflow testing, interface testing, voice interaction testing, and system performance analysis. It also discusses limitations found during implementation and how they can be improved. The chapter also introduces the ESP32-based table-side

Call/Pay unit as an additional interaction method that allows customers to request Nadel from their table for service or payment-related assistance [12], [16].

Chapter Six presents the business model of Nadel. It discusses target customers, value proposition, key resources, cost structure, revenue streams, key partners, market opportunities, and project impact. It also presents prototype cost estimation and explains how the robot can reduce operational costs and improve service quality.

Chapter Seven concludes the report by summarizing the project achievements, the main contribution of Nadel, and the lessons learned during development. It also presents future enhancements, including speech ordering, advanced computer vision, automatic charging, brushless motors, fault tolerance, Jetson Orin AGX upgrade, and multi-robot coordination.

1.5 CHAPTER SUMMARY

This chapter introduced Nadel as a fully autonomous service waiter robot designed for restaurant and café environments. It explained the motivation behind the project, including high labor cost, staff shortage, slow service, wrong deliveries, and inconsistent service quality. It also presented the main applications of the robot, the technical and operational challenges, the problem statement, and the project objectives. The next chapter reviews the literature and related work needed to support the technical direction of the project.

2 LITERATURE REVIEW AND RELATED WORK

This chapter reviews the literature and technical background relevant to the Nadel waiter robot project. It discusses service robots in hospitality environments, autonomous indoor navigation, ROS2 Nav2, mapping and localization, sensing technologies, motion control, and human-robot interaction. The chapter also reviews related service robot solutions and identifies the research gap addressed by Nadel, particularly the integration of autonomous navigation, table-side service requests, and voice or touch-screen interaction in a practical restaurant service prototype.

2.1 WAITER ROBOTS AND HOSPITALITY INDUSTRY

Service robots are robots designed to perform useful tasks for humans outside traditional industrial manufacturing environments. In hospitality, they may be used for guiding guests, delivering items, cleaning, reception services, food delivery, entertainment, or information support. Restaurants and cafés are especially suitable environments for service robots because they include repeated indoor tasks, defined destinations, structured workflows, and high demand for speed and consistency [1].

Recent hospitality research indicates that service robots are increasingly viewed as tools for improving productivity and service consistency, but adoption is affected by employee attitudes, customer acceptance, perceived usefulness, perceived risk, trust, and service context [1]. A robot may be technically capable, but if staff feel threatened by it or customers find it uncomfortable, its practical value may decrease. This is why service robot projects must consider human factors from the beginning rather than treating them as optional additions.

Customer experience is also an important part of hospitality robotics. Studies on robotic restaurants show that service quality is not limited to food quality. It also includes the interaction quality between the customer and the robot, the atmosphere created by the robotic service, perceived value, satisfaction, and intention to revisit [2]. This is directly relevant to Nadel because the robot uses a friendly body design, a face display, and voice interaction to make the service experience more engaging.

Restaurant robot acceptance also depends on whether customers understand the robot's task. A robot that performs a clear practical function, such as delivering food from the kitchen to the table, is easier to understand than a robot that attempts complex social interaction without clear value. Recent research on customers' intentions to use robot-waiter restaurants highlights that efficiency, novelty, safety, and perceived service quality influence customers' willingness to accept robotic service [3].

From the employee perspective, service robots should be positioned as assistive systems rather than direct replacements. Hospitality employees often worry that robots may reduce job opportunities or change job expectations. Systematic reviews of HRI in tourism and hospitality emphasize that adoption is most effective when robots reduce repetitive workload while allowing employees to focus on communication, problem solving, and higher-quality service [4]. This supports the Nadel project philosophy: the robot assists waiters by handling repeated delivery tasks, while human staff remain responsible for complex customer needs.

2.1.1 WAITER ROBOTS

Waiter robots are a specific category of service robots designed for restaurant environments. Their main role is usually to transport food, drinks, or utensils between preparation areas and customer tables. Some waiter robots also provide customer interaction, menu display, voice messages, or entertainment features. The development of waiter robots has been driven by several operational pressures, including labor shortage, high labor cost, need for faster service, and desire for technology-enhanced customer experience [1].

Commercial restaurant robots such as BellaBot and DINERBOT demonstrate that the waiter robot market is moving toward multi-sensor navigation, obstacle avoidance, tray-based delivery, interactive displays, and multi-robot dispatching. BellaBot uses sensors and LiDAR-based obstacle avoidance to support restaurant delivery and customer interaction [5]. KEENON's DINERBOT T9 presents a commercial design focused on route planning, task execution, and long battery operation in restaurant settings [6]. These systems show that waiter robots are no longer only experimental prototypes; they are becoming practical products in hospitality environments.

However, commercial waiter robots often provide limited public detail about their internal architecture, navigation configuration, , or speech-processing pipeline. This creates a knowledge gap for students and researchers who want to understand how a complete waiter robot system can be designed from components. Nadel contributes to this area by documenting a student-built prototype that integrates hardware, ROS2 Nav2 navigation, perception, POS workflow, Whisper Small speech-to-text, and Gemma 31B request processing [9], [18].

Restaurant automation is broader than waiter robots. It includes digital menus, self-service kiosks, kitchen display systems, inventory management, payment automation, and robotic kitchen systems. A waiter robot can only produce strong value when it is connected to this digital environment. If the robot moves manually after a staff member presses buttons on the robot itself, it may still help with delivery but it does not fully integrate with the restaurant workflow. In contrast, Nadel's design links order input, smart restaurant management, kitchen preparation, robot assignment, path planning, delivery, and tracking [1].

Table 2-1: Summary of related areas reviewed in Chapter Two.

Area	Main focus	Relevance to Nadel
Hospitality service robots	Customer experience, staff support, adoption, service quality	Explains why waiter robots are needed and how users may accept them
Waiter robots	Food delivery, tray transport, obstacle avoidance, interaction	Provides direct comparison with restaurant delivery robots
Smart restaurant systems	POS, kitchen display, order tracking, billing	Links robot movement to restaurant workflow
Voice interaction	Speech-to-text and language understanding	Supports hands-free communication using Whisper Small and Gemma 31B

2.1.2 COMPARISON WITH EXISTING SERVICE ROBOT SOLUTIONS

Existing commercial waiter robots provide useful benchmarks for Nadel. BellaBot, for example, includes a friendly cat-inspired design, trays, obstacle avoidance, and multimodal interaction [5]. BellaBot Pro adds stronger commercial features such as upgraded stability, longer runtime, and safety-focused interaction features [25]. KEENON DINERBOT models focus on restaurant delivery, route planning, battery life, and multi-robot dispatching [6], [26].

These commercial systems show that the Nadel project is aligned with current market direction. Restaurant robots are expected to navigate autonomously, avoid obstacles, carry items, support customer interaction, and fit into service workflows. However, because commercial systems are closed products, they do not always provide transparent educational detail about their internal hardware, ROS configuration, sensor fusion method, or AI interaction pipeline.

Nadel differs from commercial robots in its educational and design transparency. It is built as a graduation project that documents hardware choices, software architecture, navigation method, perception layer, POS integration, and voice interaction pipeline. It also uses accessible components such as Jetson Orin Nano, ESP32, RPLIDAR, ultrasonic sensors, encoders, ROS2 Nav2, Whisper Small, and Gemma 31B. This makes it valuable as an applied intelligent systems project [9], [13].

Table 2-3 compares Nadel with selected commercial service robots. The comparison does not claim that the academic prototype is more mature than commercial products. Instead, it shows how Nadel adopts key ideas from current service robots while focusing on open implementation, student-accessible design, and integration of restaurant workflow with ROS2 navigation and voice processing [9], [1].

Table 2-3: Comparison between existing waiter robot solutions and Nadel.

System	Main characteristics	Strengths	Relation to Nadel
BellaBot	Commercial cat-inspired delivery robot with trays and obstacle avoidance	Friendly interaction and mature commercial deployment	Nadel follows the same service goal but uses an open student-built architecture
DINERBOT T9	Commercial restaurant delivery robot with route planning and dispatch features	Long operation time and restaurant-focused design	Nadel targets similar delivery tasks with ROS2 Nav2 and accessible components
General POS + KDS systems	Digital order placement, kitchen coordination, billing updates	Improves information flow and order accuracy	Nadel connects the workflow to physical robotic delivery
Nadel	Physical prototype with ROS2 Nav2, perception, Whisper Small, and Gemma 31B	Integrated academic prototype with transparent implementation	Addresses the gap between restaurant workflow and autonomous robot movement

2.2 INDOOR AUTONOMOUS NAVIGATION

An autonomous mobile robot (AMR) is a robot capable of moving through an environment without continuous human control. AMRs differ from simple remotely controlled robots because they sense their environment, estimate their position, plan a route, and execute movement according to a goal. In restaurant service, this autonomy is essential because the robot must repeatedly travel between the kitchen and tables while avoiding customers, staff, furniture, and temporary obstacles.

Modern AMRs usually combine sensors, computation, localization algorithms, mapping, path planning, motor control, and safety behaviors. ROS2 has become an important middleware for building such systems because it provides communication between software nodes, sensor drivers, navigation components, and control interfaces.

Recent surveys of ROS2 mobile robotics algorithms describe Nav2 as one of the central projects in the ROS2 mobile robotics ecosystem, providing a next-generation navigation framework for practical mobile robots [7].

The Nadel robot fits the AMR model because it receives a table goal, interprets it as a navigation target, uses sensors to perceive the environment, and moves toward the destination autonomously. Its autonomous behavior is not only a software feature; it depends on the interaction between the physical base, encoders, LiDAR, ultrasonic sensors, Jetson Orin Nano, ESP32, and motor drivers. Therefore, literature on AMRs is directly relevant to its design [13], [14].

In indoor service environments, AMRs face several practical constraints. The robot must be compact enough to move through narrow paths. It must stop or slow down when humans are nearby. It must not collide with chairs, tables, walls, or customers. It must maintain stability while carrying items. It must also behave predictably so that staff can trust it. These requirements make restaurant AMR design more complex than moving a robot in an empty laboratory room.

Figure 2-2 summarizes the common AMR navigation pipeline. The pipeline begins with sensor input, continues through mapping and localization, then path planning and control, and finally monitoring and recovery. This pipeline is the basis for understanding how Nadel moves autonomously inside a restaurant environment [12].

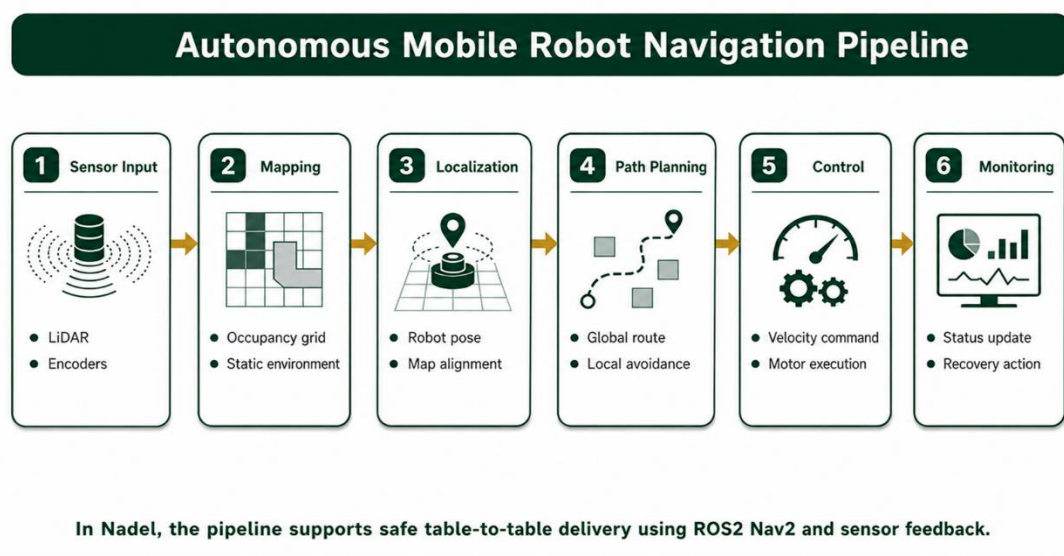


Figure 2-2: Autonomous mobile robot navigation pipeline

2.2.1 SLAM, MAPPING, LOCALIZATION

Indoor navigation is one of the most important research areas for waiter robots. Unlike outdoor navigation, indoor navigation often depends on local maps, furniture layouts, walls, narrow corridors, and dynamic obstacles. GPS is not reliable indoors, so the robot must use internal sensors and environmental perception to estimate where it is and how to move safely [1].

SLAM, or Simultaneous Localization and Mapping, allows a robot to build a map of an unknown environment while estimating its own location inside that map. For service robots, SLAM is important because the robot may need to build an occupancy grid map of the restaurant before it can navigate to tables. Nav2 documentation explains that ROS2 users can combine SLAM with Nav2 to generate occupancy grid maps and move a robot in physical environments [10].

SLAM Toolbox is a widely used ROS2 package for mapping and localization. It processes laser scans and odometry to generate maps and maintain transforms between map and odometry frames [11]. For a robot such as Nadel, this is relevant because LiDAR and wheel encoder data can be combined to support map creation and localization. Even when a map is created manually or semi-manually, the robot still needs a consistent relationship between the map, robot pose, and navigation goals.

The occupancy grid is a common representation used in indoor robot navigation. In an occupancy grid, the environment is divided into cells. Each cell represents free space, occupied space, or unknown space. Walls, furniture, and fixed objects appear as occupied cells, while corridors and movement areas appear as free cells. Restaurant tables can be represented as fixed positions with unique coordinates linked to table IDs. This is important for Nadel because the robot must translate a table ID into a physical navigation goal [12].

Mapping restaurant environments is challenging because the environment may change. Chairs may be moved, customers may stand in the path, and temporary obstacles may appear. For this reason, static mapping must be combined with dynamic obstacle detection. The static map helps the robot know the general layout, while local perception helps it react to real-time changes. Nadel follows this principle by combining

mapping and table representation with perception sensors such as RPLIDAR, ultrasonic sensors, and encoders [12], [13].

2.2.2 ROS2 NAV2, PATH PLANNING, OBSTACLE AVOIDANCE

ROS2 Nav2 is one of the most important technologies in the Nadel project. Nav2 is the ROS2 navigation framework used to move a mobile robot from one pose to another while considering maps, localization, path planning, obstacle avoidance, behavior trees, and robot control. The official Nav2 documentation describes Nav2 as a professional navigation framework for mobile robots that can operate in complex environments and complete user-defined tasks [8].

Nav2 is modular. This means that its functionality is divided into servers and plugins rather than a single fixed program. Typical Nav2 components include a map server, localization module, planner server, controller server, behavior tree navigator, costmaps, recovery behaviors, and lifecycle management. This modular structure makes it suitable for a project such as Nadel because each part of the navigation pipeline can be understood, configured, and improved independently [9].

In Nadel, Nav2 is used to compute a safe route to the selected customer table. The restaurant system provides a table ID. The system translates the table ID into a map coordinate. Nav2 plans a path from the robot's current position to the target table. During movement, the robot uses local planning and sensor feedback to avoid obstacles. This matches the general Nav2 concept of combining global planning with local control [9].

The planner component is responsible for finding a route through the map. The controller component is responsible for generating velocity commands that allow the robot to follow the planned path. The costmap represents obstacles and navigable space. If a person or object blocks the route, the robot can slow down, stop, or replan. This is critical in a restaurant because the environment is dynamic and human presence cannot be avoided.

Recent ROS2 mobile robotics literature emphasizes that Nav2 has become a major project in the ROS2 ecosystem and includes modern algorithms for planning, control, navigation behavior, and performance benchmarking [7]. Regulated Pure

Pursuit, a controller available in Nav2, was designed with attention to safety in constrained and partially observable spaces, which is highly relevant to service robots operating indoors [17].

Figure 2-3 presents a conceptual Nav2 architecture for Nadel. The figure illustrates the relationship between mapping, localization, planning, costmaps, behavior logic, control, and the robot base. This architecture helps explain why Nav2 is suitable for a waiter robot that must move safely and repeatedly between known indoor destinations [9], [12].

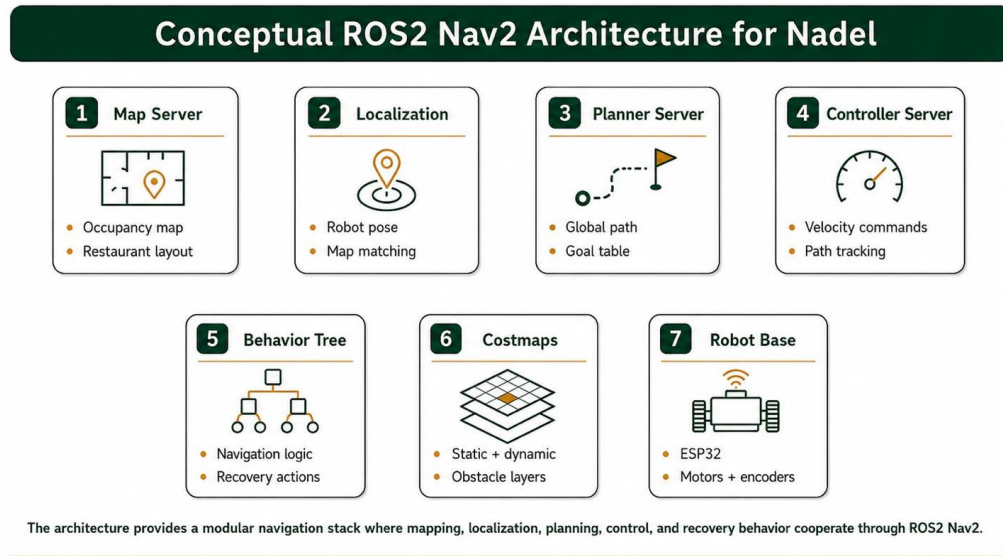


Figure 2-3: Conceptual ROS2 Nav2 architecture for Nadel. Source: Authors' diagram.

Figure 2-3: Conceptual ROS2 Nav2 architecture for Nadel.

2.3 SENSING, HARDWARE, AND MOTION CONTROL

The perception layer gives the robot awareness of its environment and movement. Without perception, the robot would not know where obstacles are, how far it has moved, or whether the path is safe. Nadel uses multiple sensing technologies because no single sensor is sufficient for all indoor navigation requirements. The main sensors include RPLIDAR A1M8, ultrasonic sensors, and motor encoders [13], [14].

RPLIDAR A1 is a 360-degree 2D laser scanner that can generate distance measurements around the robot. Its point cloud output can be used for mapping, localization, and environment modeling [12]. For Nadel, the LiDAR is important because it supports map building and obstacle awareness around the robot. A 360-

degree scan is especially useful in restaurant environments where obstacles can appear from different directions.

Ultrasonic sensors are used for short-range obstacle detection. They are especially helpful for blind spots and close-range protection where LiDAR or camera coverage may not be sufficient. Although ultrasonic sensors provide less detailed information than LiDAR or depth cameras, they are low-cost and useful for collision prevention in nearby zones [13], [14].

Motor encoders measure wheel rotation. Encoder readings allow the robot to estimate how far each wheel has moved. This data supports odometry, which estimates the robot's change in position and orientation. Encoders are essential for a differential drive robot because left and right wheel movement determines forward motion and rotation.

Sensor fusion is important because each sensor has strengths and limitations. LiDAR is strong for 2D range scanning, but it may not detect some low or overhanging objects depending on height. A depth camera provides richer 3D perception but may be affected by lighting and field of view. Ultrasonic sensors are useful for close obstacles but have limited resolution. Encoders provide motion feedback but accumulate error over time. Combining these sensors improves the reliability of the robot [13], [14].

Table 2-2: Comparison of sensors used in autonomous service robot perception.

Sensor	Strengths	Limitations	Use in Nadel
RPLIDAR A1M8	360-degree 2D scanning; useful for mapping and obstacle awareness	Limited vertical perception; may miss some overhanging objects	Mapping, localization support, obstacle detection
Ultrasonic sensors	Low cost; good for close-range detection	Low resolution and limited directionality	Blind spot and short-range collision prevention
Motor encoders	Continuous wheel motion feedback	Accumulates drift over time	Odometry and motion estimation

2.3.1 PERCEPTION SENSORS AND HARDWARE COMPONENTS

Autonomous service robots require both high-level processing and low-level control. High-level processing includes navigation, perception, decision-making, and interaction. Low-level control includes reading sensors, communicating with motor drivers, generating velocity commands, and controlling wheel movement. Nadel separates these responsibilities between the Jetson Orin Nano and the ESP32 microcontroller [15], [16].

The NVIDIA Jetson Orin Nano is designed for edge AI and robotics applications. NVIDIA presents the Jetson Orin Nano platform as a compact developer kit capable of supporting modern AI workloads and robotics development [14]. In Nadel, it acts as the main processing unit that runs the high-level software stack, including ROS2-based navigation and interaction processing.

The ESP32 is a Wi-Fi and Bluetooth-enabled microcontroller designed for embedded and wireless applications [15]. In Nadel, the ESP32 is used for wireless communication and motor-control-related tasks. It can receive velocity commands and translate them into signals for motor drivers. This division is practical because the Jetson handles computationally heavier tasks, while the ESP32 handles lower-level control and communication.

The hardware architecture must also consider power distribution, wiring, mechanical stability, sensor placement, and maintainability. A waiter robot needs a stable base, enough body space for electronics and trays, and protected wiring. The prototype body should allow access for maintenance while keeping components safe from accidental contact with customers or food items.

Compared with purely simulated robots, a physical prototype introduces more practical constraints. Motors may respond differently from theoretical models. Wheels may slip. Battery voltage may change. Sensors may produce noise. Mechanical alignment may affect odometry. Therefore, implementing Nadel as a real robot strengthens the project because it demonstrates that the design was tested beyond software simulation.

2.3.2 ODOMETRY, DIFFERENTIAL DRIVE MOTION AND EMBEDDED CONTROL

Odometry is the process of estimating a robot's motion based on sensor measurements, usually wheel encoder readings. For a differential drive robot, the left and right wheels can move at different speeds. If both wheels move at the same speed, the robot moves forward or backward. If one wheel moves faster than the other, the robot turns. This model is widely used in indoor mobile robots because it is mechanically simple and suitable for compact platforms.

In Nadel, motor encoders provide wheel tick readings. These readings are converted into wheel displacement. The displacement of the left and right wheels is used to estimate the robot's linear movement and angular rotation. The estimated pose can be represented as x , y , and θ in a two-dimensional plane. This pose estimate is then used by the navigation system to understand the robot's movement relative to the map.

Odometry has an important limitation: drift. Small errors in wheel measurement accumulate over time. Wheel slip, uneven floors, loose wheels, inaccurate wheel diameter, or mechanical imbalance can cause the estimated position to deviate from the true position. In a restaurant environment, repeated movement between tables may gradually increase this error if no correction is applied.

Because of odometry drift, odometry is usually combined with other localization methods. LiDAR-based localization, map matching, SLAM, or sensor fusion can help correct accumulated error. This is why Nadel's architecture includes both encoders and environmental sensors. Encoders provide continuous motion feedback, while LiDAR and mapping support correction and environment awareness [12], [13].

From a literature perspective, odometry is not enough for a complete waiter robot, but it is still necessary. It provides a local motion estimate and allows the control system to translate velocity commands into physical movement. In combination with Nav2, costmaps, and sensors, odometry becomes part of a larger navigation system rather than the only source of localization [9].

2.4 HUMAN-ROBOT INTERACTION AND SYSTEM INTEGRATION

Human-robot interaction is a central topic for service robots because the robot works near people. In industrial robotics, robots often operate in controlled or separated spaces. In restaurants, however, the robot operates near customers, waiters, kitchen staff, and visitors. This requires the robot to be understandable, predictable, safe, and socially acceptable [1].

Hospitality HRI research shows that customer response to robots is shaped by usefulness, ease of interaction, novelty, trust, perceived safety, robot appearance, and service quality [4]. A robot that performs the task efficiently but confuses customers may not produce a positive service experience. Similarly, a robot that looks friendly but cannot perform reliably will not support the business workflow. Therefore, Nadel combines practical navigation with a friendly customer-facing design.

The face display and visual interface are important elements of Nadel's HRI layer. A screen-based face can make the robot appear less mechanical and more approachable. The interface can also provide status feedback, such as when the robot is ready, moving, delivering, or waiting. This reduces uncertainty for customers and staff. In service robots, feedback is important because users need to understand what the robot is doing [1].

Multilingual support is another HRI factor. Restaurants may serve customers who speak different languages. Supporting Arabic and English makes the system more accessible and more suitable for local deployment in Egypt and international contexts. Multilingual interaction also increases the educational value of the project because it demonstrates awareness of real customer diversity.

HRI must also consider employee acceptance. Robots should not be introduced in a way that creates fear or resistance among workers. Public discussions of restaurant automation show that staff may worry about automation replacing jobs [22]. For this reason, Nadel should be described as a support robot that reduces repetitive walking tasks and allows staff to focus on customer service, quality control, and exception handling.

2.4.1 VOICE INTERACTION

Voice interaction can make a waiter robot more natural and accessible. Instead of requiring every interaction to occur through a touch screen, a robot can listen to spoken customer requests and interpret them. This is especially useful in restaurant environments where customers may want to ask simple questions, confirm orders, or interact without touching a device.

The Nadel project uses Whisper Small for speech-to-text. Whisper is a speech recognition model trained on a large amount of labeled audio data and designed for robust speech recognition and translation tasks [18]. In Nadel, Whisper Small converts customer speech input into text. This creates the bridge between spoken human language and the robot's software system.

After speech is converted into text, the system uses Gemma 31B to interpret the request and generate an appropriate response. Gemma 3 is a family of open models from Google DeepMind, with model sizes up to 31 billion parameters and multilingual capabilities [19], [20]. For Nadel, Gemma 31B is used as the request processing component that helps understand what the customer is asking and decides how the robot should respond within the service context.

The combination of Whisper Small and Gemma 31B creates a practical voice interaction pipeline. The customer speaks. Whisper Small transcribes the speech. Gemma 31B interprets the text and generates a response or action. The system can then use this output to support service interaction, such as acknowledging a request, confirming table information, or triggering a system response. The exact speech output method is part of the system configuration and is separate from the STT and LLM pipeline [18], [20]

Figure 2-4 shows the voice interaction pipeline used in Nadel. The pipeline is important because it connects HRI with artificial intelligence. It also shows that the robot is not only reacting to fixed buttons; it can process spoken input and generate context-aware responses.

Voice Interaction Pipeline in Nadel

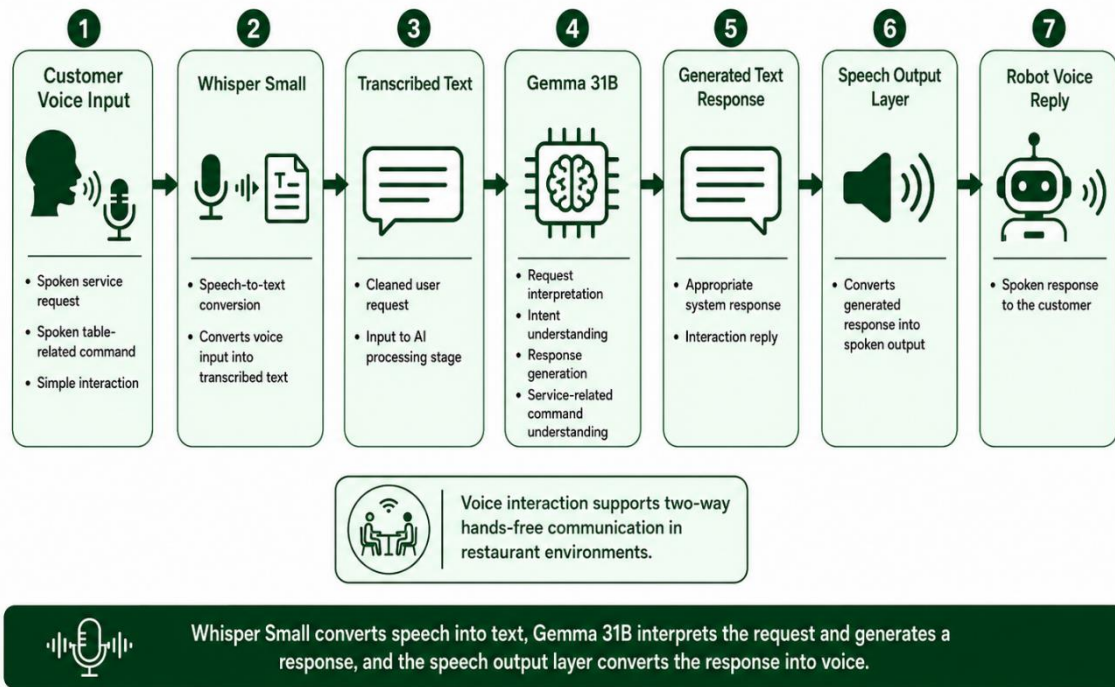


Figure 2-4: Voice interaction pipeline using Whisper Small and Gemma 31B.

2.5 SAFETY, ETHICS, AND PRACTICAL DEPLOYMENT CONSIDERATIONS

Safety is a core requirement for any robot operating near people. In restaurant environments, the robot must share space with customers, waiters, children, chairs, tables, and food items. The robot must therefore move at safe speeds, detect obstacles, stop when necessary, and avoid unpredictable behavior. Safety is not only a technical feature; it is a condition for customer trust and legal acceptability.

ISO 13482 provides safety requirements and guidelines for personal care robots, including mobile servant robots [5]. Although Nadel is an academic prototype rather than a certified commercial product, the standard is relevant because it highlights the importance of inherently safe design, protective measures, and information for use. Newer safety standard work also reflects the importance of service robots in personal and commercial applications [25].

For Nadel, safety is supported through obstacle detection, LiDAR awareness, ultrasonic short-range sensing, path planning, and controlled movement. However, future development should include additional safety mechanisms such as an emergency stop button, speed limiting in crowded zones, bumper protection, warning sounds or lights, and formal risk assessment. These features would make the robot more suitable for real restaurant deployment [13].

Ethics and privacy should also be considered. The use of cameras and microphones may raise privacy concerns. Customers should know when audio is being processed and for what purpose. If voice data is stored, clear data handling rules are needed. If camera-based perception is used for human detection or emotion recognition in the future, privacy and consent become even more important.

Another ethical concern is the social impact of automation. Restaurant robots may be perceived as replacing workers. This concern is common in public discussions of AI and robotics in hospitality [22]. To reduce this concern, the system should be presented as a collaborative support tool. Nadel handles repeated delivery tasks, while human workers remain responsible for hospitality, food quality checks, customer problem solving, and personal service.

2.6 KEY RELATED WORK AND CRITICAL REVIEW, RESEARCH GAP, AND NADEL CONTRIBUTION

A literature review should not only list technologies; it should evaluate what each related work contributes and what it does not fully solve. For Nadel, the most useful related works can be grouped into four categories: hospitality service robot studies, commercial waiter robot systems, autonomous navigation frameworks, and voice-based AI interaction systems. Each group supports part of the project, but none of them alone provides a complete model for the final system [9], [1].

Hospitality service robot studies explain why robots are useful in restaurants and hotels. They show that robots can improve consistency, provide novelty, and support employees. However, many of these studies use surveys, acceptance models, or customer perception analysis. Their limitation is that they often do not explain hardware architecture, low-level robot control, or how navigation goals are generated. Therefore, they support the motivation of Nadel but do not fully explain the implementation [1].

Commercial waiter robot systems show that the market already contains practical restaurant robots. They provide evidence that restaurants can use robots for food delivery and customer interaction. However, the technical details of commercial systems are usually hidden. Public pages describe features such as obstacle avoidance, battery life, payload, or interaction, but they do not provide a complete open design. This limitation is important for academic work because students need a transparent architecture that can be analyzed and improved.

ROS2 and Nav2 literature provides strong technical support for autonomous navigation. It explains mapping, localization, behavior trees, planners, controllers, costmaps, and recovery behaviors. However, navigation literature usually treats the robot goal as a coordinate chosen by the developer. In a restaurant waiter robot, the goal should be generated from service workflow, such as table ID and delivery assignment. Nadel connects these two sides by translating restaurant service data into robot navigation goals [9], [12].

Speech recognition and language model literature supports the interaction layer. Whisper Small provides speech-to-text capability, while Gemma 31B provides request interpretation and response generation. However, speech and language models alone do not create a robot. They become valuable only when connected to robot states, service commands, and restaurant workflow. Nadel uses them as part of the HRI layer rather than as a separate chatbot [18], [20].

Table 2-4: Critical review of key related literature and systems.

Related work area	Contribution to Nadel	Limitation identified	How Nadel responds
Hospitality service robot studies	Explain adoption, service quality, customer acceptance, and employee concerns	Often focus on surveys and perception rather than robot architecture	Nadel combines service value with a working technical prototype
Commercial waiter robots	Show practical demand for robot food delivery in restaurants	Architecture and implementation details are usually closed	Nadel documents accessible hardware and software layers
ROS2 Nav2 literature	Provides a mature framework for mapping, planning, control, and recovery	Usually does not include restaurant POS workflow	Nadel connects table ID and order assignment to navigation goals
SLAM and indoor mapping work	Supports occupancy grid generation and localization	Dynamic restaurant layouts still require real-time perception	Nadel combines mapping with LiDAR, ultrasonic sensors, and encoders
Speech-to-text and LLM literature	Supports natural voice input and request processing	Models need to be grounded in robot tasks and system state	Nadel uses Whisper Small and Gemma 31B within the restaurant service context
Safety standards and HRI studies	Emphasize human safety, trust, and predictable behavior	Academic prototypes may not meet full commercial certification	Nadel identifies safety mechanisms and future improvements

The reviewed literature shows that service robots are widely discussed in hospitality research, but much of this discussion focuses on adoption, acceptance, customer satisfaction, employee concerns, and service quality. These are important topics, but they do not always explain how a complete waiter robot can be technically designed and integrated with restaurant workflow. On the other hand, robotics literature explains

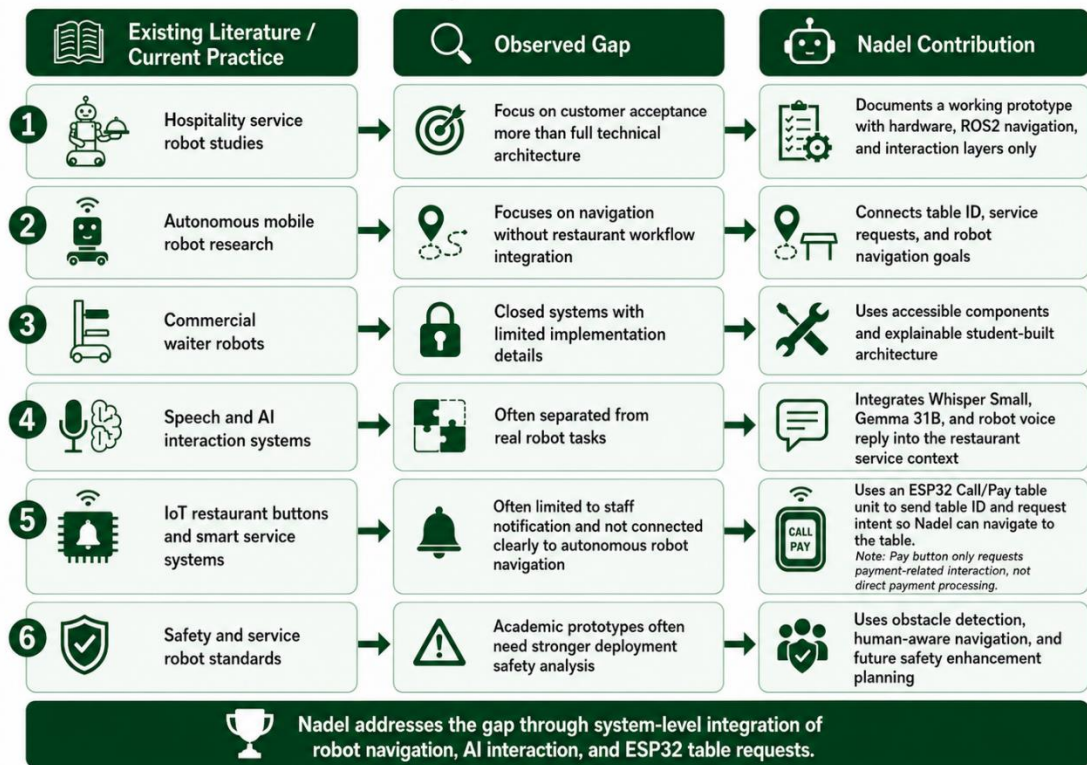
navigation, mapping, localization, and sensor processing, but it often does not address restaurant-specific order management and customer service requirements [12], [1].

This creates a gap between hospitality service research and robotics implementation. A waiter robot must satisfy both sides. It must be socially acceptable and operationally useful, but it must also navigate safely and reliably. It must also process sensor data and control motors. It must interact with users, but it must also execute delivery tasks. Nadel is positioned exactly at this intersection [1].

Nadel contributes by combining a physical robot prototype, ROS2 Nav2 autonomous navigation, LiDAR and depth sensing, encoder-based odometry, ESP32 motor control, a multilingual user interface, and a voice interaction pipeline using Whisper Small and Gemma 31B. This integration makes the project more complete than a simple robot base or a software-only restaurant ordering system [9], [13].

Table 2-6 summarizes the research gap and Nadel's contribution. Existing literature and commercial systems provide useful foundations, but the gap remains in integrated, well-documented, student-built restaurant service robots that combine navigation, sensing, and AI-based interaction. Nadel addresses this gap through practical implementation and system-level integration [1].

Literature Gap and Nadel Contribution



The reviewed studies and technical documents show that restaurant service robotics is not a single-problem field. A successful waiter robot must satisfy a chain of requirements: it must be useful for the restaurant, acceptable to customers and staff, technically capable of navigating indoors, safe around people, and integrated with the order management workflow. The literature therefore supports a system-level approach rather than a component-only approach.

A critical observation from hospitality literature is that many studies examine attitudes toward service robots but do not deeply describe how the robot is built. These studies are valuable because they explain acceptance, satisfaction, trust, perceived usefulness, and employee concerns. However, they often treat the robot as a finished product. For an engineering graduation project, this leaves unanswered questions about architecture, sensors, navigation configuration, motor control, and integration with restaurant software [1].

A critical observation from robotics literature is the opposite. Robotics papers and documentation explain navigation algorithms, SLAM, controllers, costmaps, localization, and sensor fusion, but they usually do not focus on restaurant-specific

service logic. A robot may navigate very well in a test map, yet still fail as a waiter robot if it cannot receive table IDs, understand service tasks, interact with users, or update order status. This is why Nadel combines robotics implementation with a restaurant workflow layer.

Commercial waiter robots demonstrate that the market demand already exists. Systems such as BellaBot and DINERBOT provide strong examples of delivery robots being used in hospitality settings. Nevertheless, these systems are usually closed products. Their public specifications describe features and benefits, but they do not provide a complete educational explanation of how to build a similar system from accessible components. Nadel fills this educational space by documenting the design logic behind a physical prototype [1].

The literature also shows that human factors cannot be added at the end of the project. Robot appearance, interface clarity, service predictability, and speech interaction affect whether customers and staff accept the robot. Nadel responds to this by including a friendly physical design, screen-based interaction, multilingual interface support, and a voice interaction layer based on Whisper Small and Gemma 31B. These elements help the robot behave as a service assistant rather than only a moving machine [18], [20].

Nadel can be positioned as an applied intelligent systems project that combines the main ideas found across the reviewed literature. It is not limited to one research topic. It includes autonomous mobile robotics, restaurant workflow automation, perception, embedded control, human-robot interaction, and AI-based voice understanding. This gives the project a broader scope than a simple robot car or a standard web-based restaurant ordering system.

Compared with hospitality literature, Nadel provides a practical technical prototype that demonstrates how service robot concepts can be implemented. Many hospitality studies explain the benefits and acceptance of service robots, but Nadel translates those ideas into a working robot with physical movement, sensing, navigation, and interaction. This makes the project useful for connecting theoretical service automation ideas with engineering implementation [1].

Compared with navigation literature, Nadel gives the robot a real service purpose. The robot does not navigate only because navigation is interesting; it navigates to perform order delivery. The target table is generated from the restaurant workflow, and the route is planned according to service needs. This makes the navigation system part of a complete application rather than an isolated robotics experiment.

Compared with commercial robot products, Nadel is smaller in scale but more transparent as an academic system. Its value is not only in matching commercial robot performance, but in explaining and demonstrating how the main layers can be integrated using accessible hardware and software tools. This is important for a graduation project because the report must show design, analysis, implementation logic, and learning value.

Therefore, the main contribution of Nadel in relation to the literature is integration. It brings together restaurant service requirements, ROS2 Nav2 navigation, perception sensors, embedded control, voice understanding, and POS workflow into one prototype. The literature supports each layer separately, while Nadel demonstrates their practical combination [9].

This chapter reviewed the literature and related work relevant to Nadel. The review showed that service robots are increasingly important in hospitality because they can support efficiency, reduce repetitive workload, improve consistency, and create new customer experiences. However, their success depends on both technical performance and human acceptance [1].

The chapter also reviewed autonomous mobile robots, SLAM, indoor mapping, ROS2 Nav2, sensing technologies, odometry, embedded control, HRI, voice interaction, POS integration, safety, and commercial waiter robots. The review demonstrated that each technology solves one part of the waiter robot problem, but the main challenge is integrating all parts into one reliable restaurant service system [9], [12].

The literature supports the design direction of Nadel. ROS2 Nav2 is suitable for autonomous indoor navigation. LiDAR, depth camera, ultrasonic sensors, and encoders support perception and movement estimation. Whisper Small and Gemma 31B support voice-based interaction. [9], [13].

The next chapter presents the project terminology. It defines the technical concepts used throughout the report so that the reader can clearly understand the proposed model, system architecture, implementation details, and evaluation presented in the following chapters.

The chapter also highlighted IoT-based table-side interaction and two-way voice interaction as important extensions of smart restaurant systems. In Nadel, the ESP32-based Call/Pay unit connects customer tables to the robot request workflow, while Whisper Small, Gemma 31B, and the speech output layer allow the robot to support voice-based interaction and spoken replies [16], [18].

Table 2-6: Literature gap and corresponding Nadel contribution.

Observed gap in literature or practice	Nadel contribution
Many hospitality studies focus on adoption rather than technical architecture	Nadel documents a complete technical prototype with hardware, navigation, and interaction layers
Commercial waiter robots are mature but often closed systems	Nadel uses accessible components and explainable architecture
Voice interaction is often treated separately from navigation	Nadel integrates Whisper Small and Gemma 31B into the service interaction layer
Safety and reliability are essential in public spaces	Nadel uses obstacle detection and safe navigation concepts, with future safety enhancements identified

3 PROJECT TERMINOLOGY

This chapter defines the main technical terms and concepts used throughout the Nadel project. It introduces autonomous mobile robots, service waiter robots, ROS2-based navigation, mapping, localization, sensor integration, motion control, and human-robot interaction. The chapter also explains the hardware and software terminology related to the Jetson Orin Nano, ESP32, LiDAR, ultrasonic sensors, encoders, IMU, touch-screen interaction, voice processing, and table-side Call/Pay requests. These concepts provide the technical foundation needed to understand the proposed system design and implementation.

3.1 PROJECT TERMINOLOGY AND TECHNICAL BACKGROUND

This chapter defines and explains the main technical terms used throughout the Nadel project. The project combines robotics, artificial intelligence, embedded systems, perception, autonomous navigation, user interaction, and restaurant management technologies. Therefore, a clear terminology chapter is necessary before presenting the proposed model in detail. The terms introduced in this chapter help the reader understand how the robot senses its environment, builds a map, estimates its position, plans a path, receives restaurant orders, processes spoken requests, and moves safely toward a customer table.

Nadel is not only a mechanical robot body. It is an integrated intelligent system that depends on a software stack, a sensing layer, a control layer, a human-robot interaction layer, and a restaurant workflow layer. In the current implementation, the robot uses Ubuntu 22.04, ROS2 Humble, ROS2 Nav2, SLAM Toolbox, RPLIDAR A1M8, Intel RealSense D435i, ultrasonic sensors, encoders, an IMU, NVIDIA Jetson Orin Nano, ESP32, Whisper Small for speech-to-text, Gemma 31B for request interpretation and response generation, and a touch screen interface [9], [12].

The purpose of this chapter is not to provide a full implementation description; this is presented later in Chapter Four. Instead, the chapter gives the conceptual and technical meaning of each major term and explains why it is relevant to Nadel. This

makes the later system architecture, hardware integration, navigation process, and evaluation sections easier to understand.

The project also includes an ESP32-based table-side interaction unit with Call and Pay buttons. This unit allows customers to send a table ID and request intent directly from the table. The buttons do not perform the service or payment operation directly; instead, they request Nadel to navigate to the table so that the interaction can continue through the robot interface and restaurant workflow [16].

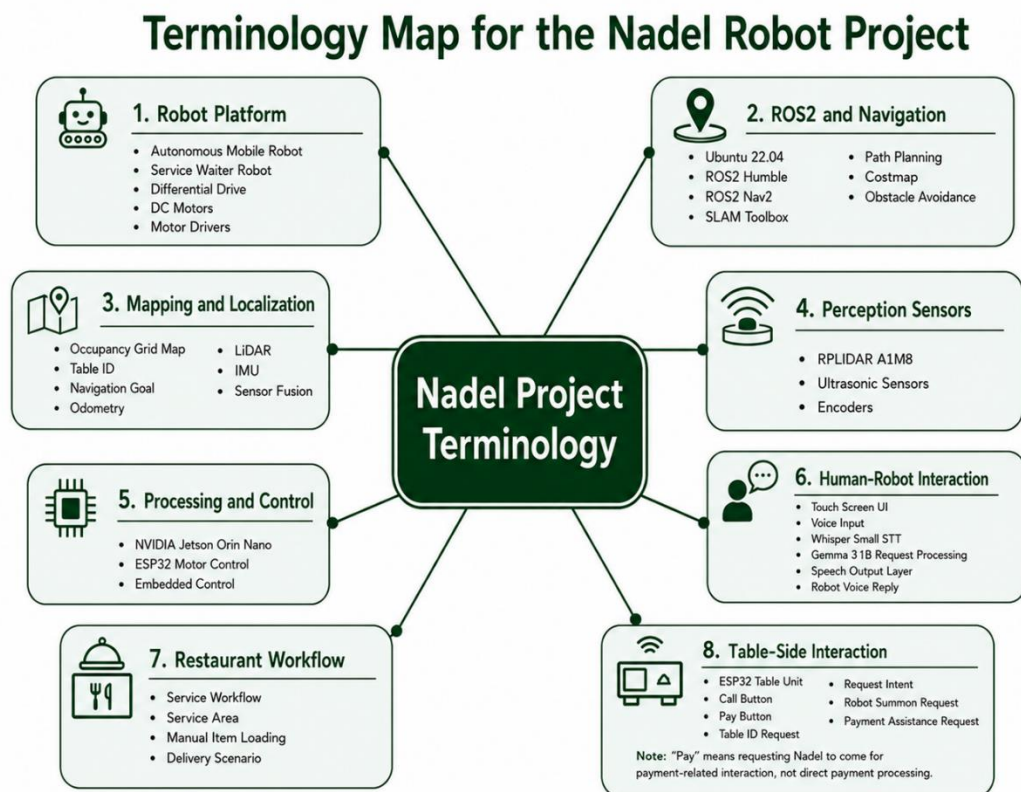


Figure 3-1: Terminology map for the Nadel robot project.

Table 3-1: Main terminology groups used in Nadel.

Term	Meaning	Role in Nadel
Autonomous Mobile Robot	A mobile robot that can move through an environment without continuous manual control.	Nadel uses autonomy to move between the kitchen and customer tables.
Service Robot	A robot designed to perform useful tasks for humans outside classical industrial automation.	Nadel is designed to support restaurant service and delivery tasks.
ROS2	A middleware framework used to connect robot software components through nodes, topics, services, and actions.	It provides the software communication foundation of the robot.
Nav2	The ROS2 navigation stack used for planning, controlling, and executing robot navigation tasks.	It allows Nadel to move to selected table goals.
SLAM	A process where a robot builds a map while estimating its own position inside the environment.	It supports mapping the restaurant environment.
Odometry	Estimation of robot motion based on wheel movement and sensor feedback.	It helps estimate Nadel's position during motion.
Human-Robot Interaction	The communication layer between people and the robot.	It includes touch screen interaction, STT, and LLM-based request processing.
Table-Side Interaction Unit	An ESP32-based module installed on customer tables with Call and Pay buttons.	It sends table ID and request intent so Nadel can navigate to the requesting table.
Pay Button	A physical button used to request Nadel for payment-related assistance.	It sends pay assistance intent and table ID to the system.
Speech Output Layer	A system layer that converts the generated text response into spoken output.	It allows Nadel to reply to the customer by voice.

3.1.1 AUTONOMOUS MOBILE ROBOTS

An autonomous mobile robot is a robotic system capable of moving from one location to another while making decisions about its route, motion, and interaction with the surrounding environment. Unlike a manually controlled robot, an autonomous robot does not require a human operator to continuously direct every movement. Instead, it uses sensors, control algorithms, maps, and navigation software to understand where it is, where it needs to go, and how to reach the target safely.

In Nadel, the autonomous mobile robot concept is applied to indoor restaurant delivery. The robot receives a target table ID, converts it to a navigation goal, plans a path through the restaurant map, avoids obstacles, and moves toward the customer table. This makes the robot suitable for repeated delivery tasks such as transporting food and drinks from the kitchen area to tables. Autonomy is essential because restaurant staff should not need to control the robot manually during every order.

The autonomy of Nadel depends on the integration of perception, mapping, localization, planning, and motion control. These functions are supported by ROS2 Humble, Nav2, SLAM Toolbox, LiDAR, encoders, an IMU, and the ESP32-based motor control layer. Therefore, the term autonomous mobile robot in this project refers to the complete ability of the system to sense, decide, and move within a restaurant environment [9], [12].

3.1.2 SERVICE WAITER ROBOTS

A service waiter robot is a type of service robot designed to assist in restaurant, café, hotel, or food service environments. Its main purpose is to support tasks that are repetitive, time-consuming, or physically demanding for human workers. These tasks may include delivering meals, transporting drinks, guiding customers, displaying menus, receiving simple commands, or providing basic interaction through screens or voice interfaces.

Nadel is classified as a service waiter robot because it is designed to deliver food and drinks inside restaurants and cafés. The robot is connected to the restaurant service workflow, where order information is linked to a target table and then used as a

navigation task. This gives Nadel a practical function within the restaurant process rather than limiting it to movement demonstration only.

The waiter robot concept also requires attention to customer acceptance. A robot that works around customers must look friendly, move safely, and communicate clearly. For this reason, Nadel includes a touch screen interface and a human-robot interaction layer. The physical appearance of the robot, the screen interface, and the service workflow are all part of the waiter robot terminology used in this project.

3.2 ROS2 NAVIGATION, MAPPING, AND LOCALIZATION

This section presents the navigation framework used in the Nadel project. It explains how ROS2 provides the software environment required for robot communication, mapping, localization, path planning, and motion control. The following subsections describe the ROS2 platform and development environment, followed by the navigation components that enable Nadel to move safely between restaurant locations.

3.2.1 ROS2 PLATFORM AND DEVELOPMENT ENVIRONMENT

Robot Operating System 2, commonly known as ROS2, is a robotics middleware framework used to build modular robot applications. It is not an operating system in the same way as Ubuntu or Windows. Instead, it provides tools, communication mechanisms, libraries, and conventions that allow different robot software components to exchange data and work together. ROS2 applications are usually built from nodes that communicate through topics, services, actions, and parameters [39].

Nadel uses ROS2 Humble on Ubuntu 22.04 as the main robotics software environment. ROS2 Humble is a long-term support ROS2 distribution and is commonly used in mobile robotics projects. In Nadel, ROS2 is responsible for connecting the navigation stack, sensor drivers, mapping modules, localization information, velocity commands, visualization tools, and interaction-related topics. This modular design makes the system easier to develop and debug because each function can be separated into a clear software component [9], [12].

ROS2 is important for Nadel because the robot requires continuous communication between sensing, planning, control, and interaction modules. For

example, LiDAR data must be published to mapping or navigation components, odometry data must be used for pose estimation, and velocity commands must be sent toward the motor control layer. ROS2 provides the communication structure that makes this integration possible [9], [12].

Ubuntu 22.04 is the operating system environment used for the Nadel implementation. It provides the Linux-based platform needed to run ROS2 Humble, robotics packages, development tools, sensor drivers, terminal commands, and supporting software. The use of Ubuntu is common in robotics because many robotics libraries and hardware drivers are designed and tested for Linux-based systems [9].

ROS2 Humble is the selected ROS2 distribution for the project. It provides the middleware and tool ecosystem used by the robot, including communication between nodes and integration with the Nav2 navigation stack. The combination of Ubuntu 22.04 and ROS2 Humble gives the project a stable software foundation and supports many commonly used robotics packages, including navigation, mapping, visualization, and sensor integration tools [39].

In Nadel, this platform choice is important because the robot requires reliable execution of navigation and perception tasks. The Jetson Orin Nano runs the high-level software environment, while low-level motor control is supported by the ESP32. This separation allows Ubuntu and ROS2 to handle the high-level robotic logic while the microcontroller handles real-time motor-related communication [9], [15].

3.2.2 NAV2, PATH PLANNING, AND OBSTACLE AVOIDANCE

Nav2 is the navigation framework used in ROS2 to enable autonomous mobile robots to move from one location to another. It provides software modules for path planning, localization, control, recovery behaviors, costmaps, and goal execution. Nav2 allows a robot to receive a navigation goal, compute a route, avoid obstacles, and send motion commands to the robot base [40].

In Nadel, Nav2 is used to navigate from the robot's starting location or kitchen area to a target customer table. The system receives a table ID, converts it into map coordinates, and passes the target pose to Nav2. Nav2 then uses the available map, sensor data, and robot configuration to calculate a safe route. During motion, the local planner updates the robot's behavior according to nearby obstacles and environmental changes [9].

Nav2 is especially relevant to restaurant environments because the robot must move through indoor spaces that include furniture, walls, customers, and staff. The navigation stack helps the robot avoid static objects represented in the map and respond to local obstacles detected by sensors. Therefore, Nav2 forms the core navigation layer of Nadel [9].

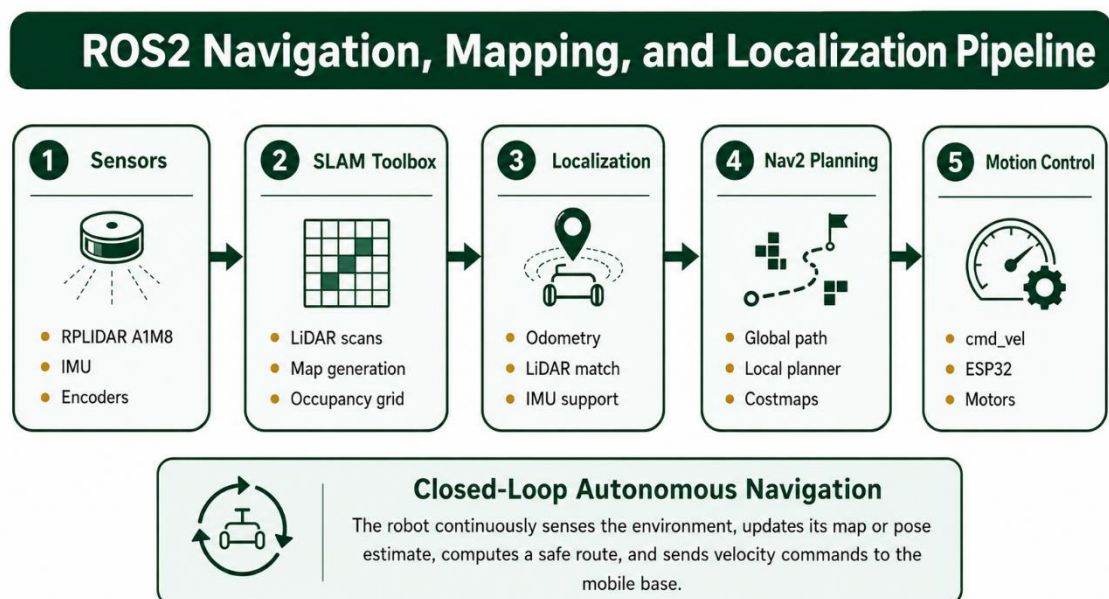


Figure 3-2: ROS2 navigation, mapping, and localization pipeline used conceptually in Nadel

Path planning is the process of finding a route from the robot's current position to a target destination. In mobile robotics, the planned route must consider obstacles, free space, robot size, and environmental constraints. A good path should be safe, feasible, and efficient. Path planning may include global planning, which uses the full map, and local planning, which responds to nearby obstacles.

In Nadel, path planning is handled through ROS2 Nav2. When the restaurant system sends a target table ID, the table ID is converted into coordinates in the map. Nav2 then calculates a path from the robot's current position to the target table. The local planner uses sensor information to avoid obstacles during motion and update the robot's behavior if the environment changes [9].

Path planning is essential in a restaurant because the robot cannot simply move in a straight line to the table. It must avoid walls, tables, chairs, customers, and staff. The path planning terminology in this project therefore refers to the complete process of route generation, obstacle-aware movement, and goal-directed navigation.

Obstacle avoidance is the robot's ability to detect objects in its path and avoid colliding with them. In indoor service environments, obstacles can be static or dynamic. Static obstacles include walls, furniture, counters, and fixed tables. Dynamic obstacles include customers, waiters, moving chairs, or temporary objects placed in the robot's path.

Nadel uses sensor information and navigation software to support obstacle avoidance. The RPLIDAR A1M8 provides 2D scan data, ultrasonic sensors provide close-range detection. [9], [13].

Obstacle avoidance is one of the most important safety terms in the project. Since Nadel works around people, it must not treat the environment as fixed and empty. The robot should continuously monitor its surroundings and adjust its motion to reduce collision risk. This makes obstacle avoidance a core requirement for the robot's deployment in restaurants.

Table 3-3: ROS2 Nav2 navigation terminology.

Term	Definition	Example in Nadel
Global Planner	Generates a route from the current position to the goal using the map.	Determines the main route to the target table.
Local Planner / Controller	Computes short-range motion commands while considering nearby obstacles.	Controls motion during movement through the restaurant.
Costmap	Represents obstacle costs and unsafe areas around the robot.	Helps the robot avoid walls, tables, and dynamic obstacles.
Recovery Behavior	An action used when navigation becomes blocked or fails.	Can help the robot recover from temporary path problems.
Navigation Goal	The target pose that the robot needs to reach.	Generated from the selected table ID.

3.2.3 SLAM, OCCUPANCY GRID MAPS, AND LOCALIZATION

Simultaneous Localization and Mapping, abbreviated as SLAM, is the process by which a robot builds or updates a map of an unknown environment while also estimating its own position within that map. SLAM is important for mobile robots because a robot needs both a map and a pose estimate to navigate effectively. Without mapping, the robot cannot understand where obstacles and free spaces are. Without localization, the robot cannot know where it is inside the environment [12].

The Nadel project uses SLAM Toolbox for mapping. SLAM Toolbox is a ROS-compatible set of tools for 2D SLAM and localization in mapped environments [41]. In the context of Nadel, the restaurant environment can be scanned using the RPLIDAR A1M8, and the resulting data can be used to create an occupancy grid map. This map represents walls, furniture, blocked areas, and free navigation spaces.

The use of SLAM Toolbox supports the practical deployment of Nadel because restaurants may have different layouts. A map can be generated for each environment, and table positions can be linked to coordinates on the map. This makes it possible to

assign table IDs to navigation goals. The concept also allows the robot to be adapted to different indoor environments after recalibration and mapping [12].

An occupancy grid map is a 2D representation of an environment where the space is divided into small cells. Each cell represents whether that area is free, occupied, or unknown. Occupied cells may represent walls, tables, chairs, counters, or other objects. Free cells represent areas that the robot can potentially move through. Unknown cells represent areas that have not been observed clearly by the robot sensors [12].

In Nadel, the occupancy grid map is used to represent the restaurant environment. The RPLIDAR A1M8 scans the environment and provides distance measurements around the robot. These measurements can be processed through the mapping system to generate a 2D map. Nav2 can then use this map to plan a safe path from the kitchen area to a selected table [9], [12].

The occupancy grid is also useful for table representation. Each table can be assigned a unique ID and fixed coordinates on the map. When an order is assigned to a table, the system can convert the table ID into a navigation goal. This connects the restaurant service workflow with the autonomous navigation system [12].

Localization is the process of estimating the robot's position and orientation within the environment. A mobile robot usually represents its pose as x , y , and θ , where x and y define its position on the map and θ defines its heading direction. Accurate localization is necessary because the robot must know where it is before it can navigate toward a target location.

In Nadel, localization is supported by odometry, LiDAR, and IMU information. Wheel encoders estimate the robot's movement by measuring wheel rotation, while the LiDAR provides environment-based information that can help compare the robot's observations with the map. The IMU supports orientation and motion feedback. Together, these sources improve the robot's ability to estimate its movement and position [13].

Localization is important in restaurants because small errors can cause the robot to stop at the wrong place, collide with obstacles, or fail to reach the target table. Using more than one source of localization information improves reliability compared with depending on wheel odometry only. This is because wheel-based odometry can drift over time due to slipping, uneven surfaces, and mechanical errors.

3.2.4 COSTMAPS AND COORDINATE FRAMES

A costmap is a map representation used by navigation systems to describe how safe or unsafe different areas are for robot movement. Instead of simply classifying cells as free or occupied, the costmap assigns values to areas based on obstacles, inflation zones, and navigation risk. Areas close to obstacles may have higher cost, which encourages the robot to choose safer paths.

In Nadel, costmaps are important because the robot must move near furniture and people without touching them. Nav2 can use global and local costmaps to represent the known map and nearby sensor observations. The global costmap helps plan routes through the restaurant, while the local costmap helps the robot respond to obstacles detected during movement [9].

Costmap terminology is important for understanding why the robot does not simply move through every visually open area. The robot has physical dimensions and safety margins. Therefore, the navigation system must consider not only whether a cell is occupied, but also whether it is safe for the robot body to pass through that area.

A coordinate frame is a reference system used to describe positions and orientations. In robotics, different coordinate frames may represent the map, the robot base, sensors, wheels, or other parts of the robot. Transformations between frames allow the robot software to understand where each component is located relative to the others.

Robot pose usually describes the robot's position and orientation. In 2D navigation, the pose is commonly represented as x , y , and θ . The x and y values represent the position on the map, while θ represents the robot's heading direction. Navigation systems use pose information to determine where the robot is and how it should move toward the goal.

In Nadel, coordinate frames and robot pose are important because the navigation stack must connect sensor data, map information, odometry, and movement commands. If coordinate frames are inconsistent, the robot may plan incorrectly or move in the wrong direction. Therefore, correct frame management is part of the technical foundation of the system [9].

3.3 MOTION CONTROL AND ROBOT HARDWARE

This section describes the motion control principles and hardware components that enable Nadel to move safely and reliably in an indoor restaurant environment. The robot uses a differential-drive mobile base supported by DC motors, motor drivers, wheel encoders, an ESP32 controller, and sensing components for feedback and safety. The following subsections explain encoder-based odometry, differential-drive kinematics, perception sensors, embedded processing, motor control, and sensor fusion within the proposed robot system.

3.3.1 ENCODER-BASED ODOMETRY AND DIFFERENTIAL DRIVE KINEMATICS

Encoder-based odometry estimates robot movement using wheel encoders. An encoder measures how much a wheel has rotated. By knowing the wheel radius, the distance between the wheels, and the number of encoder ticks, the system can estimate how far the robot has moved and how much it has turned. This information is then used to update the estimated pose of the robot.

In Nadel, encoders are used with the DC brushed motors to provide feedback for motion estimation. This is important because the robot needs to know whether it is moving as expected. Encoder feedback supports closed-loop motion control and helps the system estimate displacement during navigation. It is also used as part of the odometry information provided to the robot software stack.

However, odometry alone is not perfect. If the wheels slip, if the floor is uneven, or if the robot carries different loads, the estimated movement may differ from the real movement. This error is known as odometry drift. Therefore, Nadel combines odometry with LiDAR and IMU information to support more reliable localization [13].

Differential drive is a common mobile robot drive mechanism that uses two main driven wheels, usually one on the left side and one on the right side. The robot moves forward when both wheels rotate in the same direction at the same speed. It turns when the wheels rotate at different speeds. This design is simple, reliable, and widely used in indoor mobile robots.

Nadel uses a differential-drive-based movement concept with two DC brushed motors and supporting caster wheels. The caster wheels help balance the structure, while the two driven wheels control motion. By changing the speed of the left and right wheels, the robot can move forward, rotate, or follow curved paths. This movement model is compatible with many ROS2 mobile robot navigation configurations [9].

Differential drive kinematics connects velocity commands to wheel speeds. In ROS2, the navigation system can generate velocity commands, commonly represented as linear and angular velocity. These commands are then converted into motor control signals. In Nadel, the ESP32 supports the low-level motor control layer that receives motion-related commands and converts them into appropriate motor outputs [9], [16].

Table 3-2: Sensor terminology and function in Nadel.

Component	Type of information	Use in the project
RPLIDAR A1M8	360-degree 2D LiDAR scanning	Mapping, obstacle awareness, and navigation support
Ultrasonic Sensors	Short-range distance sensing	Blind spot coverage and close obstacle detection
Wheel Encoders	Wheel rotation feedback	Odometry and motion estimation
IMU	Inertial orientation and motion feedback	Orientation support and motion stability information

3.3.2 PERCEPTION SENSORS

LiDAR stands for Light Detection and Ranging. It is a sensing technology that measures distance by emitting laser light and measuring the reflected signal. In mobile robotics, 2D LiDAR sensors are commonly used for mapping, localization, and obstacle detection. They provide distance measurements around the robot, which can be converted into a point cloud or scan representation [12], [13].

The Nadel project uses the RPLIDAR A1M8, a 360-degree 2D laser scanner. This sensor is suitable for indoor mapping and navigation because it can provide distance information around the robot. In the project, LiDAR data is used to support environment mapping, obstacle awareness, and navigation. It helps the robot understand where walls, furniture, and obstacles are located [43].

LiDAR is especially useful in restaurant environments because the robot needs awareness of obstacles around it. Unlike a forward-facing camera, a 360-degree LiDAR can observe a wider area around the robot. This improves the robot's ability to navigate safely in indoor spaces and reduces the risk of colliding with objects that are not directly in front of the camera [13].

Ultrasonic sensors measure distance by emitting sound waves and measuring the time taken for the echo to return. They are usually used for short-range distance measurement. Although they are simpler than LiDAR or depth cameras, they are useful as additional safety sensors because they can detect nearby objects in specific directions [13], [14].

In Nadel, ultrasonic sensors are used to support short-range obstacle detection and blind spot coverage. This is important because LiDAR and cameras may not cover every close-range area depending on sensor placement and field of view. Ultrasonic sensors can provide extra information when an object is very close to the robot body [13].

The term ultrasonic sensing in this project refers to the use of these sensors as a supportive safety layer rather than the primary navigation sensor. The robot mainly depends on ROS2 Nav2, mapping, LiDAR, odometry, and other localization inputs for navigation. Ultrasonic sensors add an additional practical safety function [9], [12].

An inertial measurement unit, abbreviated as IMU, is a sensor module that measures motion-related information such as acceleration, angular velocity, and orientation-related changes. In mobile robotics, IMU data can help improve the understanding of robot movement, especially rotation and orientation changes.

In Nadel, the IMU is part of the localization and motion feedback layer. It supports odometry and LiDAR-based localization by providing inertial feedback. This is useful because wheel encoders alone may accumulate error, especially when wheels slip or when the robot moves on uneven surfaces. IMU information can help improve the estimation of orientation and motion behavior [13].

The IMU is not treated as a complete localization solution by itself. Instead, it contributes to a multi-sensor localization approach that includes odometry, LiDAR information, and inertial feedback. This is more suitable for a mobile robot working in a real indoor environment [13].

3.3.3 EMBEDDED PROCESSING AND MOTOR CONTROL

The NVIDIA Jetson Orin Nano is the main processing unit used in the Nadel project. It is an embedded AI computing platform designed for robotics, and edge AI applications. Compared with simple microcontrollers, it provides much higher processing capability and can run Linux, ROS2, perception software, and high-level control modules [44].

In Nadel, the Jetson Orin Nano runs Ubuntu 22.04 and the ROS2 Humble software environment. It is responsible for handling high-level tasks such as navigation, mapping, sensor data processing, and interaction-related software. It communicates with sensors and sends control-related commands to the lower-level motor control layer [9], [12].

The use of Jetson Orin Nano is important because the robot requires more processing power than a simple line-following or manually controlled robot. Running ROS2 Nav2, SLAM Toolbox, sensor drivers, and interaction modules requires a capable onboard computer. Therefore, the Jetson Orin Nano is considered the central computing unit of the Nadel prototype [9], [12].

The ESP32 is a microcontroller with integrated Wi-Fi and Bluetooth connectivity. It is widely used in embedded systems and Internet of Things applications because it is compact, affordable, and capable of handling low-level control and communication tasks [45]. In robotics, microcontrollers are commonly used to interface with motors, sensors, and low-level hardware.

In the Nadel project, the ESP32 is used as the low-level control interface between the high-level software and the robot's motion hardware. It receives velocity or command information from the high-level system and converts it into motor control signals through the motor drivers. Its Wi-Fi capability can also support wireless communication in the restaurant workflow [16].

This separation between the Jetson Orin Nano and ESP32 is important. The Jetson handles complex computation and ROS2-based logic, while the ESP32 handles hardware-level control. This makes the architecture more organized and allows each processing unit to perform the tasks it is best suited for [9], [15].

In addition to its role in the robot control layer, ESP32 is also used in the table-side interaction unit. Each table unit can include an ESP32 module connected to two physical buttons: Call and Pay. When the Call button is pressed, the ESP32 sends call intent and table ID. When the Pay button is pressed, the ESP32 sends pay assistance intent and table ID. The buttons do not execute the service or payment directly; they only request Nadel to come to the table so that the customer interaction can continue through the robot interface [16].

DC motors are electrical motors that convert electrical energy into rotational mechanical motion. In mobile robots, DC motors are commonly used to drive wheels and produce movement. A motor driver is an electronic circuit or module that controls the power delivered to the motors. It allows the control system to change motor speed and direction.

Nadel uses two DC brushed motors with motor drivers and encoders. The motors drive the main wheels of the robot, while the motor drivers receive control signals and supply the required motor power. Encoders provide feedback about wheel rotation, making it possible to estimate movement and support closed-loop control.

The terminology of motors and drivers is important because autonomous navigation ultimately depends on physical motion. Even if the path planning algorithm produces a correct path, the robot must still convert that plan into wheel movement. The motor control layer is therefore the final execution layer of the navigation pipeline.

3.3.4 SENSOR FUSION

Sensor fusion is the process of combining information from multiple sensors to obtain a more accurate or reliable understanding of the robot's state or environment. Each sensor has advantages and limitations. For example, encoders provide wheel movement data, LiDAR provides distance scans, [13], [14].

In Nadel, localization and environment awareness depend on more than one sensor source. The robot uses odometry, LiDAR, and IMU information for localization support. It also uses ultrasonic sensors as part of the perception layer. Combining these sources helps the system become more robust than depending on a single sensor only [13], [14].

Sensor fusion is particularly important in restaurant environments because conditions can change quickly. People move, chairs shift, objects appear, and the robot may experience small movement errors. By combining different sources of information, the robot can improve navigation reliability and safety.

3.4 HUMAN-ROBOT INTERACTION AND RESTAURANT INTEGRATION

This section explains how customers and restaurant staff interact with Nadel during service tasks. The robot supports communication through a touch-screen interface, voice interaction, and an ESP32-based table-side Call/Pay unit. The Call and Pay buttons send the table ID and request intent only, allowing Nadel to navigate to the requesting table for service or payment-related interaction. The following subsections describe the user interface, speech interaction using Whisper Small and Gemma 31B, and the table-side request mechanism used to support the restaurant service scenario.

3.4.1 TOUCH SCREEN INTERFACE AND MULTIMODAL INTEGRATION

Human-robot interaction, often abbreviated as HRI, refers to the ways humans communicate with and understand robots. In service robots, HRI is especially important because the robot works in public or semi-public environments. Customers and staff need to know what the robot is doing, how to give it commands, and how to respond to its behavior [1].

In Nadel, HRI includes the touch screen interface, the robot face display, the speech-to-text layer, and the language model response processing layer. The robot uses a touch screen UI to display information and provide interaction. It also uses Whisper Small to convert spoken customer input into text and Gemma 31B to interpret requests and generate suitable responses [18], [20].

Good HRI design improves customer acceptance. A restaurant robot should not appear confusing or difficult to use. The interface should support simple communication, clear feedback, and understandable service actions. Therefore, HRI is not an optional feature in Nadel; it is part of the complete service experience.

A touch screen user interface is an interactive display that allows users to select options, view information, and communicate with the system through direct touch. In service robots, touch screens are useful because they provide a clear and familiar way for customers or staff to interact with the robot. They can display menus, order status, messages, robot expressions, or service instructions [1].

Nadel uses a touch screen interface as part of the human-robot interaction layer. The interface can display restaurant-related information and support interaction with customers. It also contributes to the visual identity of the robot by making the robot appear more interactive and customer-friendly. This is important because customers in restaurants may prefer clear visual guidance rather than hidden system behavior.

The touch screen interface also supports the practical restaurant workflow. It can show order information, table-related details, service messages, or robot status. When combined with STT and LLM processing, the screen becomes part of a multimodal interaction system where customers can interact using both voice and visual interface elements.

Multimodal interaction means that users can interact with a system through more than one communication mode. In robots, common modes include touch screens, voice input, visual feedback, gestures, sounds, and physical movement. Multimodal interaction is useful because different users and environments may require different interaction methods.

Nadel uses a touch screen interface and voice-based interaction through Whisper Small and Gemma 31B. The touch screen provides a visual and direct interaction method, while the speech layer allows spoken requests to be converted into text and interpreted by the language model. This combination makes the robot more flexible than a system that depends on only one input method [18], [20].

In a restaurant environment, multimodal interaction is valuable because customers may prefer to interact in different ways. A staff member may use the touch interface to check order information, while a customer may speak a simple request. This improves usability and supports a more natural service experience.

3.4.2 SPEECH INTERACTION USING WHISPER SMALL AND GEMMA 31B

Speech-to-text, abbreviated as STT, is the process of converting spoken audio into written text. STT is useful in human-robot interaction because it allows users to communicate with a robot through voice instead of using only buttons or screens. This can make the system more natural, accessible, and convenient in service environments [18].

Nadel uses Whisper Small as the speech-to-text module. Whisper is a general-purpose speech recognition model trained for multilingual and multitask speech recognition [46]. In the project, Whisper Small converts customer voice input into text so that the system can process spoken service requests. This supports hands-free interaction between customers and the robot.

The STT layer is important because it connects natural speech with the robot's digital workflow. For example, a customer may ask a simple question or give a command, and the system must convert this audio input into a text representation. The transcribed text can then be passed to the language model for interpretation.

A large language model, abbreviated as LLM, is an artificial intelligence model designed to process and generate natural language. LLMs can be used for tasks such as question answering, summarization, intent understanding, dialogue generation, and command interpretation. In a service robot, an LLM can help convert a user's text request into a meaningful response or action [20].

In the Nadel project, Gemma 31B is used as the language model for request interpretation and response generation. After Whisper Small converts customer speech into text, Gemma 31B processes the transcribed request and helps generate an appropriate system response. This makes the robot interaction layer more intelligent than a fixed menu of hard-coded phrases [18], [20].

The LLM does not replace the navigation or control system. Instead, it supports the communication layer. For example, it can help interpret a customer request, generate a clear response, or connect the user's intention with available service actions. This separation is important because safety-critical robot movement should remain controlled by structured navigation and control modules, while the LLM supports language understanding and interaction.

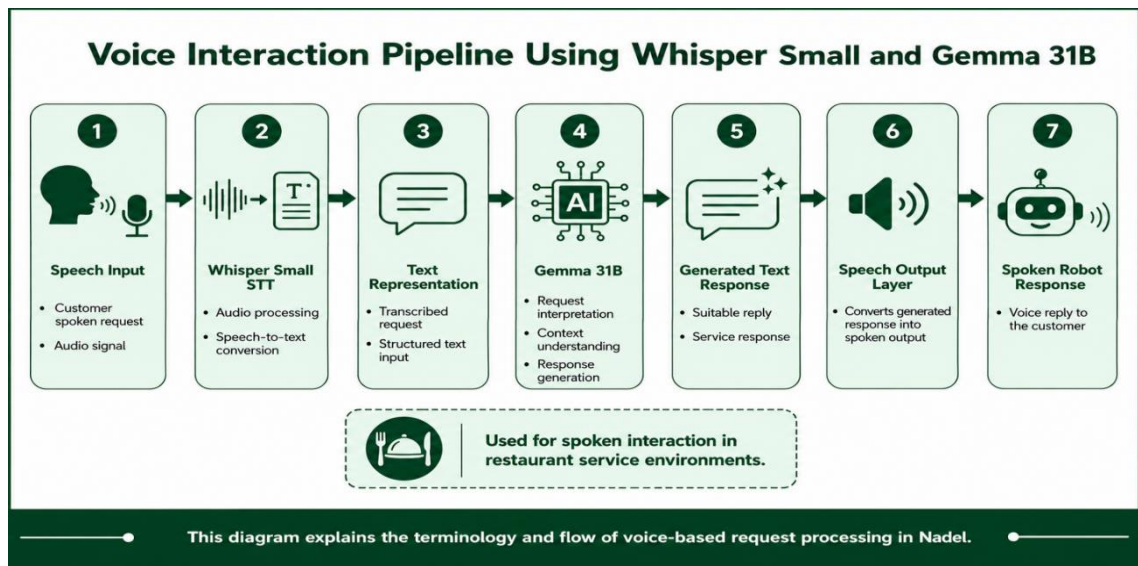


Figure 3-3: Voice interaction pipeline using Whisper Small and Gemma 31B.

3.4.3 TABLE ID AND GOAL REPRESENTATION

A table ID is a unique identifier assigned to each customer table in the restaurant. In a smart restaurant system, table IDs help connect orders to physical locations. For example, an order can be linked to Table 3, and the system can use that ID to identify where the food should be delivered.

In Nadel, the table ID is connected to the navigation goal. Each table can be represented by coordinates in the occupancy grid map. When the POS or restaurant workflow assigns a delivery task, the table ID is converted into a target pose. Nav2 can then use this target pose as the goal for autonomous navigation [9], [12].

This terminology is important because it explains the connection between restaurant management data and robot movement. The robot does not simply move to random coordinates selected by a developer. It moves according to a service task that originates from the restaurant workflow.

This chapter explained the main terminology required to understand the Nadel project. The project includes concepts from autonomous mobile robots, service robots, ROS2, Nav2, SLAM Toolbox, occupancy grid mapping, localization, odometry, differential drive, LiDAR sensing, depth cameras, ultrasonic sensors, IMU feedback, sensor fusion, embedded control, human-robot interaction, speech-to-text, large language models, touch screen interfaces, POS systems, and kitchen display systems [9], [12].

The terminology shows that Nadel is a complete intelligent system rather than only a mechanical prototype. The robot combines a physical mobile platform with a sensing layer, a navigation layer, a control layer, an interaction layer, and a smart restaurant workflow. ROS2 Humble and Nav2 support autonomous movement, SLAM Toolbox supports mapping, sensors provide environmental information, the Jetson Orin Nano provides high-level processing, the ESP32 supports motor control, and Whisper Small with Gemma 31B supports voice-based request processing [9], [12].

Chapter Four

4 PROPOSED MODEL

This chapter presents the proposed model of Nadel, a fully autonomous service waiter robot designed for smart restaurant environments. The chapter moves from the general system concept to the detailed hardware, software, communication, safety, interaction, and operational workflow layers. It explains how the physical robot, the restaurant workflow, the table-side interaction unit, the navigation stack, and the human-robot interaction layer are integrated into one working prototype [9].

The proposed model is based on a layered architecture. The high-level processing layer is implemented using the NVIDIA Jetson Orin Nano, which runs Ubuntu 22.04, ROS2 Humble, ROS2 Nav2, SLAM Toolbox, the touch screen interface, and the AI interaction components. The low-level control layer is implemented using an ESP32 robot controller, which is responsible for motor control and low-level feedback. The drive system uses two IBT-2 BTS7960 motor drivers and two RS775SH geared DC motors, while support and balance are provided by the two main drive wheels and two metallic caster wheels [9], [12].

Nadel also includes a smart restaurant interaction model. Customers can interact with the system through the touch screen UI or through voice after the robot reaches the table. Voice interaction uses Whisper Small for speech-to-text, Gemma 31B for request interpretation and response generation, and a speech output layer that allows the robot to reply by voice. The system also includes an ESP32-based table-side Call/Pay unit. The Call and Pay buttons do not directly perform service or payment operations. They only send the table ID and request intent so that Nadel can navigate to the requesting table. After arrival, the customer continues the service or payment-related interaction through the robot interface [16], [18].

4.1 SYSTEM OBJECTIVES AND CONSTRAINTS

The proposed model was designed to satisfy both technical and service-oriented requirements. Technically, the robot must move safely in indoor restaurant environments, build and use an occupancy grid map, localize itself using odometry, LiDAR, and IMU feedback, avoid obstacles, and receive table-based navigation goals. From a service perspective, the robot must support food delivery, customer interaction, table-side requests, voice-based communication, and a home-position workflow after completing the full interaction cycle [12], [13].

Table 4-1: System objectives and constraints of the proposed Nadel model.

Objective	Implementation in Nadel	Main constraint
Autonomous navigation	Use ROS2 Nav2 and SLAM Toolbox to move between kitchen, customer tables, and home position.	Indoor environment changes, narrow spaces, and dynamic obstacles.
Reliable localization	Combine odometry, LiDAR, and IMU information to estimate robot pose.	Odometry drift and sensor noise may affect long-distance motion.
Low-level motor control	Use ESP32 to translate movement commands into motor driver signals.	Motor response depends on power stability, driver behavior, and feedback quality.
Customer interaction	Allow interaction through touch screen UI and two-way voice communication.	Restaurant noise and user behavior can affect voice recognition quality.
Table-side request support	Use ESP32 Call/Pay unit to send table ID and request intent only.	Wireless reliability and correct table ID handling are required.
Safety and power control	Use power switch, emergency stop, voltage regulators, and safe drive control.	Emergency stop behavior must override normal motion commands.
Operational recovery	Return to home position after completing the full interaction.	The home position must remain reachable and accurately mapped.

4.2 OVERALL SYSTEM ARCHITECTURE

The overall architecture of Nadel is organized into five main layers: customer and restaurant interaction, AI and request processing, smart restaurant workflow, robot intelligence, and hardware/motion control. This structure separates the system into clear functional responsibilities. The customer-facing layer receives requests from the touch screen UI, voice input, and ESP32 table-side unit. The AI layer processes voice input using Whisper Small and Gemma 31B, while the restaurant workflow layer manages orders, table requests, kitchen preparation, and robot task assignment [16], [18].

The robot intelligence layer runs on the Jetson Orin Nano and contains ROS2 Humble, ROS2 Nav2, SLAM Toolbox, path planning, localization, and high-level decision logic. The sensing layer provides environmental and motion feedback through the RPLIDAR A1M8, ultrasonic sensors, encoders, and MPU6050 IMU. Finally, the motion, power, and safety layer contains the ESP32 motor controller, IBT-2 BTS7960 motor drivers, RS775SH motors, main wheels, metallic caster wheels, battery, voltage regulators, power switch, and emergency stop [9], [12].

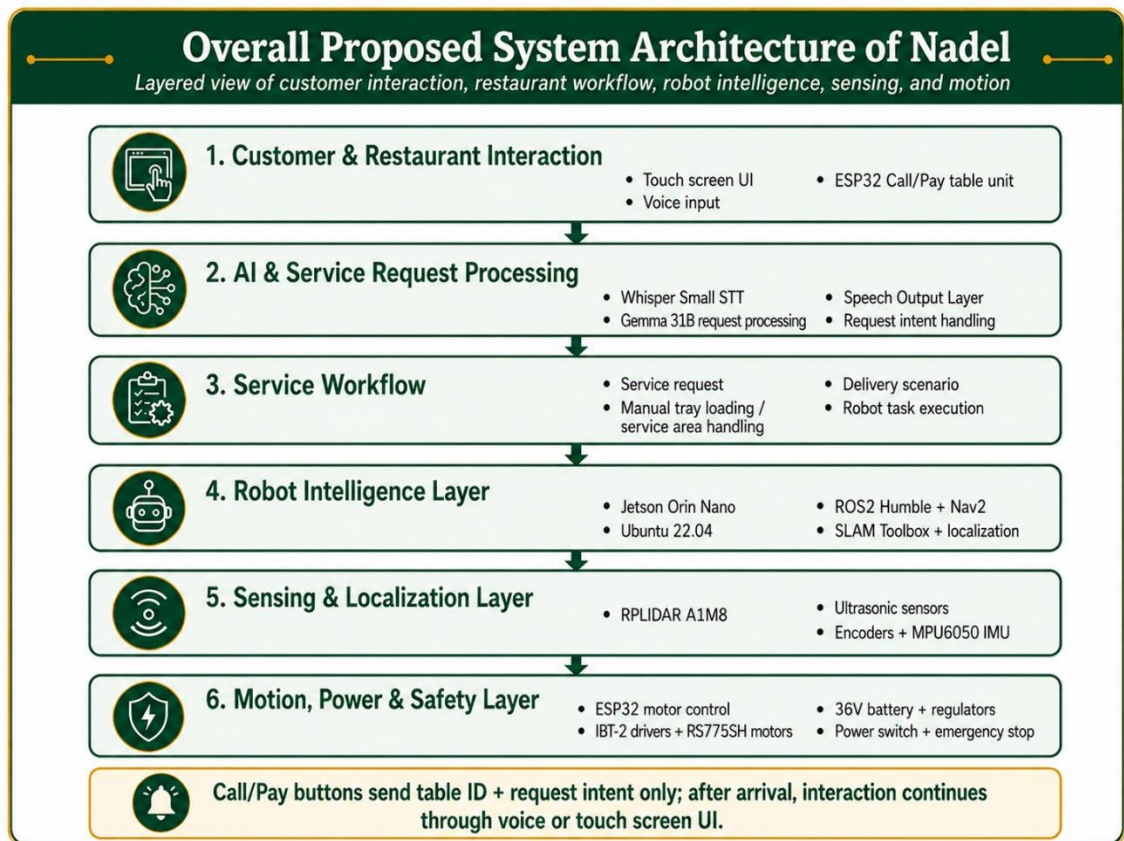


Figure 4-1: Overall proposed system architecture of Nadel. Source: Authors' diagram.

4.2.1 HARDWARE, POWER, AND SAFETY ARCHITECTURE

The hardware architecture of Nadel follows a layered control structure. The Jetson Orin Nano acts as the high-level processing unit, while the ESP32 microcontroller acts as the low-level robot controller. This separation is important because computationally intensive tasks such as mapping, localization, path planning, interface control, and AI processing are handled by the Jetson, while direct motor control and low-level feedback handling are managed by the ESP32 [12], [15].

The two RS775SH geared DC motors are used as the main actuation elements. Each motor is connected to one of the two main drive wheels, forming a two-wheel differential drive system. Two IBT-2 BTS7960 motor drivers are used, with one driver assigned to each motor. This structure allows the robot to move forward, move backward, turn, and rotate by changing the speed and direction of the left and right drive wheels. The two metallic caster wheels provide additional support and balance, especially under the robot base.

The internal electronic components are mounted on an acrylic platform inside the robot body. This platform holds the ESP32 board, voltage regulation modules, motor drivers, wiring, and power distribution elements. The internal hardware image is used as visual support for the physical arrangement only; the technical explanation in this chapter describes the system-level architecture rather than a pin-by-pin wiring layout [16].

Table 4-2: Main hardware components used in Nadel.

Component	Role	Function in Nadel
NVIDIA Jetson Orin Nano	High-level processing unit	Runs Ubuntu 22.04, ROS2 Humble, Nav2, SLAM Toolbox, UI, and AI interaction layer.
ESP32 robot controller	Low-level embedded controller	Receives motion commands, controls motor drivers, and handles feedback-related signals.
2 x IBT-2 BTS7960 motor drivers	Motor power control	Drive the left and right RS775SH motors using control signals from ESP32.
2 x RS775SH geared DC motors	Main actuation	Provide left and right wheel motion for differential drive movement.
2 main wheels	Drive wheels	Support forward, backward, turning, and rotation movement.
2 metallic caster wheels	Support and balance	Provide smooth omni-directional support under the robot base; each wheel supports up to about 30 kg.
36V battery pack	Main power source	Provides the main electrical energy for the robot power system.
2 voltage regulators	Power regulation	Provide suitable regulated voltage levels for electronic modules and peripherals.
Power switch	Normal power control	Used to turn the robot system on and off during normal operation.
Emergency stop	Safety override	Used to immediately stop robot motion in emergency conditions.
Touch screen	Customer interface	Supports menu display, order interaction, confirmation, and robot interface functions.
Speaker	Speech output	Plays the robot voice reply generated through the speech output layer.
ESP32 table-side unit	Remote table request module	Sends table ID and Call/Pay request intent only.

JETSON ORIN NANO

Compact Edge AI Computing Platform



1. Edge AI Performance

Designed for high-performance AI inference at the edge.



2. NVIDIA Ampere Architecture

Modern GPU architecture with Tensor Cores for AI workloads.



3. Compact Embedded Form Factor

Small footprint suitable for robotics and embedded systems.



4. Power-Efficient Operation

Suitable for low-power edge computing and mobile robotics.



5. Connectivity & Interfaces

Supports common development interfaces such as USB, GPIO, Ethernet, and camera integration.



6. Use Cases

Robotics, computer vision, autonomous systems, and edge AI applications.



AI Platform



Edge Computing



Embedded Robotics



Developer Friendly

Figure 4-4: NVIDIA Jetson Orin Nano used as the high-level processing platform. Source: Authors' component illustration.



ESP32 Development Board

ESP-WROOM-32 DevKit



Processor:

Dual-core Xtensa LX6, up to 240 MHz



Wireless:

2.4 GHz Wi-Fi + Bluetooth / BLE



Operating Voltage:

3.3 V



Programming Port:

Micro-USB



GPIO:

Multiple digital, analog, PWM, UART, I2C, and SPI pins



Common Uses:

IoT, robotics, sensors, and automation



Wi-Fi
2.4 GHz



Bluetooth
BLE



Dual-Core
240 MHz



IoT
Ready

Figure 4-5: ESP32 development board used for embedded robot control and table-side interaction modules. Source: Authors' component illustration.

Nadel uses a battery-based power system. The battery pack used in the prototype is a 36V 10S2P scooter battery pack with a capacity of 4400mAh and an energy rating of approximately 158.4Wh. The main battery power is distributed inside the robot body. Voltage regulator modules are used to provide suitable voltage levels for electronics and peripherals. This is necessary because motors, motor drivers, embedded controllers, sensors, screen, and audio components do not all operate at the same voltage or current level.

The power switch is used as the normal power control element. When the switch is activated, the internal power distribution layer can energize the robot electronics and allow the system to start. This is different from the emergency stop. The switch button is intended for normal startup and shutdown, while the emergency stop is intended for unsafe or urgent situations during operation.

The emergency stop provides a manual safety override. During unsafe behavior, unexpected motion, or emergency conditions, pressing the emergency stop should interrupt robot motion at the drive level or disable motor actuation. In the proposed model, the emergency stop has higher priority than normal movement commands. This means that even if the Jetson or ESP32 is sending motion commands, the safety layer can stop the robot movement immediately. After the unsafe condition is resolved, the robot can be reset according to the operating procedure [16].

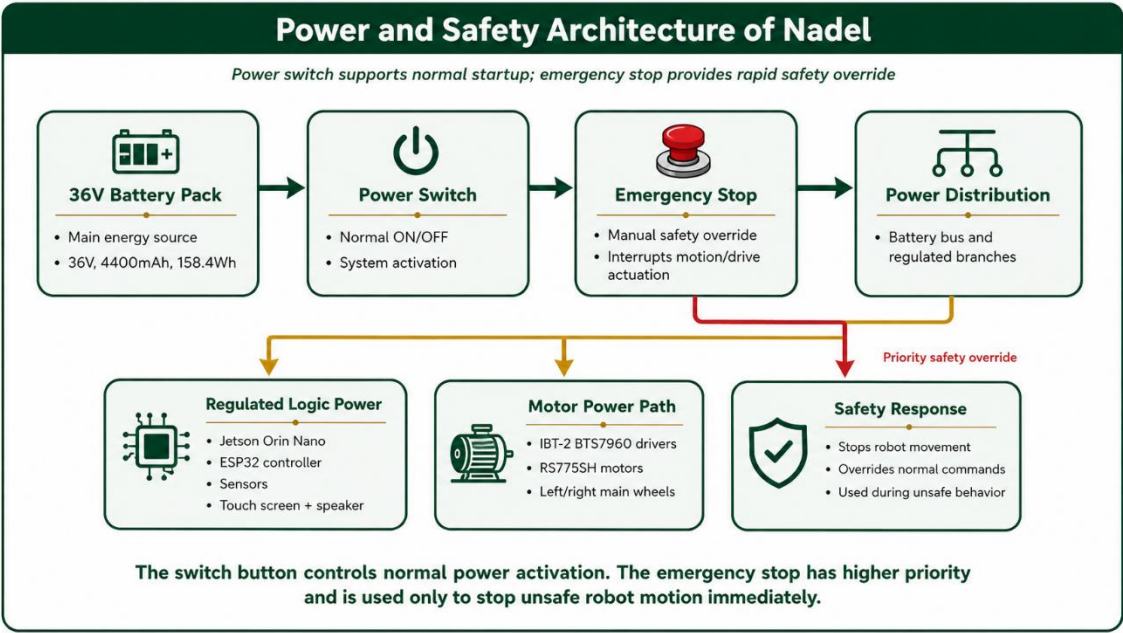


Figure 4-6: Power and safety architecture of Nadel. Source: Authors' diagram.



YONGZHI Battery Pack

10S2P Scooter Battery

	Product Name: Battery pack for 2 wheels balancing scooter
	Product Model: 10S2P
	Standard Voltage: 36V
	Capacity: 4400mAh
	Power: 158.4Wh
	Inner Cell: 18650
	Dimensions: 135 mm × 90 mm × 60 mm

**Common Use:** Self-balancing scooter / robotics power system

Figure 4-7: YONGZHI 36V battery pack used as the main power source. Source: Authors' component illustration.



Figure 4-8: Power switch used for normal robot power control. Source: Authors' component illustration.



Figure 4-9: Emergency stop button used for robot safety control. Source: Authors' component illustration.

4.2.2 SOFTWARE, COMMUNICATION AND EMBEDDING CONTROL ARCHITECTURE

The software architecture of Nadel is built around ROS2 Humble on Ubuntu 22.04. ROS2 provides the communication framework between navigation, sensing, control, and high-level logic modules. ROS2 Nav2 is used to enable autonomous navigation from one point to another inside the restaurant environment. SLAM Toolbox is used to build and use the map of the indoor environment. The robot can then use table IDs, delivery points, and the home position as navigation targets [9], [12].

The AI interaction layer is also part of the software architecture. Whisper Small is used for speech-to-text, converting customer speech into a text representation. Gemma 31B processes the text request, interprets the service context, and generates an appropriate response. The generated response is then sent to a speech output layer, which converts the response into spoken output through the speaker. This makes the voice interaction two-way: the customer speaks to the robot, and the robot replies by voice [18], [20].

The touch screen UI is another major software interaction component. After the robot reaches the customer table, the customer can interact with the robot through the screen by selecting menu items, confirming order details, or continuing service-related interaction. This UI-based interaction works alongside the voice interaction layer and provides an alternative input method for customers who prefer visual interaction or are in a noisy environment.

Table 4-3: Software tools and frameworks used in Nadel.

Software / module	Layer	Function
Ubuntu 22.04	Operating system	Base operating system running on Jetson Orin Nano.
ROS2 Humble	Robotics middleware	Node communication, data exchange, and modular robot software structure.
ROS2 Nav2	Navigation stack	Goal-based indoor navigation, planning, local control, and obstacle-aware motion.
SLAM Toolbox	Mapping and	Map generation and localization

Software / module	Layer	Function
	SLAM	support using LiDAR data.
Whisper Small	Speech-to-text	Converts customer voice input into text.
Gemma 31B	Language model	Interprets requests and generates suitable text responses.
Speech Output Layer	Voice reply output	Converts generated text response into spoken output.
Touch screen UI	Human-machine interface	Menu display, command selection, order confirmation, and interaction after arrival.
ESP32 firmware	Embedded control software	Controls motor drivers, handles low-level signals, and supports table-side request modules.

The motor control layer is responsible for converting navigation decisions into physical movement. The Jetson Orin Nano computes high-level movement requirements through ROS2 Nav2. These movement commands are sent to the ESP32 robot controller through the system communication interface. The ESP32 then translates the command into control signals suitable for the IBT-2 BTS7960 motor drivers [9]

Two IBT-2 motor drivers are used because the robot has two independently controlled drive motors. The left motor driver controls the left RS775SH motor, and the right motor driver controls the right RS775SH motor. By controlling the speed and direction of each motor independently, the robot achieves differential drive movement. If both wheels rotate forward at similar speeds, the robot moves forward. If one wheel moves faster than the other, the robot turns. If the wheels rotate in opposite directions, the robot can rotate in place.

The main wheels provide the drive force, while the metallic caster wheels provide support and balance under the robot base. The caster wheels are not powered; instead, they roll passively and help distribute the load. This combination allows the robot to maintain stability while using a two-wheel differential drive system.

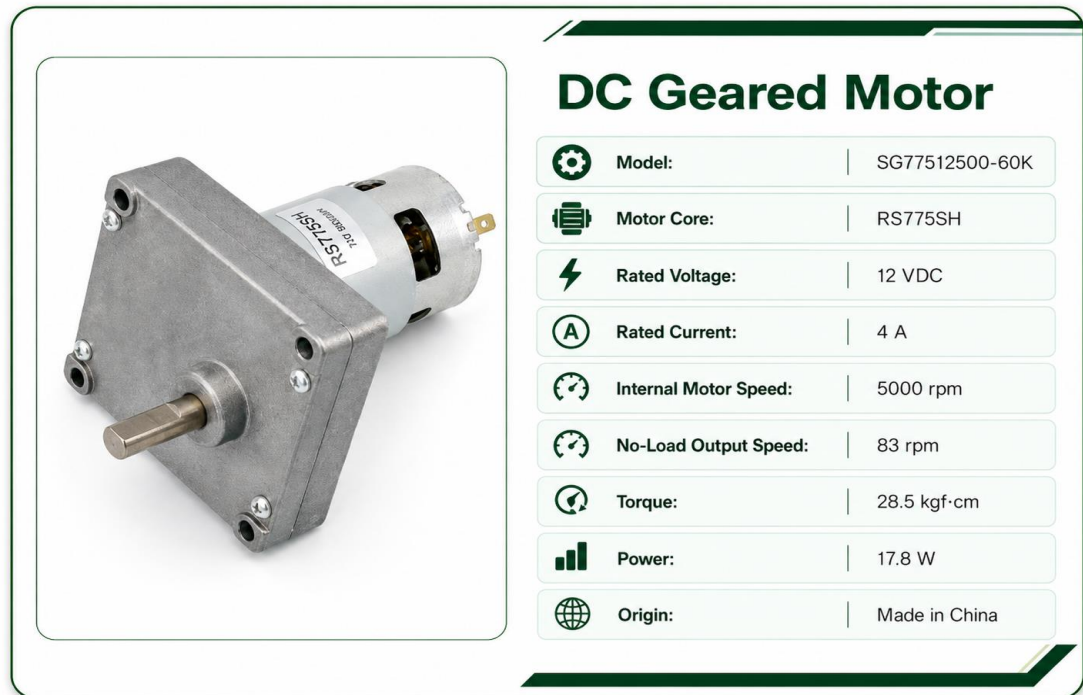


Figure 4-15: RS775SH geared DC motor used in the robot drive system. Source: Authors' component illustration.



Figure 4-16: Main drive wheels used for differential-drive movement. Source: Authors' component illustration.



Metallic Caster Wheel

Heavy-Duty Support Wheel

	Product Name	Metallic caster wheel
	Product Type	Ball transfer caster wheel
	Load Capacity	30 kg per wheel
	Structure	Metal housing with steel ball
	Motion Type	Smooth rolling support
	Rotation	Omni-directional ball movement
	Mounting Type	Bottom plate mounting
	Common Use	Robot support and balance



Common Use: Robot base support / mobility assistance / balance support

Figure 4-17: Metallic caster wheel used for base support and balance. Source: Authors' component illustration.



Figure 4-18: Bottom view of the robot showing two drive wheels and two metallic caster wheels. Source: Authors' design illustration.

The communication flow between components can be divided into four categories: sensor data flow, command flow, feedback flow, and service request flow. Sensor data flows from perception components to the high-level processing and localization layers. Command flow moves from the Jetson high-level controller to the ESP32 low-level controller and then to the motor drivers. Feedback flow moves from the encoders, IMU, and other feedback sources back toward the localization and control layers. Service request flow comes from the touch screen UI, voice interaction, or ESP32 table-side unit [16].

This communication structure allows the system to remain modular. If a customer presses the table-side Call button, the table unit sends a request intent and table ID to the smart restaurant workflow. The workflow assigns a target to Nadel. The robot then uses its navigation stack to reach the table. If the customer speaks to the robot after arrival, the voice data is processed through Whisper Small and Gemma 31B, then returned to the customer through speech output. If the customer uses the touch screen, the interaction flows through the UI and POS workflow [9], [18].

Table 4-5: Component connection and communication summary.

Component	Connected / related to	Data or signal	Purpose
RPLIDAR A1M8	Jetson / ROS2 perception layer	Laser scan data	Mapping, localization, and obstacle awareness.
MPU6050 IMU	Localization layer	Orientation/motion feedback	Supports robot pose estimation and reduces drift.
Encoders	ESP32 / odometry layer	Wheel rotation pulses	Speed estimation, wheel tracking, and odometry feedback.
Ultrasonic sensors	ESP32 / safety layer	Short-range distance readings	Near-obstacle detection and blind-spot support.
Jetson Orin Nano	ESP32 robot controller	Motion command / velocity target	High-level navigation command sent to low-level control.
ESP32 robot controller	IBT-2 motor drivers	PWM and direction control signals	Low-level motor actuation.
IBT-2 motor drivers	RS775SH motors	Motor power output	Controls left and right motor speed/direction.
Touch screen UI	Jetson / restaurant workflow	Touch commands and menu selections	Customer interaction after robot arrival.
Speaker	Speech output layer	Audio output	Plays robot voice reply.
ESP32 table-side unit	Smart restaurant workflow	Table ID + request intent only	Requests Nadel to navigate to the table.
Power switch	Power distribution layer	Normal ON/OFF control	Starts or shuts down the robot system.
Emergency stop	Safety layer / drive system	Emergency stop action	Interrupts motion or disables motor actuation during unsafe conditions.

4.3 AUTONOMOUS NAVIGATION AND PERCEPTION SYSTEM

This section presents the autonomous navigation and perception system used by Nadel to move safely inside the restaurant environment. The system combines map-based navigation, localization, path planning, obstacle awareness, and sensor feedback to guide the robot between predefined locations such as customer tables, the service area, and the home position. ROS2 Nav2, SLAM Toolbox, RPLIDAR A1M8, ultrasonic sensors, wheel encoders, and the MPU6050 IMU work together to support reliable indoor movement. The following subsections explain the mapping, localization, navigation, and obstacle detection mechanisms used in the proposed system.

4.3.1 MAPPING, LOCALIZATION, AND NAVIGATION

The navigation architecture of Nadel is designed around map-based indoor movement. The robot operates inside a restaurant-like environment where tables, kitchen area, service areas, and home position can be represented in a map. SLAM Toolbox is used to support mapping, while ROS2 Nav2 is used to plan and execute navigation paths to target locations. The target may be a customer table, a kitchen-related delivery point, or the home position [9], [12].

Localization is supported using odometry, LiDAR, and IMU information. Encoders provide wheel motion feedback, the LiDAR provides environmental scan data, and the IMU provides orientation and motion feedback. Together, these sources help the robot estimate its position and orientation inside the mapped environment. This combination is important because relying only on wheel odometry may result in drift, while relying only on environmental sensing may be affected by dynamic obstacles or changes in the environment [13].

During navigation, ROS2 Nav2 computes a path from the current robot position to the target table. The robot follows the path while continuously updating its local perception of obstacles. If a dynamic obstacle appears in front of the robot, the local planning and obstacle avoidance layer can adjust the movement behavior. After the delivery or customer interaction is completed, Nadel returns to its predefined home position [9].

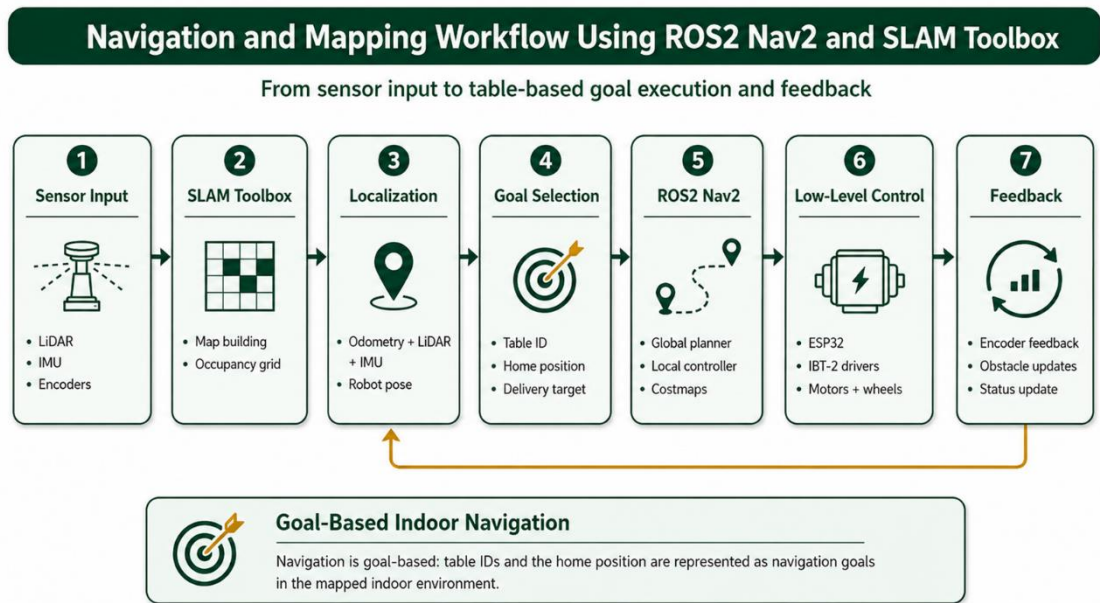


Figure 4-10: Navigation and mapping workflow using ROS2 Nav2 and SLAM Toolbox. Source: Authors' diagram.

4.3.2 PERCEPTION AND OBSTACLE DETECTION

The perception and localization layer allows Nadel to understand its surroundings and estimate its own motion. The RPLIDAR A1M8 is used as the main 2D laser scanner. It provides 360-degree scanning information that supports mapping, localization, and obstacle awareness.

Ultrasonic sensors provide short-range distance information. Their role is especially useful for near-field obstacle detection and blind-spot coverage. Encoders provide wheel rotation feedback, which is required for odometry and speed estimation. This sensor combination gives the robot multiple sources of feedback, which improves navigation reliability and supports safer movement in indoor environments.



Figure 4-11: RPLIDAR A1M8 used for 2D 360-degree scanning. Source: Authors' component illustration.

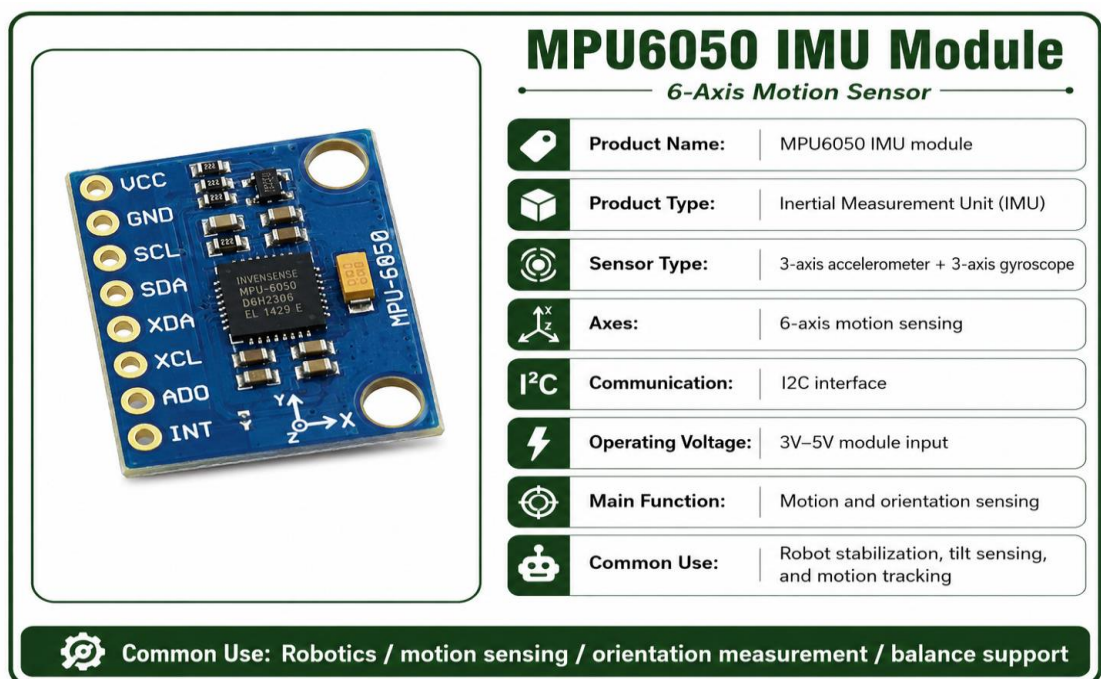


Figure 4-13: MPU6050 IMU module used for motion and orientation sensing. Source: Authors' component illustration.

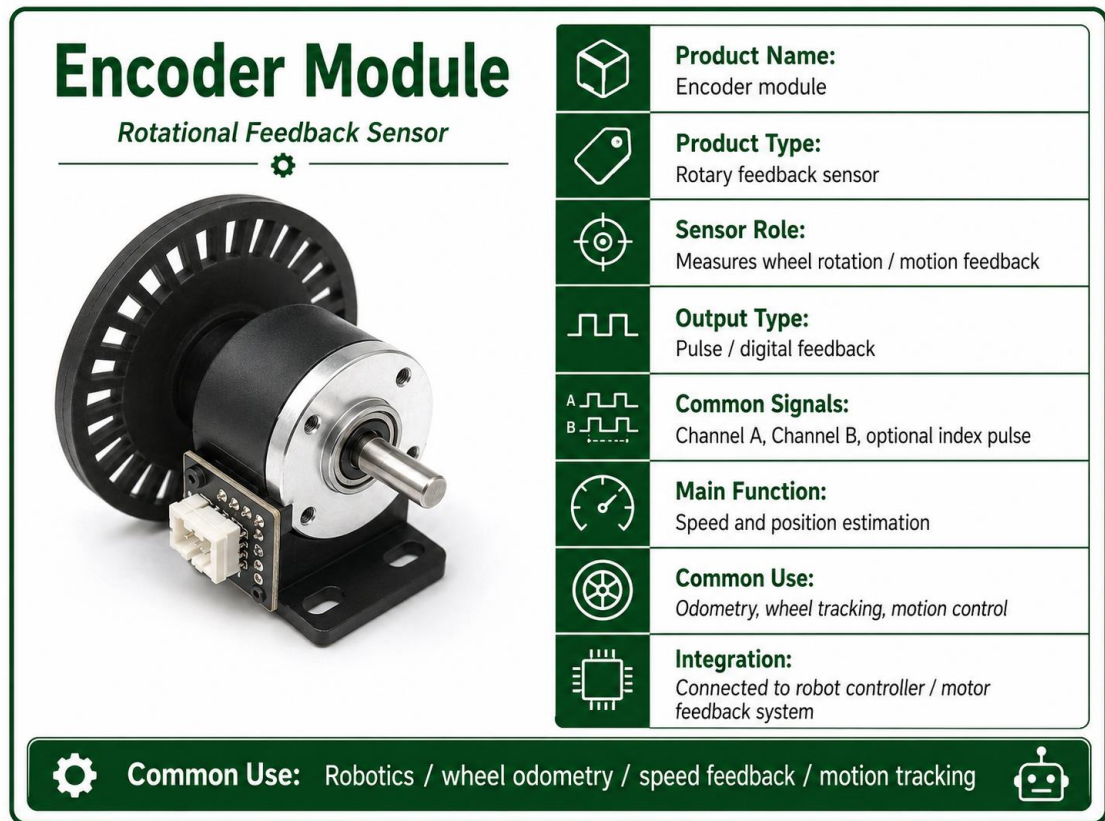


Figure 4-14: Encoder module used for rotational feedback and odometry support. Source: Authors' component illustration.

4.4 RESTAURANT SERVICE AND INTERACTION SYSTEM

This section describes the restaurant service and interaction mechanisms supported by Nadel during indoor service tasks. The system enables customers and staff to communicate with the robot through the touch-screen interface, voice interaction, and the ESP32-based table-side Call/Pay unit. These requests provide the table ID and service intent, allowing the robot to navigate to the required location and continue the interaction through its interface. The following subsections explain the table-side request module, the service workflow, and the human-robot interaction methods used in the proposed system.

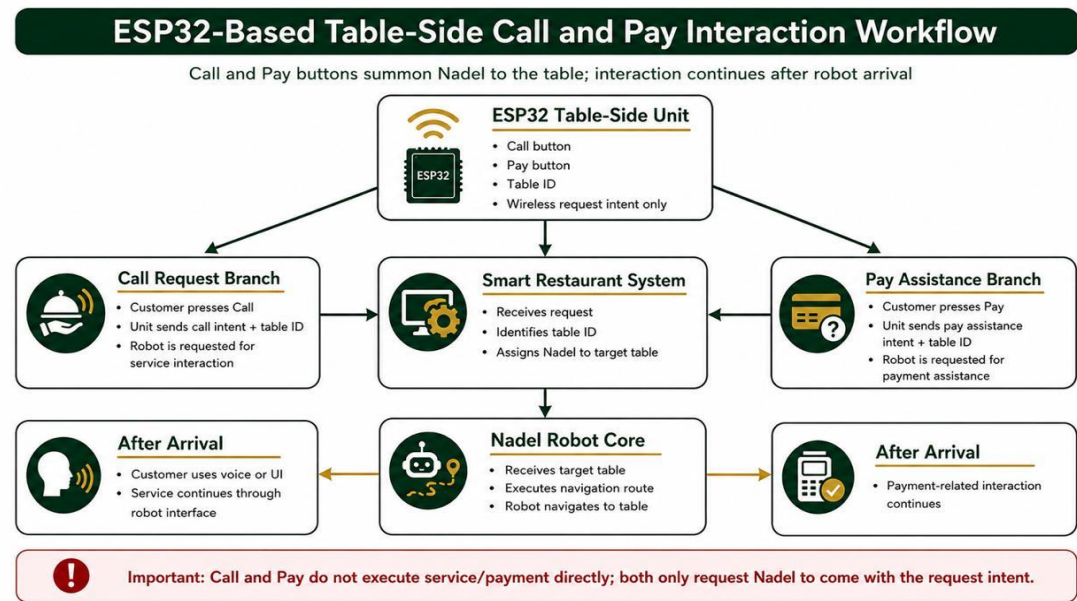
4.4.1 TABLE-SIDE CALL/PAY MODULE

The table-side ESP32 unit is an important part of the proposed smart restaurant ecosystem. It is a separate module placed at the customer table. The unit includes two physical buttons: Call and Pay. These buttons are intentionally simple so that customers

can request the robot without using a full interface or waiting for a waiter to notice them [16].

The Call button sends the table ID and call intent to the smart restaurant system. This does not perform the service directly. Instead, it informs the system that a customer at a specific table needs service interaction. The system can then assign Nadel to navigate to the requesting table. After the robot arrives, the customer can interact with the robot using voice or the touch screen UI.

The Pay button works in the same general way, but it sends pay assistance intent instead of call intent. It does not process payment, calculate the bill, or complete billing directly. It only requests Nadel to come to the table for payment-related interaction. After the robot arrives.



ESP32 Table-Side Call / Pay Unit

Restaurant Request Interaction Module

	Product Name:	Nadel table-side Call / Pay unit
	Product Type:	ESP32-based request module
	Main Function:	Summons the robot from the table
	Request Types:	Call request / Pay assistance request
	User Interface:	Two physical buttons with Call and Pay icons
	Communication:	Wireless table-to-robot / system communication
	Power / Charging:	Rechargeable unit with USB-C charging port
	Control:	Integrated power switch
	Common Use:	Restaurant table-side robot request system

Common Use: Table-side robot summon / service request / payment assistance request

Figure 4-20: ESP32 table-side Call/Pay unit used to summon Nadel from customer tables. Source: Authors' component illustration.

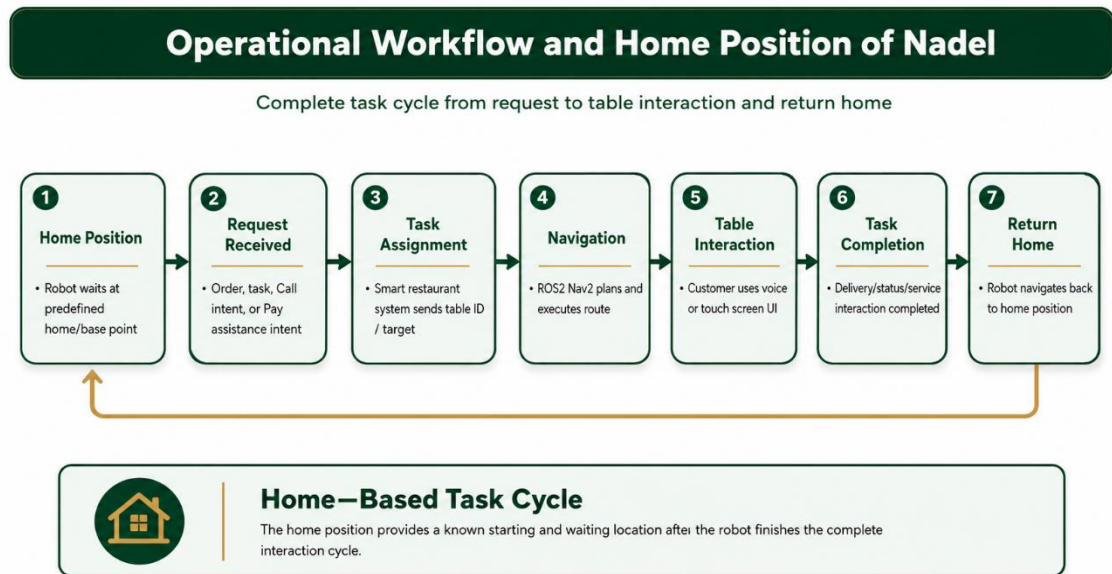


Figure 4-23: Operational workflow and home position of Nadel. Source: Authors' diagram.

4.4.2 HUMAN-ROBOT INTERACTION AND VOICE LAYER

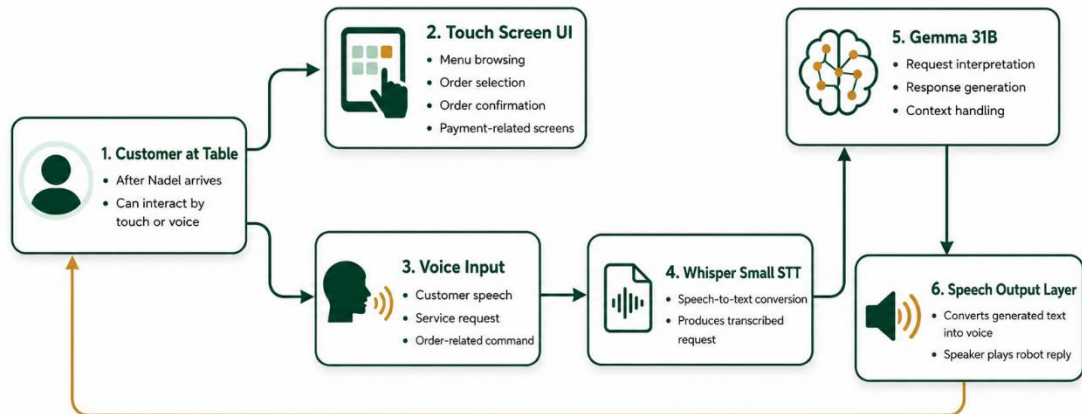
The human-robot interaction layer allows customers to interact with Nadel after it reaches the table. The system supports two main interaction modes: touch screen UI and voice interaction. The touch screen UI provides a visual interface for menus, selections, order confirmation, and service interaction. This is useful when the customer prefers visual input, when the environment is noisy, or when precise item selection is required.

The voice interaction layer provides a more natural interaction method. The customer speaks to the robot, Whisper Small converts the speech into text, and Gemma 31B interprets the request and generates a suitable response. The speech output layer then converts the generated response into spoken output, allowing the robot to reply by voice through the speaker. This makes the interaction two-way and more human-like [18], [20].

Gemma 31B does not directly generate sound. It generates a text response, and the speech output layer converts that response into audible speech. The report therefore refers to this output component generally as the speech output layer [20].

Human-Robot Interaction Layer in Nadel

Customer interaction after robot arrival can use touch screen UI or two-way voice interaction



The interaction layer supports both touch-based ordering and two-way voice communication after the robot reaches the table.

Figure 4-21: Human-robot interaction layer in Nadel. Source: Authors' diagram.



Figure 4-22: Touch screen used for customer interaction and robot interface display. Source: Authors' component illustration.

4.5 PROTOTYPE INTEGRATION, DESIGN CONSIDERATION, AND LIMITATIONS

The physical prototype of Nadel combines mechanical structure, electronics, sensing, control, power, and interaction modules. The robot body provides the physical frame for the tray structure, electronic compartment, wheels, and interface components. The upper section includes the interaction components such as the touch screen and speaker, while the lower body contains the internal control and power electronics. The lower mechanical base contains the main drive wheels and metallic caster wheels that support movement and balance.

The prototype synthesis process shows that the system was not developed as a single isolated component. Instead, it was assembled from multiple hardware and software subsystems. The Jetson handles high-level intelligence, the ESP32 handles low-level motor control, the sensors provide environmental awareness, the power system supplies and regulates energy, and the interface modules enable customer communication. Together, these modules create the final working restaurant service robot [16].

Table 4-6: Description of main Nadel system modules.

System module	Main components	Description
High-level processing	Jetson Orin Nano	Runs ROS2, Nav2, SLAM, UI, and AI interaction modules.
Embedded control	ESP32 robot controller	Executes low-level motor control and feedback handling.
Motor actuation	IBT-2 drivers + RS775SH motors	Provides left/right wheel actuation for differential drive.
Sensing	RPLIDAR, ultrasonic, encoders, IMU	Supports mapping, localization, obstacle awareness, and odometry.
Power and safety	Battery, regulators, switch, emergency stop	Supplies power and provides manual safety control.
Customer interaction	Touch screen, microphone, speaker, speech output layer	Allows customer interaction through UI and two-way voice communication.

Although the Nadel prototype is implemented and operational, several design issues must be considered. Indoor navigation can be affected by dynamic obstacles such as customers, chairs, waiters, and changing table arrangements. Odometry drift can also occur because wheel-based feedback may accumulate error over time. For this reason, odometry is supported by LiDAR and IMU feedback [13].

Power distribution is another important issue. The robot includes motors, motor drivers, embedded controllers, sensors, screen, speaker, and processing hardware. These components have different voltage and current requirements. Voltage regulation and careful power distribution are therefore essential for stable operation. The emergency stop and switch button also need to be treated as safety-critical elements rather than simple accessories.

Human-robot interaction also has practical limitations. Voice recognition can be affected by noise in the restaurant, distance between the customer and the microphone, and unclear speech. The touch screen UI provides a useful alternative interaction method in such cases. The table-side Call/Pay unit adds convenience, but wireless communication and table ID accuracy must be reliable so that the robot goes to the correct table.

Table 4-7: Design issues and proposed mitigation strategies.

Design issue	Possible effect	Mitigation in Nadel
Odometry drift	Wheel feedback may accumulate error during movement.	Combine odometry with LiDAR and IMU localization.
Dynamic obstacles	Customers and furniture may change the environment.	Use LiDAR, ultrasonic sensors, and local planning.
Power stability	Different modules require different voltage/current levels.	Use battery distribution and voltage regulators.
Emergency safety	Unexpected robot motion may create risk.	Use emergency stop as manual safety override.
Voice noise	Restaurant noise may reduce STT accuracy.	Support touch screen UI as alternative input method.
Wireless table requests	Table-side requests may be delayed or duplicated.	Send table ID and request intent clearly and verify request handling.
Mechanical balance	Loaded trays and movement may affect stability.	Use main drive wheels and metallic caster wheels for support.
Home return reliability	Robot must reach home after completing task.	Represent home position as a known navigation goal in the map.

This chapter presented the proposed model of Nadel. The chapter explained the overall system architecture, hardware architecture, power and safety architecture, software architecture, navigation and mapping workflow, perception and localization layer, embedded motor control layer, table-side Call/Pay interaction module, human-robot interaction layer, communication flow, home position behavior, and design limitations [12].

The proposed model shows that Nadel is not only a mobile robot but an integrated restaurant service ecosystem. The Jetson Orin Nano runs the high-level intelligence, ROS2 navigation, AI interaction, and user interface. The ESP32 robot controller handles low-level motion control through the IBT-2 motor drivers and RS775SH motors. The sensing layer supports mapping, localization, obstacle detection, and motion feedback. The table-side ESP32 Call/Pay unit allows customers to summon the robot using table ID and request intent only. After the robot arrives, customers can continue the interaction through voice or the touch screen UI [9], [12].

The chapter also clarified the roles of the power switch, emergency stop, voltage regulators, battery pack, main drive wheels, metallic caster wheels, and home position. These elements are essential for building a practical and safe service robot prototype that can operate reliably in an indoor restaurant environment.

5 PROJECT SIMULATION AND PERFORMANCE EVALUATION

This chapter presents the evaluation and testing of Nadel after the proposed model was implemented as a functional service waiter robot. The purpose of the evaluation is to verify that the main system modules can work together in a realistic service sequence. The testing focuses on robot navigation, table-side request handling, customer interaction, order collection, delivery behavior, return-to-home behavior, and public demonstration performance.

The evaluation is based on practical functional testing rather than laboratory-grade statistical benchmarking. This is suitable for the current stage of the project because Nadel is a graduation prototype that was built, integrated, and demonstrated as a working system. The robot was tested internally in indoor environments and was also demonstrated publicly during ICT and TransMEA participation under the Arab Academy representation. These demonstrations provided practical evidence that the robot could operate in front of real observers, handle a service scenario, and present the concept in a dynamic public environment.

The main tested scenario started from the home position. A customer pressed the table-side request button, and Nadel moved from its home location to the customer table. After arrival, the customer interacted with the robot using the touch screen interface and/or the voice interaction layer. The robot then moved to the kitchen or service area, collected the requested item after it was placed on the tray, returned to the customer table, delivered the item, and finally returned to its home position after the full interaction was completed.

The chapter reports successful functional behavior while also discussing limitations honestly. The purpose is not to claim that the system is free from errors or ready for commercial deployment without further work. Instead, the chapter shows that the major prototype functions operated successfully under practical conditions, while identifying issues that require additional testing, refinement, and long-term validation.

5.1 TESTING FRAMEWORK AND METHODOLOGY

This section defines the objectives and environment of the testing process used to evaluate the Nadel prototype. The testing focuses on verifying the main functions of the robot, including autonomous navigation, table-side Call/Pay request handling, touch-screen and voice interaction, obstacle awareness, service-area movement, delivery tasks, and return-to-home behavior. The evaluation is conducted in an indoor restaurant-like environment with predefined tables, service locations, navigation paths, and common obstacles. The following subsections describe the testing goals, setup conditions, and criteria used to assess the reliability of the proposed system.

5.1.1 TESTING OBJECTIVES AND ENVIRONMENT

The testing objectives were defined according to the main functions of the Nadel system. Since the robot combines mechanical motion, embedded control, ROS2 navigation, table-side request handling, touch interaction, voice interaction, and restaurant workflow logic, the evaluation had to cover both individual modules and the complete integrated scenario [9].

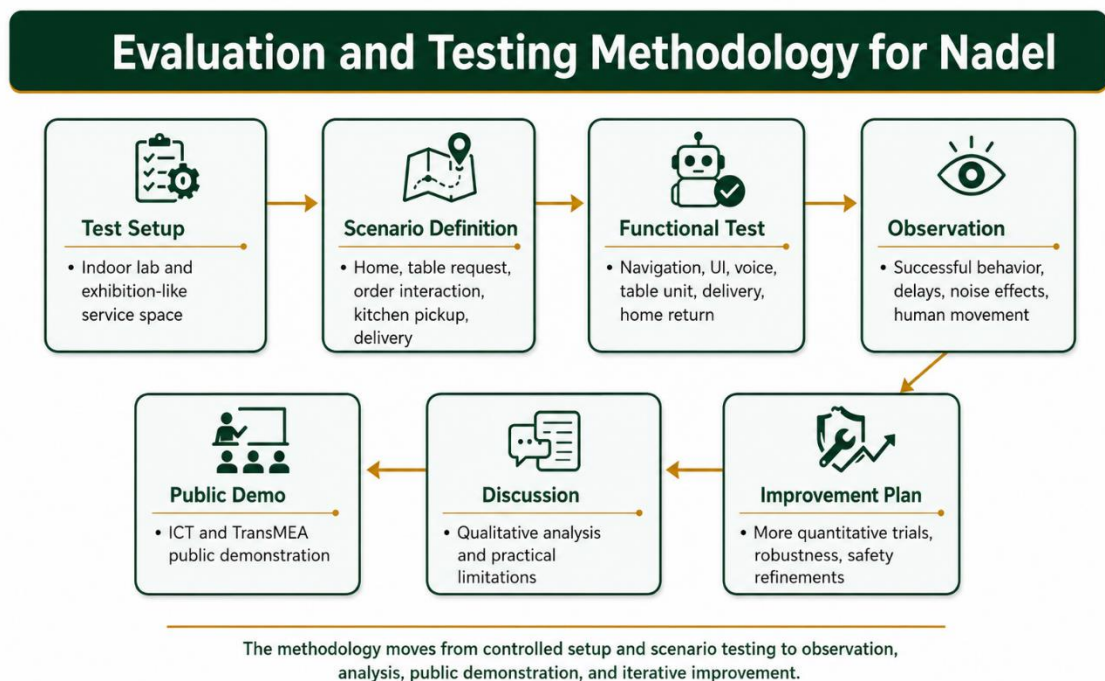


Figure 5-1: Evaluation and testing methodology used for Nadel. Source: Authors' diagram.

Table 5-1: Main testing objectives of Nadel.

Objective	Description	Related System Module
Verify autonomous navigation	Confirm that the robot can move from the home position to a target table and service area.	ROS2 Nav2, SLAM Toolbox, localization layer
Verify table-side request behavior	Confirm that the Call/Pay unit can request the robot using table ID and request intent.	ESP32 table-side unit
Verify customer interaction	Confirm that customers can interact with the robot after arrival using the touch screen UI and voice interaction.	Touch screen UI, Whisper Small, Gemma 31B, speech output layer
Verify delivery workflow	Confirm that the robot can move to the service area, receive the item on the tray, and return to the customer table.	Navigation, tray structure, operational workflow
Verify return-to-home behavior	Confirm that the robot can return to a predefined home position after the full interaction.	Robot task sequence and navigation goals
Evaluate public demonstration readiness	Observe the robot behavior in front of visitors and representatives during exhibition participation.	Complete integrated system

The testing environment was indoor and similar to the expected deployment environment of a restaurant service robot. The floor was smooth and suitable for wheeled motion. Tables, chairs, cabinets, people, and temporary objects were present in the operating space. These conditions are important because a restaurant robot is not expected to move in a completely empty environment; it must operate around furniture, customers, staff, and changing obstacles.

In addition to the internal test area, Nadel was presented in public demonstration environments. The ICT and TransMEA participation exposed the robot to a more dynamic context, including a public audience, noise, human movement, and exhibition layout constraints. This helped evaluate the practical presentation of the system, although it was not a controlled scientific benchmark.

Table 5-2: Testing environments used during Nadel evaluation.

Environment	Purpose	Observed Conditions	Evaluation Value
Indoor laboratory / service-like area	Functional testing of navigation, table request, UI, and delivery sequence.	Smooth floor, furniture, tables, human presence, controlled supervision.	Allowed repeated testing of the complete service workflow.
ICT public demonstration	Demonstrate the robot concept in a public setting.	Audience presence, movement around the robot, higher noise, exhibition layout.	Showed that Nadel could be presented and operated outside the development area.
TransMEA public demonstration	Present the project under Arab Academy representation.	Public observers, representatives, exhibition booth, dynamic surroundings.	Provided practical validation and exposure to real feedback.

5.1.2 TESTING METHODOLOGY AND EVALUATION CRITERIA

The evaluation used scenario-based functional testing. Each test case was built around an expected action, such as pressing the Call button, moving to the customer table, receiving an order, going to the service area, delivering the item, or returning to the home position. The observed behavior was compared with the expected behavior, and the result was recorded as successful, partially successful, or requiring improvement.

Since the goal of this stage was to validate a working graduation prototype, the evaluation relied mainly on qualitative observation supported by photographs, screenshots, and public demonstrations. Exact timing values, large-scale repeated trials, and statistical success rates were not formally recorded during all demonstrations. Therefore, the results are reported as functional validation rather than precise quantitative benchmarking.

The complete service sequence was treated as the most important evaluation scenario because it combines all major modules. A successful end-to-end test required the table-side unit, robot navigation, user interface, voice interaction layer, service area movement, delivery behavior, and home return behavior to work together in one connected workflow.

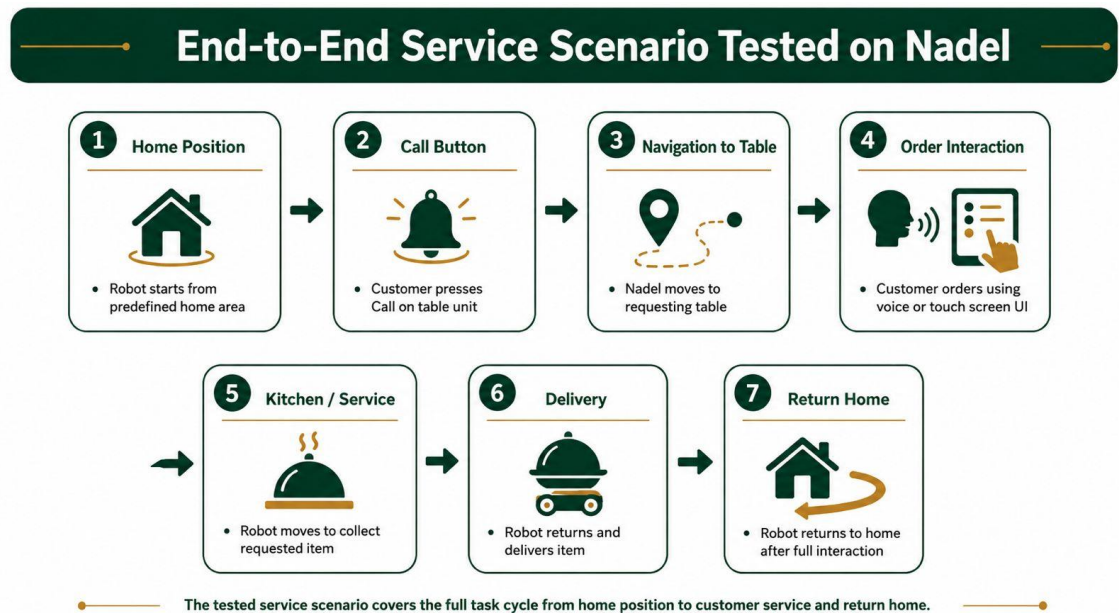


Figure 5-2: End-to-end service scenario tested on Nadel. Source: Authors' diagram.

Table 5-3: Main functional scenarios evaluated in Chapter Five.

Scenario	Expected Behavior	Evaluation Method	Status
Call request scenario	The robot leaves home and navigates to the requesting table.	Press table-side Call button and observe movement.	Successful
Customer interaction scenario	The customer interacts with the robot after arrival.	Use touch screen UI and voice interaction.	Successful
Order-taking scenario	The robot receives the customer request through UI and/or voice.	Observe order interaction after the robot reaches the table.	Successful
Service area scenario	The robot moves to the kitchen/service area after the order.	Observe movement from table to service area.	Successful
Delivery scenario	The robot carries the item and returns to the customer table.	Place item on tray and observe delivery movement.	Successful
Home return scenario	The robot returns to the predefined home position after completion.	Observe final navigation after interaction ends.	Successful
Public demonstration scenario	The robot operates and presents the concept in public event conditions.	Observe behavior during ICT and TransMEA participation.	Successful with practical limitations

5.2 NAVIGATION, LOCALIZATION AND SAFETY

This section evaluates the navigation, localization, and safety performance of the Nadel robot during indoor service tasks. The evaluation focuses on the robot's ability to follow planned routes, estimate its position on the map, avoid static and nearby obstacles, reach predefined tables and service locations, and return safely to its home position. The tests rely on ROS2 Nav2, SLAM-based mapping, RPLIDAR A1M8, ultrasonic sensors, wheel encoders, and IMU feedback to support safe and reliable movement.

5.2.1 NAVIGATION AND MAPPING TESTING

Navigation and mapping testing was conducted to verify that Nadel could move within an indoor environment using its ROS2-based navigation system. The robot uses ROS2 Humble, ROS2 Nav2, and SLAM Toolbox with localization support from odometry, LiDAR, and IMU data. The RPLIDAR A1M8 provides 360-degree scanning, while the IMU and wheel-related feedback support motion estimation and localization stability [9], [12].

The navigation test focused on the robot ability to move from the home position to a table location, then to the service area, and back to the customer table. The robot was able to complete the required movement during functional testing. This indicates that the navigation architecture was sufficiently integrated to support the service workflow. However, navigation performance was evaluated qualitatively, and a large numerical dataset of repeated travel time, positioning error, and success rate was not recorded at this stage.

Table 5-4: Navigation and mapping test summary.

Test Item	Expected Result	Observed Result	Status
SLAM and map support	Robot uses a map of the indoor environment to support navigation.	The system operated with ROS2 Nav2 and SLAM Toolbox in the implemented prototype.	Successful
Home-to-table navigation	Robot moves from home position to the customer table.	Robot moved from the home position toward the requesting customer table.	Successful
Table-to-service-area navigation	Robot moves from table to kitchen/service area after receiving the request.	Robot moved toward the service area to collect the requested item.	Successful
Service-area-to-table navigation	Robot returns to the same customer table to deliver the item.	Robot returned to the customer table after item placement.	Successful

5.2.2 OBSTACLE AWARENESS AND SAFETY TESTING

Obstacle awareness and operational safety are essential for indoor service robots. Nadel uses multiple sensing sources, including LiDAR, ultrasonic sensors, encoders, and IMU feedback. These sensors support navigation awareness, obstacle detection, localization, and motion estimation. During practical testing, the robot operated in environments that included furniture, tables, humans, and public audience movement [13], [14].

The safety layer also includes a power switch and emergency stop. The power switch is used for normal system activation and shutdown, while the emergency stop provides a manual safety override. In case of unexpected motion or unsafe behavior, the emergency stop can be used to stop robot motion at the drive level. This is especially important when demonstrating the robot in front of people.

Table 5-8: Safety and obstacle awareness testing.

Safety / Sensing Element	Purpose	Testing Observation	Result
LiDAR-based awareness	Supports mapping, localization, and obstacle awareness.	Used as a main navigation sensing source in the implemented system.	Successful
Emergency stop	Provides manual safety override during unsafe behavior.	Included as a physical safety control on the robot.	Successful
Power switch	Controls normal robot power activation.	Used for normal startup and shutdown of the system.	Successful
Public environment operation	Test practical operation around people and observers.	Demonstrated in exhibition conditions with supervision.	Successful with caution

5.2.3 HOME POSITION AND TASK COMPLETION TESTING

A home position was used as the starting and ending point of the service scenario. The home position represents the robot base location where it waits before receiving a new task. In the evaluated workflow, Nadel starts from the home position, responds to a customer request, completes the table interaction, moves to the service area, delivers the item, and returns home after the full interaction is completed.

The return-to-home behavior is important for restaurant operation because it prevents the robot from remaining in random locations after finishing a task. It also makes the next task easier to manage because the robot begins from a known location. During testing, Nadel was observed returning to its home location after completing the full service sequence.

Table 5-9: Home position and task completion evaluation.

Item	Expected Behavior	Observed Behavior	Status
Home start	Robot begins the task from a known home location.	The robot started from the home position before moving to the customer.	Successful
Home-to-table request	Robot moves from home to the requesting table.	The robot moved to the table after the request button was pressed.	Successful
Task completion	Robot completes order interaction, service area movement, and delivery.	The complete service scenario was demonstrated.	Successful
Return home	Robot returns to home after the full interaction.	The robot returned to the home location after completion.	Successful
Operational readiness	Robot becomes ready for another task after returning home.	The home position supports repeated service operation.	Successful concept validation

5.3 HUMAN-ROBOT INTERACTION AND RESTAURANT WORKFLOW

This section evaluates the human-robot interaction and restaurant service workflow supported by Nadel. The evaluation focuses on how customers and staff communicate with the robot through the touch-screen interface, voice interaction, and the ESP32-based table-side Call/Pay unit. It also examines the robot's ability to receive service requests, navigate to the requesting table, support customer interaction after arrival, move between the service area and customer tables, and complete delivery-related tasks. The following subsections present the tests for table-side requests, touch-screen and voice interaction, manual service-area loading, and delivery workflow execution.

5.3.1 TABLE-SIDE CALL / PAY REQUEST TESTING

The table-side unit was evaluated as the starting point of the customer service scenario. The unit is based on an ESP32 and includes Call and Pay buttons. In the tested workflow, the buttons do not directly perform service or payment operations. Instead, they send the table ID and request intent to the system so that Nadel can be assigned to the requesting table [16].

During the functional test, the Call button was pressed by the customer. The system received the request, and the robot moved from its home position toward the requesting table. This confirmed that the table-side unit could act as a practical trigger for robot attendance. The Pay button follows the same concept, but its request intent indicates that the customer needs payment-related assistance. In both cases, the actual interaction continues after the robot reaches the table, using the touch screen UI, voice interaction layer.

Table 5-5: Table-side Call/Pay request testing.

Test Case	Expected Result	Observed Result	Status
Press Call button	System receives call intent and table ID, then sends Nadel to the table.	The robot was requested from the table and moved from home to the customer side.	Successful
Press Pay button	System receives pay assistance intent and table ID, then sends Nadel to the table.	The Pay function is treated as a robot request for payment-related interaction after arrival.	Successful concept validation
Request intent handling	System distinguishes between service request and payment assistance request.	The request type is used to define the intended interaction after the robot arrives.	Successful
Table ID concept	The request is linked to the requesting table location.	The robot navigated to the customer table as part of the scenario.	Successful
Safety of interpretation	Buttons do not directly execute payment or service actions.	The system uses the buttons only to summon the robot and start later interaction.	Successful

5.3.2 TOUCH SCREEN UI AND VOICE INTERACTION TESTING

After Nadel reached the customer table, the customer could interact with the robot using two methods: the touch screen interface and the voice interaction layer. The touch screen interface allows the customer to visually select items, review options, and continue the order process. The voice interaction layer allows the customer to speak to the robot, where Whisper Small converts speech into text, Gemma 31B interprets the request and generates the response, and the speech output layer allows the robot to reply by voice [18], [20].

Testing confirmed that the interaction layer is suitable for the expected service flow. The customer can place or continue an order after the robot arrives at the table. This is important because the table-side Call/Pay unit only summons the robot; it does

not replace the ordering or payment interaction itself. The actual interaction happens through the robot interface after arrival.

Table 5-6: Customer interaction testing results.

Interaction Method	Expected Behavior	Observed Behavior	Status
Touch screen UI	Customer can select or confirm order-related options visually.	The customer interacted with the robot through the screen interface during testing.	Successful
Voice input	Customer can speak to the robot after arrival.	Voice interaction was part of the implemented two-way HRI layer.	Successful
Gemma 31B processing	The system interprets the spoken or text request and generates a response.	Gemma 31B was used for request understanding and response generation.	Successful
Speech output layer	The generated response is converted into a spoken reply.	The robot was able to reply by voice through the speech output layer.	Successful
Multimodal interaction	Customer can interact through UI or voice depending on the situation.	Both interaction paths were considered valid in the implemented workflow.	Successful

5.3.3 ORDER COLLECTION AND DELIVERY WORKFLOW TESTING

The delivery workflow test evaluated the complete service behavior after the robot reached the customer table. Once the customer interaction was completed, Nadel moved toward the kitchen or service area. The requested item was then placed on the robot tray. After receiving the item, Nadel returned to the same customer table and delivered it. This scenario demonstrates the main service purpose of the project: reducing repeated human movement between the customer area and kitchen/service area while maintaining a controlled restaurant workflow.

The photos taken during the testing show the robot operating near tables and carrying items on its shelves. These observations support the functional validation of the robot body, tray structure, movement system, and table-to-service-area navigation sequence. The test also demonstrated that the robot can be used as a service carrier rather t

Table 5-7: End-to-end order collection and delivery test summary.

Stage	Expected Action	Observed Result	Status
Home position	Robot waits at a predefined home area before receiving a task.	Nadel started from its home location before the customer request.	Successful
Move to customer table	Robot reaches the requesting customer table after Call/Pay intent.	Nadel moved to the customer table after the request.	Successful
Receive order	Customer interacts with Nadel through UI or voice.	Order interaction was completed after arrival.	Successful
Move to service area	Robot goes to the kitchen/service area to collect the item.	Nadel moved toward the service area after receiving the request.	Successful
Load item	Requested item is placed on the robot tray.	The item was placed on the robot tray/shelf during testing.	Successful
Deliver item	Robot returns to customer table and delivers the item.	Nadel returned to the table and completed delivery.	Successful
Return home	Robot returns to home after the full interaction ends.	Nadel returned to its home position after completing the scenario.	Successful

5.4 SYSTEM INTEGRATION AND DEMONSTRATION

This section evaluates the overall integration of the Nadel prototype and its performance during complete service demonstrations. It examines how the main hardware, software, navigation, sensing, interaction, and table-side request components operate together as one system. The evaluation includes end-to-end scenarios in which Nadel receives a service request, navigates safely between predefined locations, supports customer interaction through the touch screen or voice interface, performs delivery-related movement, and returns to its home position. The following subsections present the overall system integration testing and the public demonstration evaluation of the proposed prototype.

5.4.1 OVERALL SYSTEM INTEGRATION TESTING

The most important result of the evaluation was the successful integration of multiple system layers. Nadel is not only a moving robot. It combines a table-side request module, smart restaurant workflow, autonomous navigation, embedded motor control, touch screen interface, voice interaction, item carrying, and task completion logic. The complete test scenario confirmed that these elements can operate together.

The end-to-end service scenario provides the strongest validation of the proposed model. The robot responded to a customer table request, moved from the home position, allowed the customer to interact through the robot interface, moved to the service area, received the item, returned to the customer table, delivered the item, and returned home. This confirms that the project achieved.

Table 5-11: Overall integration testing matrix.

Integrated Layer	Tested Function	Integration Result
Table-side interaction	Call/Pay request with table ID and request intent.	Integrated with robot assignment and service workflow.
Robot navigation	Home, table, service area, delivery, and return-home navigation.	Integrated with the task sequence.
Embedded control	ESP32, motor drivers, motors, encoders, and motion execution.	Supported robot movement during tests.
Customer UI	Touch screen interaction after robot arrival.	Supported order and interaction workflow.
Voice interaction	Whisper Small, Gemma 31B, and speech output layer.	Supported two-way voice interaction concept.
Service workflow	Customer request, service area pickup, delivery, and home return.	Successfully demonstrated as one connected scenario.
Public demonstration	ICT and TransMEA event demonstration.	Confirmed practical presentation of the integrated prototype.

5.4.2 PUBLIC DEMONSTRATION EVALUATION

In addition to internal testing, Nadel was demonstrated in public exhibition environments, including ICT and TransMEA participation under the Arab Academy representation. These public demonstrations are important because they exposed the robot to non-laboratory conditions. The robot was presented in front of visitors, representatives, and public audience members, and the team explained the system concept and its service workflow.

Public demonstration does not replace controlled experimental benchmarking. However, it provides valuable practical validation. It shows that the robot can operate as a visible, interactive prototype in a realistic public setting. It also demonstrates that the project can be presented as an applied service robot concept rather than only as a laboratory model.



Participation in ICT and TransMEA Exhibitions

The Arab Academy for Science, Technology and Maritime Transport (AASTMT) participated in the ICT and TransMEA exhibitions held at the Egypt International Exhibition Center in Cairo, presenting a range of innovative projects that reflect the Academy's focus on artificial intelligence, robotics, and smart technologies. Among the displayed projects, the Nadel autonomous service robot was selected as one of the main projects representing the College of Artificial Intelligence at the Academy's booth.



During the exhibition, the project engineer presented Nadel to distinguished visitors and senior representatives who showed considerable interest in the project. The discussion focused on the robot's main concept, its hardware design, and its potential role as an autonomous service robot in smart restaurant environments. The visitors asked about how Nadel receives customer requests, navigates inside the restaurant, and supports service operations through its interactive interface. This direct presentation provided an important opportunity to explain the practical value of the project and demonstrate how artificial intelligence, robotics, and embedded systems were integrated into a functional physical prototype.



The project also received media attention, as several television channels conducted interviews and filmed the Nadel prototype during the exhibition.

Table 5-10: Public demonstration evaluation summary.

Evaluation Aspect	Observed Evidence	Interpretation
Public presentation	Nadel was presented in exhibition environments in front of visitors, representatives, media coverage, and senior AAST representatives.	The prototype was sufficiently complete to demonstrate its concept publicly.
System visibility	The robot body, interface, trays, and service concept were visible to observers.	The system communicated its service role clearly.
Dynamic surroundings	The exhibition environment included people, movement, noise, and limited space.	The demonstration provided practical non-laboratory exposure.
Team explanation	The team explained the robot functionality and service workflow.	The project could be presented as an applied intelligent system.
Practical limitations	The public test was not a controlled statistical benchmark.	Further repeated quantitative evaluation is still required.

5.5 RESULTS, DISCUSSION AND OBSERVED LIMITATIONS

The evaluation shows that Nadel achieved the main functional requirements of the project. The robot was able to support a realistic restaurant service sequence starting from a customer request and ending with delivery completion and return to home. The Call/Pay unit was useful as a simple interaction trigger because the customer did not need to use a complete interface before requesting the robot. After arrival, the customer could interact with Nadel using voice or the touch screen UI.

The navigation and delivery results demonstrate that the mechanical structure, differential drive base, motor control layer, and ROS2 navigation stack were integrated sufficiently for prototype operation. The tray structure allowed items to be carried, and the robot could move between service points. The human-robot interaction layer also increased the practical value of the system because it made the robot more than a delivery platform; it became an interactive service unit [9].

At the same time, the results should be interpreted realistically. The evaluation confirmed functional success, but it did not include a large number of repeated quantitative trials. Exact timing, positioning error, battery endurance curves, and long-term reliability measurements were not formally recorded for all tests. Therefore, the results are best understood as successful prototype validation rather than final commercial certification.

Table 5-12: Summary of testing results and interpretation.

Evaluation Area	Result	Interpretation
Navigation	Successful functional movement between home, table, and service area.	The navigation system was suitable for the tested indoor scenario.
Table request	Successful robot request using table-side unit.	The table unit can act as a practical request trigger.
Touch interaction	Successful customer interaction through the screen.	The touch UI is suitable for table-side ordering and confirmation.
Voice interaction	Successful two-way interaction concept using STT, LLM processing, and speech output.	Voice adds natural interaction but needs more testing under noise.
Delivery	Successful item carrying and delivery workflow.	The robot can support the main service use case.
Home return	Successful return to predefined home position.	The robot can reset its operational location after completion.
Public demonstration	Successful public presentation at ICT and TransMEA.	The prototype was sufficiently mature for demonstration, but formal benchmarking remains future work.

Although the prototype operated successfully, several limitations were identified or remain expected for future testing. These limitations do not invalidate the project; instead, they define the next engineering steps required to improve reliability, robustness, and readiness for real restaurant deployment.

Table 5-13: Testing limitations and recommended improvements.

Limitation	Impact	Recommended Improvement
Limited quantitative measurements	The evaluation relies mainly on qualitative observation and public demonstration evidence.	Record repeated trials, arrival time, delivery time, path error, and success rate.
Voice noise sensitivity	Public environments may include background speech, music, and exhibition noise.	Test voice recognition under controlled noise levels and improve microphone placement.
Dynamic human movement	Crowded environments can affect navigation and safety.	Perform more obstacle avoidance trials with moving people and furniture.
Wireless request reliability	Table-side requests depend on wireless communication.	Test multiple tables, different distances, network delays, and repeated requests.
Odometry drift over time	Wheel-based odometry may accumulate errors during long operation.	Improve sensor fusion and repeated localization correction using LiDAR/IMU.
Battery endurance not fully benchmarked	Long service duration was not measured systematically.	Record operating time under repeated navigation and delivery cycles.
Prototype body refinement	Prototype dimensions and wiring may require refinement for commercial use.	Improve internal cable routing, enclosure accessibility, and maintenance design.

The main future testing direction is to convert the current qualitative validation into a more formal quantitative evaluation. Future experiments should measure arrival accuracy, average response time after pressing the table button, delivery time, obstacle avoidance success rate, voice recognition success rate in noisy environments.

This chapter presented the evaluation and testing of Nadel. The testing confirmed that the robot could complete the main service scenario: starting from home, responding to a table-side request, navigating to the customer table, supporting customer interaction through voice or touch screen UI, moving to the kitchen/service

area, receiving the requested item, delivering it to the customer, and returning to home after completion.

The system was also demonstrated publicly during ICT and TransMEA participation, which provided practical exposure to non-laboratory conditions. These demonstrations showed that Nadel could operate as a complete service robot prototype in front of real observers. At the same time, the chapter discussed limitations honestly, including the need for more quantitative testing, stronger noise evaluation, wireless reliability testing, battery endurance measurement, and wider dynamic obstacle testing.

Overall, the evaluation supports the conclusion that Nadel achieved the main functional goals of the graduation project. It operated as an integrated smart restaurant service robot combining autonomous navigation, table-side request handling, human-robot interaction, embedded control, delivery workflow, and public demonstration readiness.

6 BUSINESS MODEL

6.1 BUSINESS MODEL CANVAS

A business model describes how a proposed product creates value, delivers value to customers, and captures value in a sustainable way. For Nadel, the business model is important because the project is not only a functional graduation prototype; it is also a potential smart restaurant service product that can be developed into a commercial solution. The Business Model Canvas is used in this chapter to organize the business logic of the project into customer segments, value propositions, channels, customer relationships, revenue streams, key resources, key activities, key partners, and cost structure [68].

The business model of Nadel is based on a simple idea: many restaurants and hospitality environments need support for repetitive service tasks, especially food delivery, table service requests, and customer interaction during busy periods. Nadel responds to this need by combining autonomous indoor navigation, table-side request buttons, touch screen interaction, voice interaction in one service robot system. The robot is designed to support staff rather than replace them completely. This makes the proposed business direction more realistic, because restaurants can use Nadel as an assistant that improves workflow efficiency and customer experience [1].

Business Model Canvas for Nadel



Figure 6-1: Business Model Canvas for Nadel robot.

Figure 6-1 summarizes the business model of Nadel. The canvas shows that the project can serve restaurants, cafes, hotels, food courts, smart restaurants, university cafeterias, and other indoor hospitality environments. The main value of Nadel comes from autonomous delivery support, direct table-side Call/Pay request, touch and voice interaction, local customization, and lower prototype cost compared with many commercial waiter robots. The cost side is mainly related to hardware, body manufacturing, software development, testing, installation, and maintenance, while the revenue side can be based on direct sales, leasing, Robot-as-a-Service, maintenance contracts, software subscriptions, and customization fees [1].

Table 6-1: Summary of the Business Model Canvas for Nadel.

Canvas Element	Nadel Business Interpretation	Business Importance
Customer Segments	Restaurants, cafes, hotels, food courts, university cafeterias, smart restaurants, and hospitality businesses inside and outside Egypt.	Defines the main buyers and users of the robot.
Value Proposition	Autonomous delivery, table-side robot request, voice and UI interaction, POS support, local customization, and cost-efficient prototype development.	Explains why customers may choose Nadel.
Channels	Direct demonstrations, ICT and TransMEA exhibitions, academic innovation events, social media, restaurant visits, and partnerships with POS providers.	Explains how Nadel can reach target customers.
Customer Relationships	Installation, staff training, maintenance, software updates, customization, and technical support.	Builds trust and supports long-term use.
Revenue Streams	Robot sale, monthly leasing, RaaS, software subscription, maintenance contracts, spare parts, and customization fees.	Explains how the project can generate income.
Key Resources	Robot hardware, ROS2 software stack, Jetson Orin Nano, ESP32 modules, sensors, POS workflow, engineering team, and restaurant testing data.	Identifies the assets needed to operate and scale.
Key Activities	Manufacturing, integration, mapping, deployment, testing, maintenance, training, and	Defines the activities required to deliver the product.

Canvas Element	Nadel Business Interpretation	Business Importance
	continuous software improvement.	
Key Partners	Hardware suppliers, manufacturing workshops, restaurants, POS providers, maintenance partners, academic incubators, and investors.	Supports production, deployment, and market expansion.
Cost Structure	Sensors, motors, body manufacturing, Jetson, screen, power system, software development, testing, support, and marketing.	Shows the main cost drivers of the business.

This section explains the business model components in more detail and connects them to the actual Nadel prototype, testing results, public demonstrations, cost structure, and future commercial potential. The analysis is written as a realistic business view: Nadel has strong potential as a service robot product, but commercialization still requires manufacturing refinement, longer reliability testing, safety validation, after-sales support planning, and stronger integration with real restaurant operations.

6.2 MARKET OPPORTUNITY AND VALUE PROPOSITION

This section examines the market opportunity for Nadel and the value it offers to restaurants, cafés, hotels, food courts, and similar hospitality environments. Many service businesses face pressure to improve delivery speed, reduce repetitive staff workload, and maintain a more consistent customer experience during busy periods. Nadel provides a practical autonomous service-support solution through indoor navigation, table-side Call/Pay requests, touch-screen and voice interaction, and delivery assistance between service areas and customer tables. The following subsections discuss the target market, customer segments, and the main value proposition of the proposed robot.

6.2.1 Business Opportunity and target market

The business opportunity for Nadel comes from the increasing interest in service robots and automation in hospitality environments. The International Federation of Robotics reported that professional service robot sales increased worldwide, with more than 205,000 units registered in 2023 [69]. Market studies also indicate strong growth expectations for hospitality and restaurant robots, driven by service efficiency, labor pressure, and demand for automation-enabled customer experiences [70].

Nadel fits this opportunity because it targets a clear service problem. In busy restaurants, waiters repeatedly move between the kitchen and customer tables, customers may wait before receiving assistance, and service quality may vary during peak hours. Nadel helps by handling repetitive delivery and table-response tasks, while human staff remain responsible for hospitality, supervision, and high-value customer care. This makes the business opportunity realistic because the robot is positioned as a staff-support tool rather than a complete replacement for restaurant employees [1].

Nadel can target both the Egyptian market and international markets. The initial market can include restaurants, cafes, hotels, food courts, university cafeterias, and event venues in Egypt. These environments are suitable because they are indoor spaces with repeated service routes and identifiable tables. After further technical refinement, the solution can also be expanded to the MENA region and other international hospitality markets [1].

Restaurants and cafes that want to reduce repetitive waiter movement and improve customer experience.

Hotels and hospitality venues that need modern service support and guest-facing technology [1].

Food courts and large dining areas where delivery distance and staff workload are high.

University cafeterias and institutional dining environments that operate in controlled indoor spaces.

Smart restaurants that already use POS systems and are ready to integrate robotic service support.

6.2.2 Customer Segments and Value Proposition

The value proposition of Nadel is stronger than simple food delivery because the robot combines delivery, customer interaction, table-side request, and restaurant workflow support. Many waiter robots focus mainly on carrying plates from one point to another. Nadel adds local customization, table-side Call/Pay request, voice interaction, touch screen interaction, and a smart restaurant workflow around the robot [1].

Table 6-2: Value propositions offered by Nadel.

Value Proposition	Explanation
Service efficiency	Nadel can reduce repetitive walking tasks by moving between home, customer tables, and kitchen/service area.
Customer convenience	Customers can request the robot from the table using Call or Pay assistance buttons.
Multimodal interaction	Customers can interact using the touch screen UI or voice interaction after the robot arrives.
Modern restaurant image	The robot provides a smart and attractive customer-facing technology experience.
Workflow integration	Nadel connects table requests, ordering, delivery, and service status within a restaurant workflow.
Local customization	The prototype can be adapted to local restaurant layouts, language requirements, and service scenarios.
Lower development cost	The prototype cost is significantly lower than many imported commercial waiter robots, supporting local development potential.
Market exposure	The robot was demonstrated publicly in ICT and TransMEA environments, supporting business visibility and validation.

6.3.1 Channels

The main channels for Nadel should depend on demonstration and trust-building. Restaurants are more likely to adopt a robot if they can see it moving, carrying items, responding to customers, and returning to its home position. Therefore, direct demonstrations, exhibition participation, restaurant pilot programs, and partnerships with POS providers are important channels.

Direct visits and demonstrations to restaurant and cafe owners.

Public exhibitions such as ICT and TransMEA to show the robot to potential stakeholders.

University innovation events and entrepreneurship competitions.

Social media videos showing the robot performing a full service workflow.

Pilot deployment programs with selected restaurants to collect feedback and build credibility.

6.3.2 Customer Relationships

A service robot business cannot depend only on selling the robot. Customers need installation, mapping, staff training, maintenance, and software support. For Nadel, customer relationships should be built around technical support and operational reliability. Restaurant staff should be trained to use the table-side unit, assign tasks, load trays safely, and use the emergency procedures when needed [12].

Possible customer relationship models include installation support, staff onboarding, maintenance contracts, periodic software updates, replacement parts, and customization of the interface, language, menu, and table map. This relationship model is important because restaurants usually need continuous support after deployment.

6.4 OPERATIONAL MODEL AND KEY PARTNERSHIPS

This section describes the operational model required to support the deployment and maintenance of the Nadel service robot in hospitality environments. The model focuses on how Nadel can assist staff through table-side service requests, autonomous navigation, manual loading at the service area, delivery support, customer interaction, and routine maintenance. It also identifies the key partnerships needed for practical deployment, including hardware suppliers, manufacturing workshops, restaurant partners, academic incubators, and maintenance providers. The following subsections explain the key resources, activities, and partnerships that support the operation and future scalability of the proposed system.

6.4.1 Key Resources and Activities

The key resources of Nadel include physical hardware, embedded electronics, software modules, integration knowledge, and testing experience. The main hardware resources are the Jetson Orin Nano, ESP32 controllers, LiDAR, ultrasonic sensors, encoders, IMU, motor drivers, motors, battery, touch screen, speaker, and the table-side ESP32 Call/Pay unit. The main software resources are Ubuntu 22.04, ROS2 Humble, ROS2 Nav2, SLAM Toolbox, Whisper Small, Gemma 31B, the speech output layer, touch screen UI, and the POS workflow [9], [12].

Another important resource is the practical experience gained from building and demonstrating the prototype. Public exposure during ICT and TransMEA supports the credibility of the project and can help attract restaurants, academic partners, investors, and innovation sponsors.

The key activities required to commercialize Nadel include hardware manufacturing, body design improvement, robot assembly, electronics integration, software development, POS integration, restaurant mapping, testing, installation, staff training, maintenance, and customer support. If the product is scaled to multiple robots, additional activities will be needed, such as fleet management, cloud dashboard development, spare parts management, and remote diagnostics [12].

6.4.2 Key Partners

Nadel will require partnerships to move from prototype to commercial product. Hardware suppliers are needed for sensors, batteries, motor drivers, motors, screens, and embedded computers. Manufacturing partners are needed to produce the robot body with higher consistency and durability. POS providers can help integrate Nadel with existing restaurant systems. Restaurants and cafes are also important pilot partners because they provide real feedback about operational constraints, user behavior, and service expectations.

Hardware and robotics component suppliers.

Local manufacturing and body fabrication workshops.

Restaurant and cafe pilot partners.

POS and restaurant software providers.

Maintenance and after-sales service partners.

Academic incubators, accelerators, investors, and innovation programs.

Exhibition organizers and technology events for market visibility.

6.5 REVENUE MODELS AND COST ANALYSIS

This section presents the revenue models and cost analysis of the Nadel project. It examines the financial aspects of developing, deploying, and maintaining an autonomous service robot for hospitality environments. The analysis considers prototype hardware costs, sensors, embedded controllers, motors, power components, body manufacturing, and software development requirements. It also discusses possible revenue models such as direct robot sales, monthly leasing, Robot-as-a-Service, maintenance contracts, and software support subscriptions. The following subsections explain the expected revenue streams and the estimated cost structure of the proposed system.

6.5.1 REVENUE MODELS AND COST ANALYSIS

Nadel can generate revenue through more than one model. A single model may not suit all customers because small cafes, large restaurants, hotels, and event venues have different budgets. Therefore, the proposed business model can combine hardware sales, leasing, Robot-as-a-Service, software subscription, maintenance, and customization.

Table 6-3: Possible revenue streams for Nadel.

Revenue Stream	Description	Suitable Customer Type
Direct sale	Restaurant buys the robot as a capital asset.	Large restaurants, hotels, food courts
Monthly leasing	Restaurant pays a monthly amount instead of full purchase.	Small and medium restaurants
Robot-as-a-Service	Robot is provided with service, maintenance, and updates under a recurring fee.	Customers who prefer operational expense model
Maintenance contract	Periodic inspection, repair, and replacement parts.	All deployed customers
Customization fees	Custom body, UI design, language, table map, or workflow integration.	Hotels, branded restaurants, special venues

6.5.2 Cost Structure and Prototype Cost

The prototype cost is a key business advantage of Nadel. The project cost is estimated at approximately 1,500 to 2,000 USD based on the main components and manufacturing cost. This estimate should be treated as a prototype development cost, not as the final commercial selling price. A commercial version would also include manufacturing quality control, packaging, warranty, technical support, safety testing, spare parts, software maintenance, marketing, and profit margin.

Table 6-4: Prototype cost breakdown used for the business model.

Cost Item	Estimated Cost
RPLIDAR A1M8 360-degree LiDAR	\$177
Wheels, motors, wheel assemblies, 2 motor drivers, and 2 caster wheels	\$220
10-inch touch screen display	\$240
NVIDIA Jetson Orin Nano development kit / edge module	\$250
Model body and 3D/body manufacturing	\$400
Additional electronics, wiring, battery, regulators, ESP32 units, and integration overhead	Included within total project estimation
Total prototype cost	\$1500 - \$2,000

6.6 MARKET VALIDATION AND COMPETITIVE ANALYSIS

This section evaluates the market readiness of Nadel and compares its proposed capabilities with existing service robot solutions. The analysis considers customer interest, practical feedback from demonstrations, expected deployment needs, and the suitability of Nadel for hospitality environments such as restaurants, cafés, hotels, and food courts. It also compares Nadel with commercial service robots in terms of navigation, table-side service requests, interaction methods, hardware accessibility, prototype cost, and local customization. The following subsections present the competitor comparison, market validation activities, and SWOT analysis of the proposed system.

6.6.1 Competitor and Price Comparison

Commercial waiter robots already exist in the international market. Examples include PUDU BellaBot, KEENON DINERBOT, and Bear Robotics Servi+. These products demonstrate that robotic restaurant service is a real market direction, but they are often expensive and may be sold through distributors, leasing, or quote-based pricing. This creates an opportunity for a locally developed robot such as Nadel, especially if it can provide a lower-cost and more customizable alternative for local and regional restaurants [1].

PUDU BellaBot Pro is a commercial delivery robot with a 40 kg maximum load capacity, 10 kg per tray, about 11 hours of no-load runtime, and VSLAM/LiDAR navigation options [71]. RobotLAB lists BellaBot direct purchase from about 15,900 USD or RaaS leasing from about 2,430 USD per month [72]. Bear Robotics Servi+ is presented as a restaurant robot that can navigate tight spaces and reroute around obstacles; RobotLAB lists Servi+ at about 16,000 USD or RaaS from about 293 USD per month [73], [74]. KEENON DINERBOT T8 and T9 provide restaurant delivery features such as stereo vision, route planning, and long battery-life options, but pricing is usually dependent on distributor quotation and deployment package [75], [76].

Table 6-5: Indicative comparison between Nadel and selected waiter robots.

Robot	Indicative Cost / Pricing Model	Main Features	Strengths	Nadel Position
Nadel	Prototype cost around \$1500-\$2,000	ROS2 Nav2, SLAM Toolbox, LiDAR, touch UI, voice interaction, table-side Call/Pay request	Locally developed, customizable, low prototype cost, POS/table request integration	Cost-efficient prototype with strong development potential
PUDU BellaBot / BellaBot Pro	Commercial purchase listed from about \$15,900 by RobotLAB; other regional prices vary	Multi-tray delivery, LiDAR/VSLAM navigation, obstacle avoidance, interactive design	Mature commercial robot with polished industrial design	Nadel can compete through lower cost and local customization
Bear Robotics Servi+	Commercial purchase listed around \$16,000 by RobotLAB; RaaS option available	Restaurant delivery, obstacle detection, rerouting, multiple trays	Strong commercial deployment model and service support	Nadel needs further reliability and support maturity to compete
KEENON DINERBOT T8/T9	Often quote-based depending on distributor and deployment package	Stereo vision, path planning, multi-tray delivery, touch screen, long battery options	Established commercial ecosystem and fleet capabilities	Nadel can differentiate through local restaurant workflow and table-side request unit

The comparison shows that Nadel is not yet equivalent to mature commercial products in industrial finish, certification, long-term reliability, or after-sales support. However, it has a strong advantage as a locally developed and lower-cost prototype. The main business strategy should not claim that Nadel already replaces global products.

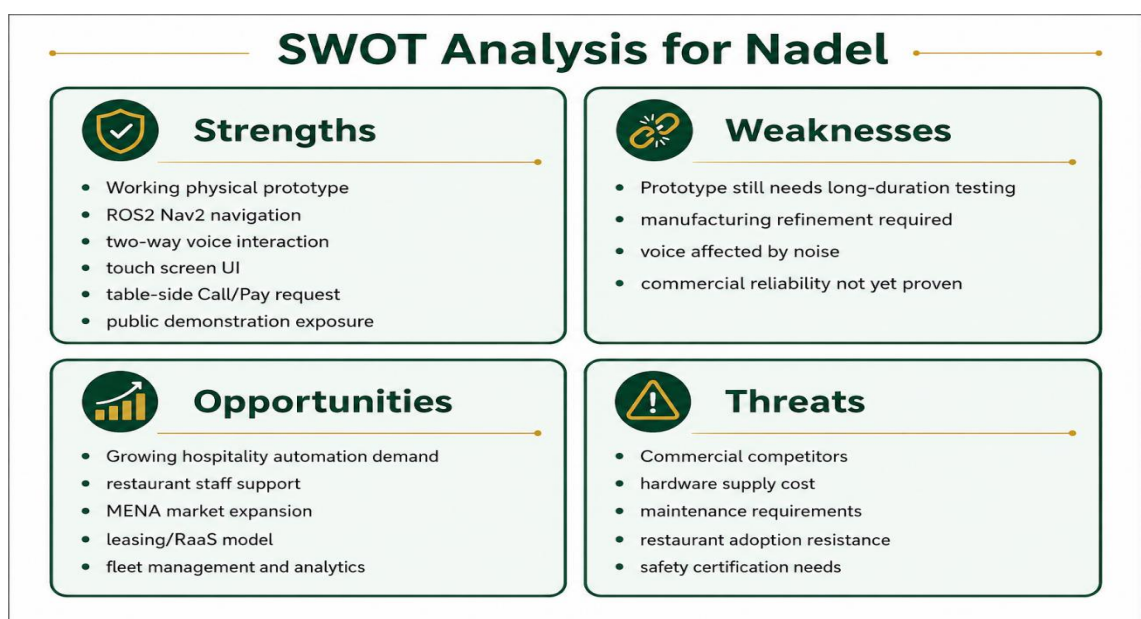
6.6.2 Market Validation through ICT and TransMEA

Nadel was not only tested internally; it was also demonstrated in public exhibition environments, including ICT and TransMEA participation under the Arab Academy representation. This public exposure is important for business validation because it shows that the robot concept can attract attention from visitors, representatives, and technology audiences. While public demonstration does not equal commercial market proof, it provides early validation that the project has customer-facing appeal and can be presented as a serious business-oriented prototype.

The exhibition demonstration helped position Nadel as more than a classroom prototype. It showed the robot in a realistic public context, where it could be observed by people outside the project team. This strengthens the business argument because restaurants and technology partners often need to see practical evidence before considering adoption. The demonstration also supports future fundraising, incubation, and pilot deployment discussions.

6.6.3 SWOT Analysis

A SWOT analysis helps identify the internal strengths and weaknesses of Nadel, as well as the external opportunities and threats that may affect commercialization. The purpose is not to claim that the project is already market-ready, but to evaluate how it can move from a graduation prototype toward a business product.



6.7 FEASIBILITY, SCALABILITY, AND FUTURE ENHANCEMENTS

Nadel has strong feasibility as a prototype and promising scalability as a future business. The feasibility is supported by the fact that the robot was built, tested, and publicly demonstrated. It can move from home position to the requesting table, interact with the customer, return to the kitchen/service area, deliver the requested item, and return home after completing the workflow. The scalability is supported by the possibility of creating multiple units, improving the body design, adding autonomous charging, and connecting the robot fleet to a central restaurant dashboard.

However, successful business scaling depends on several conditions. The robot must become more robust under long-duration operation, crowded environments, and noisy spaces. Manufacturing quality should be improved so that every unit has consistent mechanical and electrical reliability. The system should also include clear installation procedures, mapping services, safety documentation, and after-sales support. These requirements are essential before Nadel can be deployed commercially across restaurants in Egypt or internationally [12].

The future development plan aims to improve both the technical and business readiness of Nadel. Hardware improvements can include upgrading to Jetson Orin AGX, using brushless motors, adding an automatic charging station, refining the mechanical body, and improving battery endurance. Software improvements can include enhanced speech interaction, swarm or multi-robot coordination, fault tolerance, recovery behavior, remote monitoring, and analytics.

From a business perspective, the most important future enhancements are autonomous charging, fleet management, stronger fault recovery, improved voice interaction under noise, and better restaurant analytics. These features can convert Nadel from a single prototype into a scalable restaurant automation platform. If developed further, Nadel can be offered as a direct sale product, a leasing package, or a managed Robot-as-a-Service solution.

The business model of Nadel is based on providing restaurants with an intelligent autonomous service robot that supports delivery, table-side request, voice and touch interaction, and smart workflow integration. The main customers include restaurants, cafes, hotels, food courts, and smart hospitality environments in Egypt and abroad. The main value proposition is a locally developed, customizable, lower-cost service robot that can support staff, improve customer experience, and modernize restaurant operations [1].

The prototype cost of approximately 2,000 to 2,500 USD gives Nadel a strong development advantage when compared with many imported commercial waiter robots, which may cost above 10,000 to 16,000 USD depending on brand, distributor, and deployment package. Nevertheless, the chapter presents the business potential realistically: Nadel still needs more reliability testing, manufacturing refinement, formal safety validation, and customer support planning before full commercial deployment. With these improvements, the project has strong potential to achieve success in the coming period as a locally developed smart restaurant robotics solution [1].

7 CONCLUSION AND FUTURE WORK

This chapter concludes the Nadel graduation project by summarizing the main achievements, technical outcomes, practical contribution, limitations, and future development directions of the system. The purpose of the chapter is not only to close the report, but also to show how the implemented prototype can be extended into a stronger and more commercially scalable restaurant service robot.

Nadel was developed as a fully autonomous service waiter robot that combines mobile robotics, restaurant workflow automation, touch screen interaction, voice interaction, and table-side request support. The system was not limited to a simulation or a conceptual proposal. A physical prototype was built, tested, demonstrated, and used to execute a complete service scenario in which the robot moved from its home position to a customer table, received interaction from the customer, returned to the kitchen or service area, collected the requested item, delivered it back to the customer, and then returned to its home position.

The project therefore demonstrates the integration of robotic navigation, embedded control, sensing, human-robot interaction, and smart restaurant service workflow in one working prototype. It also provides a strong foundation for future improvement toward a more advanced commercial version.

7.1 OVERALL CONCLUSION AND ACHIEVEMENTS OF PROJECT OBJECTIVES

The main conclusion of this project is that Nadel successfully achieved its core objective: building an autonomous restaurant service robot capable of supporting customer service through movement, interaction, and restaurant workflow integration. The robot was designed to operate in indoor environments and to support a practical restaurant scenario rather than only demonstrating isolated technical modules.

From the interaction side, Nadel provided multiple ways for the customer to communicate with the robot. The customer can interact through the touch screen interface or through voice. In the voice layer, Whisper Small converts speech to text, Gemma 31B interprets the request and generates a text response, and the speech output layer allows the robot to reply by voice. This makes the interaction more natural and supports a more realistic service experience [18], [20].

From the restaurant service side, Nadel was connected to a workflow in which table requests, order interaction, delivery tasks, kitchen or service-area movement, and return-to-home behavior are organized as one complete process. The table-side Call/Pay unit is important because it allows the customer to request the robot from the table. The buttons do not directly perform service or payment operations; instead, they send table ID and request intent so that Nadel can navigate to the customer and continue the interaction through the robot interface and restaurant workflow.

The project achieved the main objectives defined in the earlier chapters. The robot was able to move, interact, respond to requests, support table-side service flow, and demonstrate the feasibility of an autonomous restaurant service prototype. Table 7-1 summarizes the main achievements of the project.

Table 7-1: Summary of main project achievements.

Project Objective	Implementation in Nadel	Achievement Status
Build a working restaurant service robot	A physical mobile prototype was built with shelves/tray area, touch screen, sensors, motor system, power system, and safety controls.	Achieved
Enable autonomous navigation	ROS2 Humble, ROS2 Nav2, SLAM Toolbox, LiDAR, odometry, and IMU-based feedback were used to support navigation and movement.	Achieved
Support customer interaction	The robot supports touch screen interaction and two-way voice interaction using Whisper Small, Gemma 31B, and speech output.	Achieved
Support table-side request	An ESP32-based Call/Pay table unit was used to request the robot with table ID and request intent.	Achieved
Complete service workflow	The robot demonstrated a full sequence from home to table, order interaction, kitchen/service area, delivery, and return home.	Achieved
Demonstrate real-world potential	The project was presented and demonstrated in public exhibition environments including ICT and TransMEA participation.	Achieved with further development required

The achievement status should be understood in the context of a graduation prototype. The system demonstrated the intended functional behavior, but it still requires extended quantitative testing, manufacturing refinement, and safety validation before being considered ready for commercial deployment.

7.1.1 VALIDATION OF THE COMPLETE SERVICE SCENARIO

One of the strongest results of the project is that Nadel was tested in a complete service scenario instead of only testing separate hardware modules. The service scenario started when the customer pressed the table-side request button. The request indicated the target table and request intent. Nadel then moved from its home position toward the customer table. After the robot arrived, the customer was able to interact with it through the touch screen interface and/or voice interaction layer.

After receiving the order or service interaction, Nadel moved to the kitchen or service area to collect the requested item. Once the item was placed on the robot tray, the robot returned to the customer table and completed the delivery. After the service interaction was completed, the robot returned to its predefined home position. This confirms that the system was able to connect table-side request, navigation, interaction, service-area movement, delivery, and return-home behavior into one operational workflow.

Table 7-2: End-to-end service scenario validated by Nadel.

Stage	Action	Purpose	Result
1	Customer presses Call/Pay button	Request Nadel from the table	Robot receives request intent and target table
2	Robot leaves home position	Move from idle/home location to customer table	Robot navigates toward the requester
3	Customer interacts with robot	Use voice or touch screen interface	Order/service interaction is completed
4	Robot goes to kitchen/service area	Collect requested item	Item is placed on robot tray
5	Robot returns to customer table	Deliver requested item	Delivery is completed
6	Robot returns home	Prepare for the next request	Robot returns to predefined home position

7.1.2 TECHNICAL,PRACTICAL, AND BUISNESS CONTRIBUTIONS

The technical contribution of Nadel comes from the integration of multiple robotic and intelligent system layers into one working prototype. Many student projects focus on one isolated module, such as navigation, hardware design, or user interface. Nadel combines these layers and demonstrates how they can work together in a real service scenario.

The high-level processing layer is centered on the Jetson Orin Nano, which supports ROS2-based navigation, AI-related computation, user interaction, and system coordination. The low-level control layer is based on ESP32, motor drivers, geared motors, encoders, voltage regulation, and safety controls. The sensing layer includes LiDAR, depth camera, ultrasonic sensors, encoder feedback, and IMU feedback. The interaction layer includes touch screen UI, voice input, Gemma 31B response generation, and speech output. The restaurant workflow layer includes table-side requests, POS/service interaction, kitchen/service-area movement, delivery, and home return [9], [13].

This layered design gives the project a clear engineering structure. It also makes future upgrades easier because hardware, navigation, interaction, and business workflow can be improved independently without redesigning the whole system from the beginning.

Integration of ROS2 Humble, Nav2, and SLAM Toolbox in a working service robot prototype [9], [12].

Use of Jetson Orin Nano for high-level robotics, interaction, and AI-related processing [15].

Use of ESP32 for embedded robot control and table-side IoT request handling [16].

Combination of touch screen UI and two-way voice interaction in the same robot.

End-to-end service workflow from table request to delivery and return home.

Practical demonstration in public exhibition environments, not only internal laboratory testing.

Beyond the technical implementation, Nadel has a practical business contribution. The project addresses a real need in restaurants and hospitality environments: repetitive movement between tables and service areas, delay during busy periods, customer waiting time, and the need for more organized service workflows. By automating part of the delivery and customer interaction process, Nadel can support staff rather than replace them completely [1].

The public demonstration of Nadel during ICT and TransMEA activities provided an important early form of market validation. These demonstrations allowed the robot to be presented in front of visitors, representatives, and a non-laboratory audience. Although public demonstration is not a replacement for formal commercial pilot testing, it provided evidence that the idea is understandable, presentable, and attractive as a service robot concept.

The project also has strong commercial development potential. The current prototype already includes key features required by smart restaurant solutions: autonomous movement, table request support, touch screen ordering, voice interaction, delivery capability, and POS/service workflow integration. With further engineering refinement, improved manufacturing, stronger safety validation, and business deployment planning, Nadel can be developed into a commercial product for restaurants, cafes, hotels, food courts, university cafeterias, and similar indoor service environments in Egypt and outside Egypt.

7.2 LIMITATIONS AND LESSONS LEARNED

Although Nadel achieved its main functional objectives, the project still has limitations. These limitations are expected in a graduation prototype and should be treated as future improvement directions rather than project failures. A realistic discussion of limitations makes the project more credible and helps define the next engineering stage.

Table 7-3: Main limitations and lessons learned from the project.

Area	Observed Limitation	Lesson Learned / Future Direction
Voice interaction	Background noise can affect speech recognition and response reliability.	Improve speech processing and microphone/noise handling.
Navigation	Crowded spaces and dynamic human movement can affect smooth navigation.	Increase testing in realistic restaurant conditions and improve recovery behavior.
Odometry and localization	Long operation may introduce drift or small localization errors.	Improve sensor fusion and repeated calibration procedures.
Power system	Battery endurance needs longer-duration testing under real service load.	Upgrade battery capacity and add automatic charging.
Mechanical body	The prototype body is functional but still needs commercial manufacturing refinement.	Develop a more robust, lighter, and service-ready body.
Quantitative testing	Some results were observed qualitatively without full statistical benchmarking.	Record repeated trials, time measurements, success rates, and failure modes.

The most important lesson learned is that a restaurant service robot requires more than movement. It must combine reliable navigation, safe motion, user-friendly interaction, workflow integration, power management, mechanical stability, and maintenance planning. Nadel successfully demonstrated the foundation of this integrated system, and the remaining limitations identify the direction for its next stage.

7.3 FUTURE WORK

The future work of Nadel focuses on transforming the current working prototype into a more advanced, reliable, scalable, and commercially ready service robot. The planned improvements can be grouped into hardware upgrades, software and AI improvements, operational scalability, and commercial development.

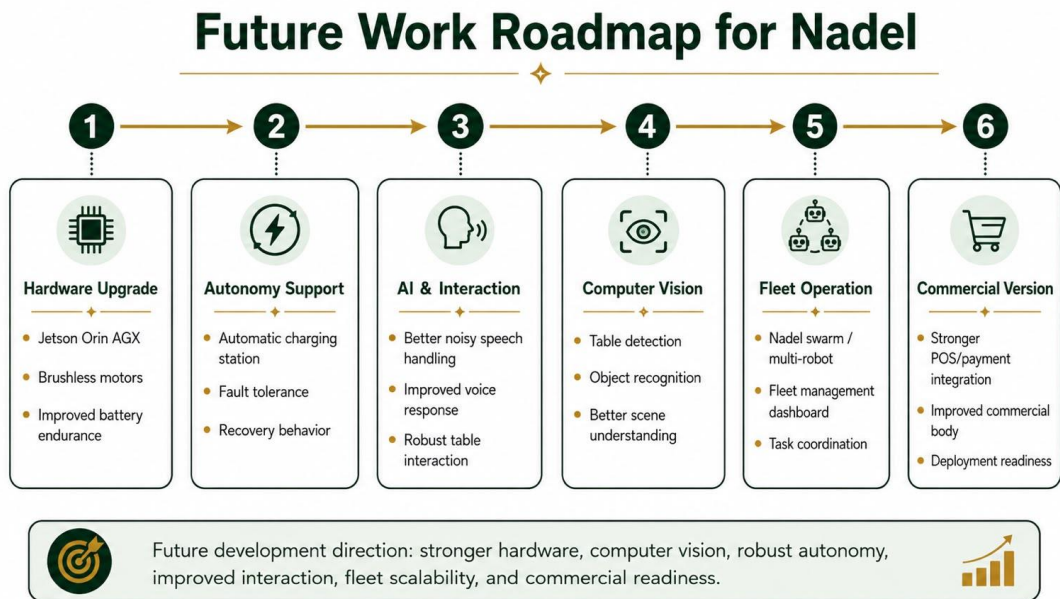


Figure 7-2: Future work roadmap for Nadel. Source: Authors' diagram.

7.3.1 Hardware Future Work

The first future direction is improving the hardware platform. The current Jetson Orin Nano provides a strong base for the prototype, but a future commercial version can be upgraded to Jetson Orin AGX to provide stronger AI and perception performance. This upgrade would support more demanding computer vision, multi-sensor processing, and advanced real-time decision-making [15].

Another important hardware improvement is replacing the current motor system with brushless motors. Brushless motors can improve efficiency, durability, torque control, and long-term reliability. This is important because commercial restaurant robots are expected to operate for long periods and repeat many delivery cycles per day.

The power system should also be improved through better battery endurance and an automatic charging station. Instead of relying only on manual charging, Nadel can

return to a docking station when the battery level becomes low. This would improve operational independence and make the robot more suitable for daily restaurant use.

Upgrade the main processing unit to Jetson Orin AGX for stronger edge AI performance.

Use brushless motors to improve efficiency, durability, and motion reliability.

Improve battery endurance for longer operating time.

Add an automatic charging station so the robot can dock and recharge without manual intervention.

Improve the commercial body design to make the robot lighter, stronger, safer, and easier to manufacture.

7.3.2 Software and AI Future Improvements

The second future direction is improving the software and AI layer. Voice interaction should be improved for noisy environments because real restaurants can include customer conversations, kitchen noise, music, and movement. Better microphone placement, noise reduction, voice activity detection, and speech-processing optimization can improve the reliability of spoken interaction.

Fault tolerance and recovery should also be added. A commercial robot should not stop permanently when it faces a navigation issue, communication delay, sensor reading problem, or temporary obstacle. Instead, it should detect the fault, recover safely, inform the system, and continue operation when possible. This would make Nadel more reliable in real deployment.

A fleet management dashboard is another important future feature. If multiple Nadel robots are used in one restaurant, hotel, or food court, the system should assign tasks, monitor robot status, distribute workload, track battery level, and prevent route conflicts between robots. This would transform Nadel from a single robot into a scalable service robot platform.

Improve speech interaction in noisy restaurant environments.

Add fault tolerance and recovery behavior for navigation, communication, and system errors.

Make Nadel swarm / multi-robot system to allow multiple robots to work together.

Add a fleet management dashboard for monitoring robot location, battery, task status, and service queue.

Improve POS/payment integration so payment-related interaction can become smoother and more connected to restaurant operations.

7.3.3 Commercial and Operational Future Development

The third future direction is commercial development. The project already showed a strong proof of concept and public interest during exhibition demonstration. However, moving from graduation prototype to commercial product requires additional steps. These include repeated testing, safety refinement, cost optimization, maintenance planning, spare part availability, customer training, and restaurant pilot deployment.

The commercial body should be improved to match restaurant expectations. It should be durable, easy to clean, visually attractive, safe around customers, and simple to maintain. The tray system can also be improved to support different dish sizes, cup holders, and anti-slip surfaces. The table-side Call/Pay unit can be further developed into a more polished customer service accessory.

Future business development should also test Nadel in real restaurants, cafes, university cafeterias, and hotel service areas. Pilot trials would allow the team to collect delivery time, success rate, customer feedback, staff feedback, battery endurance, and maintenance data. This data would help convert Nadel from a successful prototype into a validated product.

Table 7-4: Future work plan for Nadel.

Future Improvement	Purpose	Expected Benefit
Jetson Orin AGX upgrade	Increase AI and perception processing capacity.	Better performance for advanced vision and multi-task processing.
Brushless motors	Improve motion efficiency and reliability.	Smoother movement and better long-term durability.
Automatic charging station	Allow the robot to recharge autonomously.	Longer daily operation with less manual intervention.
Noisy-environment speech improvement	Improve voice interaction in real restaurants.	More reliable customer communication.
Nadel swarm / multi-robot system	Allow several robots to work together.	Scalable deployment in larger restaurants and hotels.
Fault tolerance and recovery	Allow the system to recover from errors safely.	Higher reliability and safer operation.
Fleet management dashboard	Monitor and assign tasks to multiple robots.	Better operational control and business scalability.
Stronger POS/payment integration	Improve order and payment-related workflow.	Smoother restaurant operations and better user experience.
Improved commercial body	Refine manufacturing, safety, cleaning, and appearance.	Higher market readiness and stronger product identity.

Nadel represents a successful graduation project because it moved beyond theory and produced a working robot that can be demonstrated in real service scenarios. The project combined mechanical design, embedded systems, ROS2 navigation, AI-based interaction, touch screen UI, table-side request handling, delivery workflow, safety controls, and public demonstration. This level of integration shows that the project is not only an academic exercise, but also a strong foundation for a future product [9].

The system is not claimed to be a final commercial robot yet. It still needs longer testing, stronger safety validation, commercial body refinement, better noisy-environment speech handling, improved battery endurance, and more formal market trials. However, the current prototype proves that the idea is technically feasible, practically demonstrable, and commercially promising.

With the planned upgrades, Nadel can be developed into a scalable smart restaurant service platform. Its ability to combine autonomous delivery, table-side request, touch interaction, voice interaction, POS workflow, and public-ready demonstration.

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APPENDIX A: PROJECT PROTOTYPE EVIDENCE

This appendix provides a concise record of the main prototype evidence used throughout the report. The Nadel robot was implemented as a physical waiter robot prototype with a mobile base, tray structure, touch screen interface, sensing layer, embedded controller, power system, and table-side Call/Pay request unit.

The prototype evidence includes front, side, and rear views of the robot body; component illustrations for the Jetson Orin Nano, ESP32, RPLIDAR A1M8, MPU6050 IMU, encoder module, motor system, wheels, battery pack, power switch, and emergency stop; and workflow diagrams showing the restaurant service sequence.

All figures included in the main chapters should be treated as authors' diagrams, authors' prototype photographs, or authors' component illustrations unless a different source is explicitly stated in the caption.

APPENDIX B: TESTING AND DEMONSTRATION EVIDENCE

The main testing evidence for Nadel is based on internal functional testing and public demonstration. The tested service sequence included the table request, robot movement from home to the customer table, customer interaction, movement to the kitchen or service area, item loading, delivery back to the customer, and return to home.

Nadel was demonstrated during ICT and TransMEA participation under the Arab Academy representation. These demonstrations were used as practical market and public validation because the robot was shown in a dynamic non-laboratory environment with visitors, observers, and public interaction.

The public demonstrations do not replace controlled statistical benchmarking. However, they provide strong evidence that the prototype was complete enough to be presented and operated as an applied intelligent service robot system.

APPENDIX C: FINAL TECHNICAL CONFIGURATION

Configuration Item	Final Project Configuration
Main processor	NVIDIA Jetson Orin Nano
Operating system	Ubuntu 22.04
Robotics middleware	ROS2 Humble
Navigation stack	ROS2 Nav2
Mapping	SLAM Toolbox
Localization support	Odometry, LiDAR, and IMU
Main perception sensors	RPLIDAR A1M8, ultrasonic sensors, encoders, and MPU6050 IMU
Embedded control	ESP32 robot controller
Drive system	Two RS775SH geared DC motors, two IBT-2 BTS7960 motor drivers, two main wheels, and two metallic caster wheels
Voice interaction	Whisper Small for STT, Gemma 31B for request interpretation and response generation, and a speech output layer for voice reply
Customer interface	Touch screen UI and table-side ESP32 Call/Pay unit
Safety and power	36V battery pack, voltage regulators, power switch, and emergency stop